2

Design of Low Dropout (LDO) Regulators

Low dropout (LDO) regulators are widely used in portable electronic devices because they occupy small chip and printed circuit board (PCB) areas. The performance advantages of these regulators include low quiescent current and wide bandwidth (BW), which result in fast transit response. Unlike switching regulators (SWRs), LDO regulators convert voltage through a linear operation, and thus a well-regulated output voltage can be derived without the occurrence of output voltage ripples. However, LDO regulators suffer from an inherent disadvantage, namely poor power conversion efficiency (PCE) if the ratio of the output voltage to the input voltage is small, because a large voltage stress over the pass transistor causes significant power loss. When considering the small silicon area, compact PCB, and reduced discrete component costs, an LDO regulator is one candidate that can provide a conversion function for regulated and scaled-down voltages. LDO regulators are frequently utilized as post-regulators in series with SWRs to suppress voltage ripples from the switching operation of SWRs caused by their large open-loop gain. In general, the combination of an SWR in series with an LDO regulator can be viewed as a simple and efficient power management module if the ratio of the output voltage to the input voltage, as well as the open-loop gain, can remain large in the LDO regulator. An LDO regulator is regarded as a voltage buffer for decreasing voltage ripples at the cost of slightly reduced PCE.

LDO regulators have structures that are defined according to their controlling methods and compensated skills. As illustrated in Figure 2.1, LDO regulators may either be analog-low dropout (A-LDO) regulators or digital-low dropout (D-LDO) regulators. The controller of the former is designed with analog circuits, whereas that of the latter is designed with digital circuits. A-LDO regulators have two major categories, namely dominant pole compensation, which involves a large compensation capacitor C_{out} located at the output node, and a capacitor-free (*C-free*) structure, which involves a dominant pole generated by Miller compensation capacitance.

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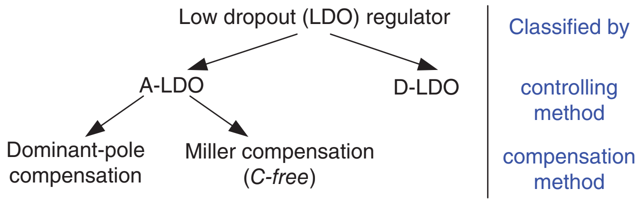

Figure 2.1 Classification of LDO regulators

Applications determine the suitable type of LDO to use. Various multimedia and portable devices demand Soc integration. Soc typically requires a compact-size, fast-transient, low-noise supply voltage. A *C-free* structure can be made compact at the expense of a large quiescent current and transient voltage variation. An A-LDO regulator with dominant pole compensation improves the transient voltage variation and reduces the quiescent current at the cost of a large output capacitor. When considering a D-LDO regulator as the post-regulator, an improvement in transient response and regulator size leads to the feasibility of integration; however, a trade-off occurs between regulation performance and quiescent current.

In this chapter, the basic LDO regulator is first introduced. Then, concerns over compensation for loop stability are presented to develop dominant pole compensation and *C-free* structures. Design flow and tips are also illustrated to understand the specifications and performance in A-LDO regulators. The characteristics of A-LDO and D-LDO regulators are then discussed and compared. LDO regulators with specific features are introduced to satisfy the requirements of various applications. Furthermore, low-power techniques, including novel dynamic-voltage scaling (DVS) and analog dynamic-voltage scaling (ADVS), are introduced for compatible utilization in Soc applications.

2.1 Basic LDO Architecture

In general, the voltage drop across power transistors in LDOs results in unavoidable heat in the power transistors. LDOs are appropriate for low-power applications because of heat dissipation. A simple LDO consists of an error amplifier (EA) and power transistors to eliminate ripples and regulate output voltage. LDO regulators have the smallest layout and footprint area, and a low quiescent current to provide small volume and high current efficiency. Figure 2.2 illustrates the architecture of a basic LDO regulator. The dropout voltage across the pass transistors is defined as the voltage difference between the input supply voltage and the output voltage. A small dropout voltage can lead to satisfactory PCE. However, decreasing the dropout voltage of an LDO may decrease the gain of the last stage, and thus deteriorate the regulation performance. A trade-off occurs between PCE and regulation performance. Designers generally use large power transistors to reduce the dropout voltage effectively and take advantage of its low on-resistance. However, large power transistors increase the silicon area, and thus the chip cost also increases. Moreover, the transient response is slowed down by the limited slew rate designed in the EA when driving at the gate of the power transistor. Given the differences between the reference voltage V_{REF} and the feedback voltage V_{FB}, the output voltage of the EA

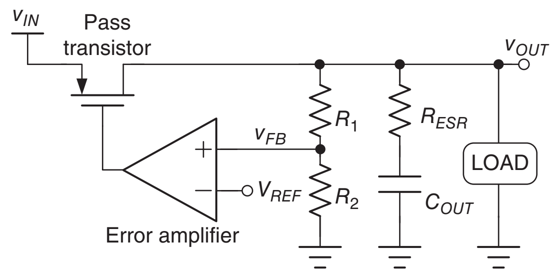

Figure 2.2 Structure of a basic LDO regulator

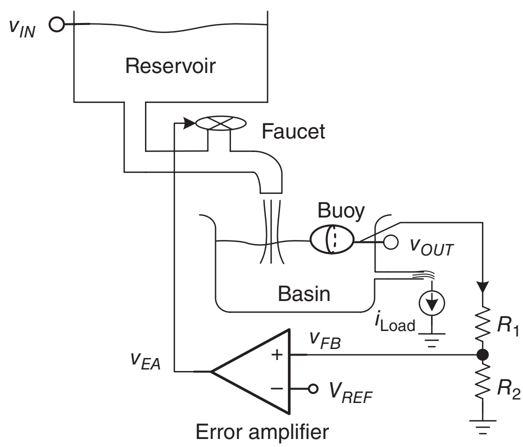

Figure 2.3 Operation of LDOs explained using an emulated reservoir and basin system controlled by a buoy with closed-loop system

controls the gate voltage of the pass transistors, and thus modulates the current flowing through the pass transistors.

Given the negative feedback control, LDO regulators can regulate the output voltage irrespective of load current and input voltage variations if the loop gain is sufficiently large. Any switching disturbance from a pre-regulator, such as SWRs, can be effectively reduced by an effective loop gain. However, an increased loop gain may cause stability deterioration because high-frequency poles will be higher than the crossover frequency. Hence, another trade-off occurs between the regulation performance and system stability.

As illustrated in Figure 2.3, the operation of LDOs can be explained by an emulated reservoir and basin system, which is controlled by a buoy with a closed-loop system. The input voltage V_{IN} is emulated by the reservoir. The power transistor functions as a faucet, and the charge stored in an output capacitor can be regarded as the water stored in a basin. The pipe aperture of the faucet implies driving capability. The height of the water in the basin represents the level of the output voltage. The load current can be viewed as water flowing out at the bottom of a basin. To control the water level at a constant height, similar to a regulated voltage level labeled

V_{OUT} in LDOs, a closed-loop system should be formed using the buoy to detect the water level. The detected water level is scaled down and compared with the predefined reference voltage V_{REF} to obtain an error control signal V_{EA}. That is, V_{EA} can be used to determine how tight or loose the faucet is. Water flowing out of the faucet can correspond to the value of V_{EA}. Under a dynamic equilibrium condition, the water flowing into and out of a basin remains in dynamic balance, which is maintained by the negative feedback system. In the emulated reservoir and basin system, only the water level that is similar to the output voltage level in LDOs is monitored. Thus, the system is called a voltage-mode control system.

How fast the faucet can be turned tightly or loosely determines how fast the water level can reach the desired level in case of a sudden change in output load current. In the emulated reservoir and basin system, the water level is controlled by the negative feedback closed-loop system, wherein the water level is sensed by the buoy. A large pipe aperture of the faucet can deliver high current to the basin, but the faucet experiences driving difficulty of the EA output. Thus, turning the faucet to an adequate position requires good driving capability of the EA. Furthermore, the water level is easily maintained at a constant level by a basin with a large volume, because a sudden considerable change in loading current cannot instantly disturb the water level of a large basin even if the faucet has not been turned to an adequate position. Moreover, if the output load current changes from high to low, then the water level will remain high for a short period because of the absence of an extra path for surplus water to flow out of the basin. To reach the desired water level, an extra path where water can rapidly flow out of the basin is required, which leads to wastage of valuable water. Evidently, this approach is not energy efficient and not environmentally friendly. If cost and energy efficiency are considered, a bulky basin is not the appropriate design to keep the water level constant.

In summary, two concerns are worth mentioning. First, the driving capability of the EA depends on the faucet size, and affects the rapid delivery of the current from the reservoir to the basin. Second, the basin size determines the volume of water stored in it and influences the maintenance of the water level in case of any sudden change in output load current. These two issues can be mapped into similar LDO designs, and thus two possible large capacitances should be considered carefully in LDO design. The first large capacitance is located at the output node and the second is located at the output of the EA, which can be mapped into two capacitances at V_{OUT} and V_{EA} in Figure 2.3, respectively. The values of these capacitances determine the types of suitable LDO topology. Therefore, application requirements have to be considered first. One of the possible topologies has to be selected to satisfy all specifications. In the following sections, compensation and advanced techniques are introduced to enhance LDO performance.

2.1.1 Types of Pass Device

The faucet is the power transistor. Thus, the type of power transistor suitable for LDO designs should be identified. All possible pass transistor designs are shown in Figure 2.4. Considering the type of bipolar transistor (BJT) as pass transistor, NPN Darlington, NPN, and PNP shown in Figure 2.4(a)–(c), respectively, can be used to obtain the advantage of high driving capability because of the high current gain of BJTs. However, BJTs have two obvious disadvantages. First, the dropout voltage required by the voltage headroom, which consists of base/emitter and collector/emitter voltages, to keep the BJT working in the active region

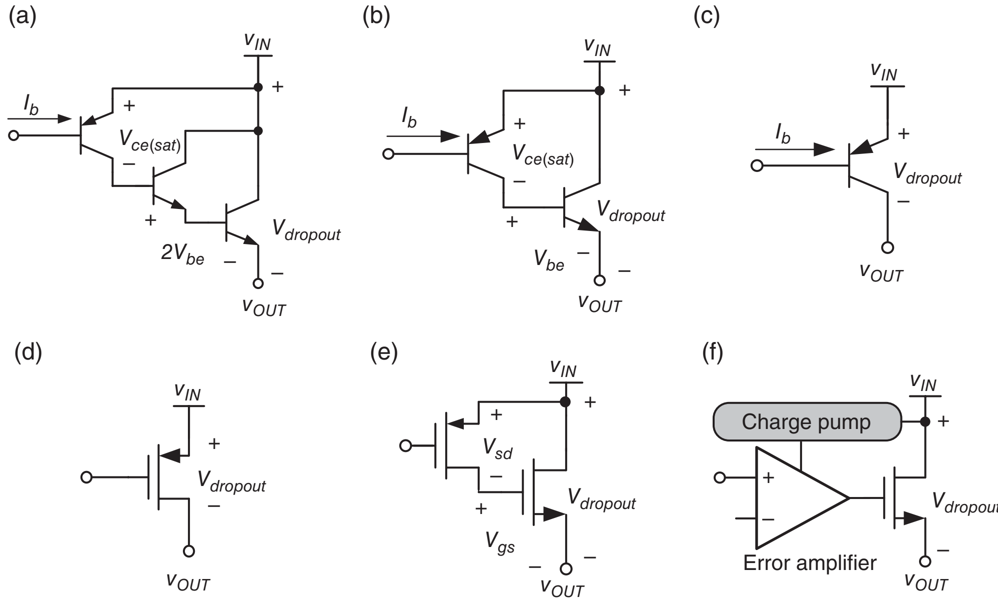

Figure 2.4 Pass devices include (a) NPN Darlington, (b) NPN, (c) PNP, (d) N-MOSFET, (e) P-MOSFET, and (f) N-MOSFET with one charge pump circuit

is large. In Figure 2.4(a)–(c), the dropout voltage is $V_{ce(sat)} + 2V_{be}$, $V_{ce(sat)} + V_{be}$, and $V_{ec(sat)}$, respectively. Thus, in low-quiescent and low-power Soc applications, Darlington NPN and NPN configurations are unsuitable because of their large dropout voltage. By contrast, the PNP configuration is the most appropriate configuration because its minimum dropout voltage is only $V_{ec(sat)}$, which is typically 0.4 V. The second critical disadvantage is the large leakage current at the base terminal. The base current is basically proportional to the value of I_C/β_n or I_C/β_p, where I_C is the collector current and β_n, β_p are the current gains of the NPN and PNP transistors, respectively. In general, β_n and β_p are within the ranges of 50–100 and 5–10, respectively. The leakage current caused by the non-zero base current is strongly affected by current gain. If a LDO voltage is required, then the most appropriate BJT is the PNP transistor; however, this transistor suffers from a large leakage current because of its low β_p compared with the NPN transistor, which has a large β_n.

A MOSFET is selected to replace the BJT as the power transistor to eliminate these two critical and obvious disadvantages, as shown in Figure 2.4(d)–(f). In general, no DC flows into or out of the gate terminal of the MOSFET. Thus, the leakage current problem can be solved completely and low quiescence is guaranteed. However, the driving capability of a MOSFET is considerably smaller than that of a BJT, and thus a large aspect ratio of the MOSFET is required to generate a similar driving current to that provided by a BJT. An obvious disadvantage is the large silicon area occupied by the MOSFET.

In Figure 2.4(d) the p-channel MOSFET (P-MOSFET) with large aspect ratio has a lower dropout voltage than its counterpart PNP transistor, which has a minimum dropout voltage limited to approximately 0.2 V. That is, if the LDO regulator requires LDO voltage and low quiescent current, then the P-MOSFET is the best choice among the various configurations.

Table 2.1 Comparison of different pass devices

	NPN Darlington	NPN	PNP	NMOS	PMOS
I_{Load}	High	High	High	Medium	Medium
I_q	Medium	Medium	Large	Low	Low
$V_{dropout}$	$V_{ce(sat)} + 2V_{be}$	$V_{ce(sat)} + V_{be}$	$V_{ec(sat)}$	$V_{gs} + V_{ds(sat)}$	$V_{sd(sat)}$
Speed	Fast	Fast	Slow	Medium	Medium
Compensation	Easy	Easy	Complex	Easy	Complex

However, the P-MOSFET also has several disadvantages. For example, it requires a larger silicon area to compensate for the lower mobility μ_p than in the n-channel MOSFET (N-MOSFET). Moreover, the P-MOSFET works as a common source (CS) stage in an LDO regulator. The gate-to-drain capacitance C_{gd} is amplified by the Miller effect and increases the difficulty of loop compensation. By contrast, the N-MOSFET shown in Figure 2.4(e) works as a source follower (SF) stage that functions as a buffer stage in an LDO regulator. Thus, the compensation complexity of an LDO regulator with N-MOSFET is considerably lower than that of an LDO regulator with P-MOSFET. Considering the dropout voltage of an N-MOSFET, a large voltage headroom equal to $V_{sd} + V_{gs}$ is the critical disadvantage, which is similar to that in the BJT. To address this problem in commercial products, a charge pump circuit can be used to decrease the dropout voltage, as shown in Figure 2.4(f). The boosted supply voltage for the EA can minimize the dropout voltage, but an LDO regulator suffers from switching noise caused by the charge pump circuit. A trade-off appears to occurs between noise suppression and dropout voltage. An off-chip low-pass filter is inserted between the control signal and the gate of the N-MOSFET, because an on-chip low-pass filter is difficult to apply. Consequently, the disadvantages are extra off-chip components and a large PCB area.

The products listed in Table 2.1 use P-MOSFETs as power transistors because of their low quiescence and LDO voltage at the cost of complex compensation skills. In the following section, compensation skills are introduced by assuming that a P-MOSFET is used as the power transistor. Before ending this subsection, the *I–V* curve of the power P-MOSFET is observed, as shown in Figure 2.5. The P-MOSFET works similarly to the faucet shown in Figure 2.3. Controlling the value of the source-to-gate voltage V_{SG}, which is similar to the tightness or looseness of the faucet, can determine the current flowing through the power P-MOSFET. At light loads (e.g., i_{D1} in Figure 2.5), a small V_{SG} indicates less current flowing out of the input supply voltage V_{IN}. By contrast, increasing V_{SG} at heavy loads (e.g., i_{D2}) can loosen the faucet to force a high current to flow through the power P-MOSFET. In case of a load change from light to heavy (e.g., from i_{D1} to i_{D2}), a large $V_{dropout}$ can guarantee that the power P-MOSFET will work in the saturation region for high gains in CS configuration, where $V_{dropout}$ is the source-to-drain voltage of the P-MOSFET.

A high gain in the system loop leads to a well-regulated output voltage. However, as mentioned previously, a large $V_{dropout}$ implies poor PCE of an LDO regulator. Ensuring that the power P-MOSFET at the saturation region has a constant $V_{dropout}$ and a large $V_{dropout}$ is required at heavy loads, but redundant and power inefficient at light loads because of the small V_{SG}. If $V_{dropout}$ can be adjusted according to the output load current conditions, then the PCE can be kept high over a wide load range, which is referred to as the ADVS technique. ADVS and digital-dynamic voltage scaling (DDVS) are introduced later.

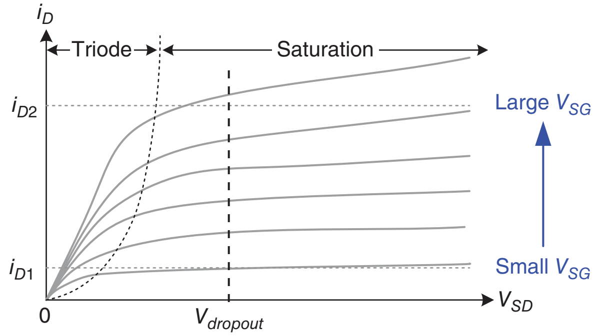

Figure 2.5 *I–V* curve of the power P-MOSFET

2.2 Compensation Skills

As illustrated in Figure 2.2, the regulation of an LDO regulator is structured with a negative feedback loop, and thus the output voltage can be regulated at a desired voltage level. Therefore, the frequency response should be designed carefully not only for stability, but also for transient response. In general, compensation skills should ensure that an LDO regulator has various capabilities, including low quiescent current at light loads, high current efficiency, low voltage operation, and high driving capability. Different compensation skills present various challenges that correspond to different specifications of systems and applications. In this section the characteristics of poles and zeros provide the concept of compensation skills for LDO regulators. Pole distribution is introduced first.

2.2.1 Pole Distribution

First, in the design of the LDO regulator shown in Figure 2.6(a), the existence of poles can be clearly identified at P_0 and P_1, located at the LDO output and the first stage output, respectively. Without considering the Miller effect on the gate-to-drain capacitance and by adding output capacitors at the output node, P_1 is below P_0, as shown in Figure 2.6(b), because a large parasitic capacitance C_{par} and a large output resistance $r_{OUT(ea)}$ are observed at the gate of the power MOSFET and at the output of the EA, respectively. $r_{OUT(ea)}$ is large because of the low quiescent current requirement. Given that the power P-MOSFET should be large to provide high driving capability, the gate-to-drain capacitance C_{GD} is sufficiently large to be a Miller capacitance with the gain contributed by the power P-MOSFET. Thus, P_1 and P_0 are split by the Miller capacitance C_{GD}; that is, P_0 moves toward high frequencies as a new high-frequency pole P_0' while P_1 moves toward the origin as a new low-frequency pole P_1'. The root locus of the pole-splitting effect that results from a large C_{GD} is illustrated in Figure 2.6(b). Consequently, system stability can be ensured if two poles can be widely split by C_{GD}. That is, a low-frequency pole P_1' at the gate of the power MOSFET can be regarded as the system-dominant pole without being affected by other low-frequency poles because the first non-dominant pole P_0' is now located at high frequencies.

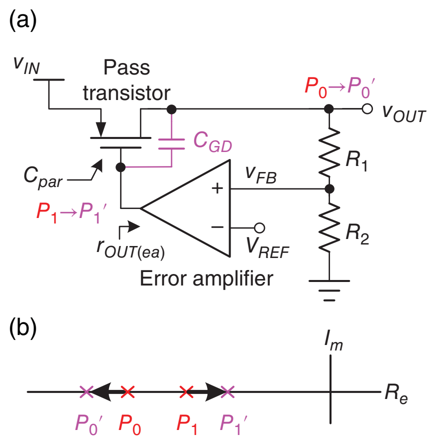

Figure 2.6 Basic LDO regulator. (a) Schematic. (b) Root locus showing the pole-splitting effect with Miller capacitance C_{GD}

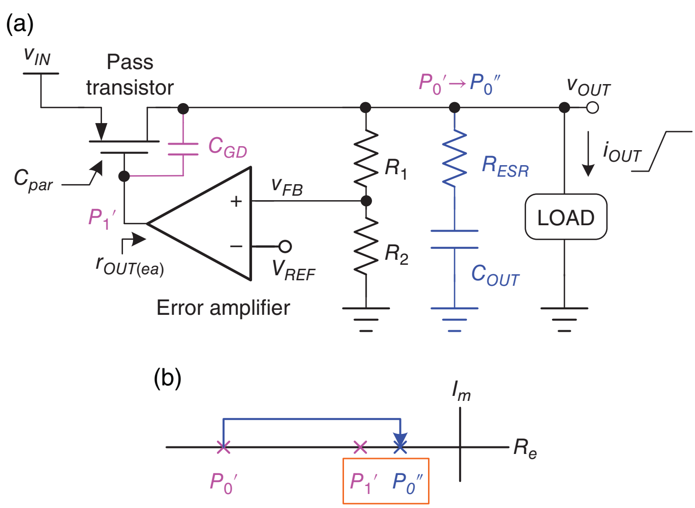

Figure 2.7 Basic LDO regulator with a large output capacitor to deal with significant sudden loading current change. (a) Schematic. (b) Root locus

When checking the requirements of LDO regulators, experiencing the sudden load changes required for system operation is necessary. As previously mentioned, a sudden load change cannot immediately change the water level of a big basin in the emulated reservoir and basin system. Thus, connecting a large output capacitor at the output to sustain a well-regulated output voltage is easy, as shown in Figure 2.7(a). A large output capacitor solves the sudden load change problem, but the stability decreases because the large output capacitor forces the output pole toward the origin with the role of dominant pole. Meanwhile, the pole at the gate of the power P-MOSFET is not the dominant pole anymore. The position of the output pole changes from P_0' to P_0'', where P_0'' is considerably lower than P_1' because of the large output capacitor. P_0'' becomes the new dominant pole of the system when a large output capacitor is used. However, we should remember the subsequent problem that two low-frequency poles, namely P_0'' and P_1' (the output pole and the pole at the gate of the power P-MOSFET, respectively) exist in the LDO regulator. Consequently, the phase margin (PM) is deteriorated by two low-frequency poles, as shown in Figure 2.7(b). Given that the output pole is now the dominant pole, the previous non-dominant pole is located at the gate of the power MOSFET and should move toward high frequencies to increase the system stability. Obviously, we should determine how to push P_1' toward high frequencies or add a compensation zero to cancel its effect. If this problem is reviewed carefully, we can determine that the output resistance of EA is large because of the low quiescent current. The resistor/capacitor (RC) time, which is constant at the gate of the power P-MOSFET, is too large. Thus, the EA cannot directly drive large parasitic capacitance at the gate of the power P-MOSFET if a low quiescent current is still required.

A useful technique to alleviate large RC time constants occurring at the gate of the power P-MOSFET is to add a buffer stage between the output of the EA and the gate of the power MOSFET, as illustrated in Figure 2.8(a). Consequently, the driving capability and the PM can be simultaneously enhanced by the inserted buffer stage. The buffer stage splits a low-frequency pole P_1' into two high-frequency poles, namely P_1^* and P_1^{**}, as shown in Figure 2.8(b). P_1^* and P_1^{**} have small RC time constants, that is $C_{par}{}^* r_{OUT(buf)}$ and $C_{IN(buf)}{}^* r_{OUT(ea)}$, respectively, where $C_{IN(buf)}$ and $r_{OUT(buf)}$ are the input capacitance and output resistance of the buffer stage, respectively. $C_{IN(buf)}$ and $r_{OUT(buf)}$ are small because of the characteristics of the voltage buffer. The large RC time constant stops increasing when the buffer stage is used. P_1^* is below P_1^{**} because of the large parasitic capacitance C_{par} at the gate of the power P-MOSFET. P_1^* becomes the first non-dominant pole after the buffer stage is inserted.

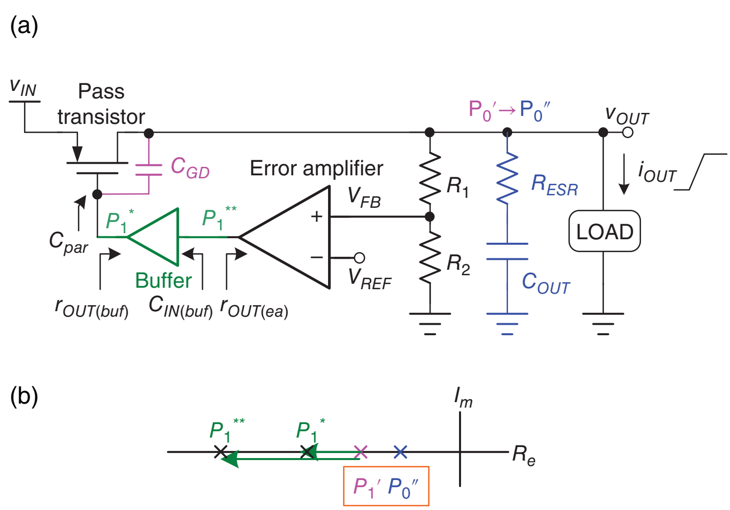

Figure 2.8 Basic LDO regulator with a buffer stage. (a) Schematic. (b) Root locus

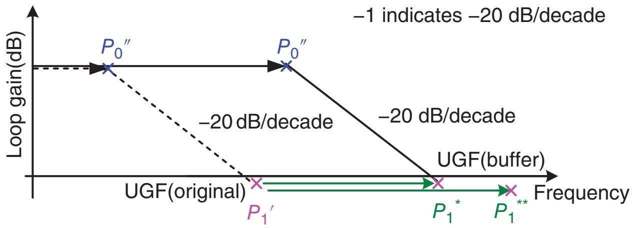

Figure 2.9 Relationship between the dominant pole and non-dominant poles in the frequency domain before and after the buffer stage is inserted

After the buffer stage is inserted, the low-frequency pole $P_1{}^*$ gains a small RC time constant. If $P_1{}^*$ is set at the unity gain and no low-frequency pole exists below $P_1{}^*$, excluding the output pole $P_0{}''$, a line with slope −20 dB/decade can be drawn to determine the highest position of $P_0{}''$, as illustrated in Figure 2.9. Similarly, before the buffer stage is inserted, the original position of $P_0{}''$ can be determined by drawing a line with slope −20 dB/decade from the location of $P_1{}'$, which is also found at unity gain. Figure 2.9 shows the possible working range of $P_0{}''$. In conclusion, if $P_0{}''$ is located at the highest position, then the RC time constant of $P_0{}''$ can be reduced because $P_1{}^*$ is above $P_1{}'$. The advantages of a smaller RC time constant of $P_0{}''$ are twofold. First, the LDO regulator can tolerate a high loading current, which indicates a small equivalent output resistance. Second, the LDO regulator can remain stable if a small output capacitor is used at medium loads instead of the conventional design without using the buffer stage.

Basically, the buffer can be a simple SF, as depicted in Figure 2.10(a), with an equivalent output resistance of $1/g_{m1}$. However, $1/g_{m1}$ is not sufficiently small to push the pole toward high frequencies. Thus, the design target is to reduce the equivalent output resistance. The analysis can involve a test voltage v_t applied at source M_1 when the equivalent output resistance is measured, as depicted at the top left of Figure 2.10(b). The small signal current i_t ($\approx g_{m1}v_t$) is redirected to the base terminal of an additional BJT Q_1 to generate an amplified current βi_t. The advantage of this technique is that the small signal i_t is enlarged by the current gain β of the BJT. Given the local negative feedback, the equivalent output resistance of the buffer can be attenuated by a factor of (β + 1), which can be expressed as:

$$r_{OUT(buf)} \approx \frac{1}{g_{m1}} \cdot \frac{1}{(\beta+1)} \tag{2.1}$$

In general, only lateral PNP or NPN BJT is provided in the standard CMOS process. Thus, the designer can use a bipolar-CMOS-DMOS (BCD) process to facilitate design at the expense of cost and silicon area. To solve this issue, the function of the BJT device can be substituted by combining MOSFETs from M_{U1} to M_{U4}, as shown in Figure 2.10(c). A new equivalent output resistance is expressed in Eq. (2.2), which indicates that the output resistance is reduced by the ($g_{m2}r_{o1} + g_{m3}g_{m4}r_{o1}r_{o4}$) contributed by the local negative feedback control of MOSFETs from M_{U1} to M_{U4} without any additional process:

$$r_{OUT(buf)} \approx \frac{1}{g_{m1}} \cdot \frac{1}{(g_{m2}r_{o1} + g_{m3}g_{m4}r_{o1}r_{o4})} \tag{2.2}$$

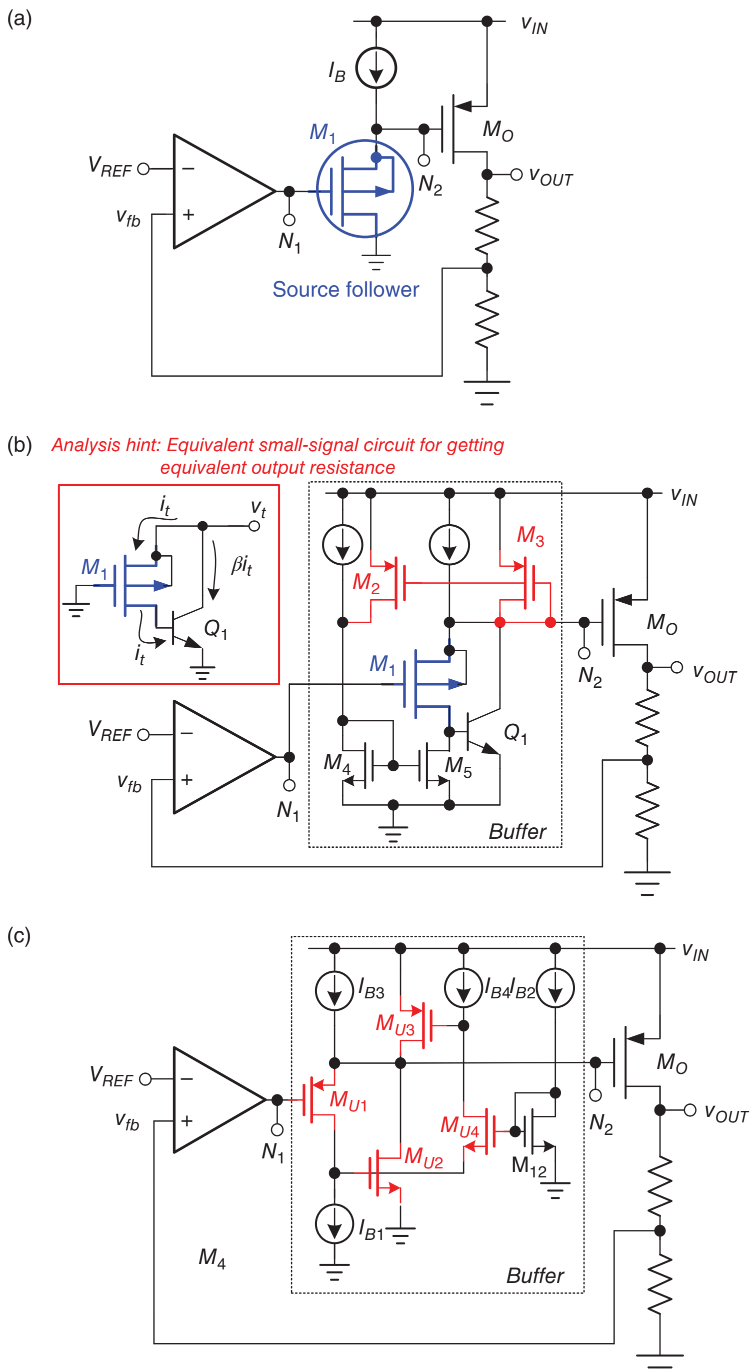

Figure 2.10 Structure of the LDO regulator with a buffer stage, which can be implemented with (a) an SF configuration, (b) an SF with a local negative feedback implemented by BJT, and (c) an SF with a reduced output resistance implemented by CMOS devices

As previously mentioned, the size of the output capacitor can be reduced if a buffer stage is used. The output capacitor cannot be implemented on the same silicon because P_1^* in Figure 2.9 should be pushed higher than P_1', because the acceptable size of an on-chip capacitor is within the range of several pico-farads compared with an external output capacitor that measures several micro-farads. That is, the advantage of the buffer stage is that it allows the external output capacitor to be smaller than that of the conventional LDO design without a buffer stage. Consequently, the external output capacitor remains too large to become an on-chip capacitor even if a buffer stage is used.

To obtain a compact-sized solution for the LDO regulator, the output capacitor shown in Figure 2.7 is used because the large charge stored in a large output capacitor can deal with instant load changes. If this large output capacitor is removed, then the power transistor should be rapidly turned on or off because the small charge stored in the parasitic output capacitance cannot satisfy the load change requirements of the system. If the system bandwidth is sufficiently large, the LDO regulator has the capability to handle a sudden load change even when the large output capacitor is removed. The Miller compensation skill, which is completely contrary to the dominant-pole compensation skill, can be applied to LDO regulators without adding any large external output capacitor. The Miller compensation skill can have a large bandwidth if a gain-stage amplifier is inserted between the EA and the power transistor. Thus, a high bandwidth is achieved but Miller capacitors are necessary. The inherent gate-to-drain capacitance C_{GD} can be regarded as a Miller capacitor. Only one Miller capacitor C_m is necessary, as shown in Figure 2.11(a). The characteristics and features of the two aforementioned compensation skills should be examined to identify which skill is more suitable for Soc applications.

Figure 2.11(a) shows that a Miller capacitor C_m is added and a large output capacitor C_{OUT} is removed. Consequently, Figure 2.11(b) shows that the pole at V_{OUT} moves toward high frequencies from P_0' to P_0''', while the pole at the output of the EA moves toward low frequencies from P_1' to P_1''. P_1'' replaces P_0' in the dominant-pole compensation skill as the new dominant pole because of the Miller effect. The advantage of this process is that the Miller capacitor C_m is considerably smaller than the original C_{OUT}, because the Miller effect amplifies the equivalent capacitance. Moreover, another Miller capacitance C_{GD} can split P_2 and P_0''' into high

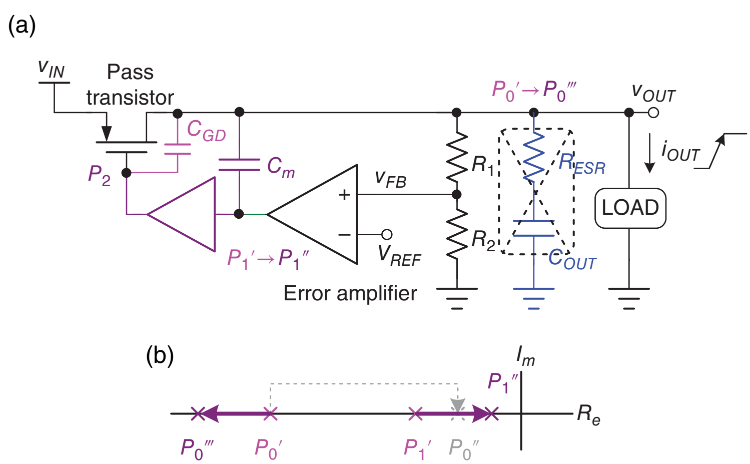

Figure 2.11 Basic LDO regulator with Miller compensation. (a) Schematic. (b) Root locus

frequencies. However, without a large C_{OUT}, any load transient response will cause a larger output transient voltage variation than the dominant-pole compensation skill if two compensation skills are assumed to have the same unity gain frequency (UGF), because the small basin cannot accommodate large water system requirements if the bandwidth does not increase.

When the load changes from heavy to light, the equivalent output impedance decreases, and then P_0''', which is the pole at V_{OUT}, moves toward a lower frequency. To maintain the performance of the transient response, extending the UGF is a challenge, particularly under ultra-light load conditions.

2.2.2 Zero Distribution and Right-Half-Plane (RHP) Zero

LDO regulators experience a load change from Soc applications. Thus, the bandwidth of LDO regulators should be sufficiently high to have a good transient load response. To extend the bandwidth further, a left-half-plane (LHP) zero is required to cancel the effect of the first non-dominant pole. We provide several common techniques to generate LHP zero, without being affected by right-half-plane (RHP) zero. The first well-known technique, as shown in Figure 2.12(a), is to use an equivalent series resistance at the output capacitor to form an LHP zero according to the equivalent output impedance:

$$Z_{OUT} = R_{ESR} + \frac{1}{sC_{OUT}} = \frac{1 + sR_{ESR}C_{OUT}}{sC_{OUT}} \tag{2.3}$$

The ESR value is temperature dependent, and thus determining an exact value in the design is difficult. Moreover, in case of a loading current change, any current flowing through the ESR will cause a large IR-drop voltage. The transient voltage variation becomes large, which is not desired in power management design. Thus, low ESR output capacitors are favored in converter designs.

In general, in a CS configuration, the Miller capacitor C_m will contribute an RHP zero, which is not desired in regulator designs. RHP zero is caused by a feedforward path through C_m from v_i to v_o. RHP zero significantly deteriorates the system stability, because it contributes +20 dB and −90° per decade. Thus, removing the RHP zero effect is necessary. As shown in Figure 2.12(b), the second technique is to convert RHP zero into LHP zero through the null resistance R_Z in series with C_m. If $1/g_{m1}$ is considerably smaller than R_Z, then the expression of LHP zero (ω_Z) is shown as:

$$\omega_Z \approx \frac{1}{C_m\left(g_{m1}^{-1} - R_Z\right)} \tag{2.4}$$

where g_{m1} is the transconductance of MOSFET M_1. Furthermore, given that the implementation of R_Z occupies a large silicon area, conventional MOSFET resistance working in the triode region can be used. However, its equivalent resistance value suffers from perturbations resulting from output voltage variation. Figure 2.12(c) shows an effective technique to remove RHP zero by adding a SF in series with a Miller capacitor C_m on the feedback path from v_o to v_i, which indicates that the voltage difference across C_m remains at $v_i - v_o$. However, the input capacitance observed at v_i is the series equivalent capacitance generated by C_m and C_{gs2}. Consequently, the equivalent capacitance becomes smaller than that of the original design. That

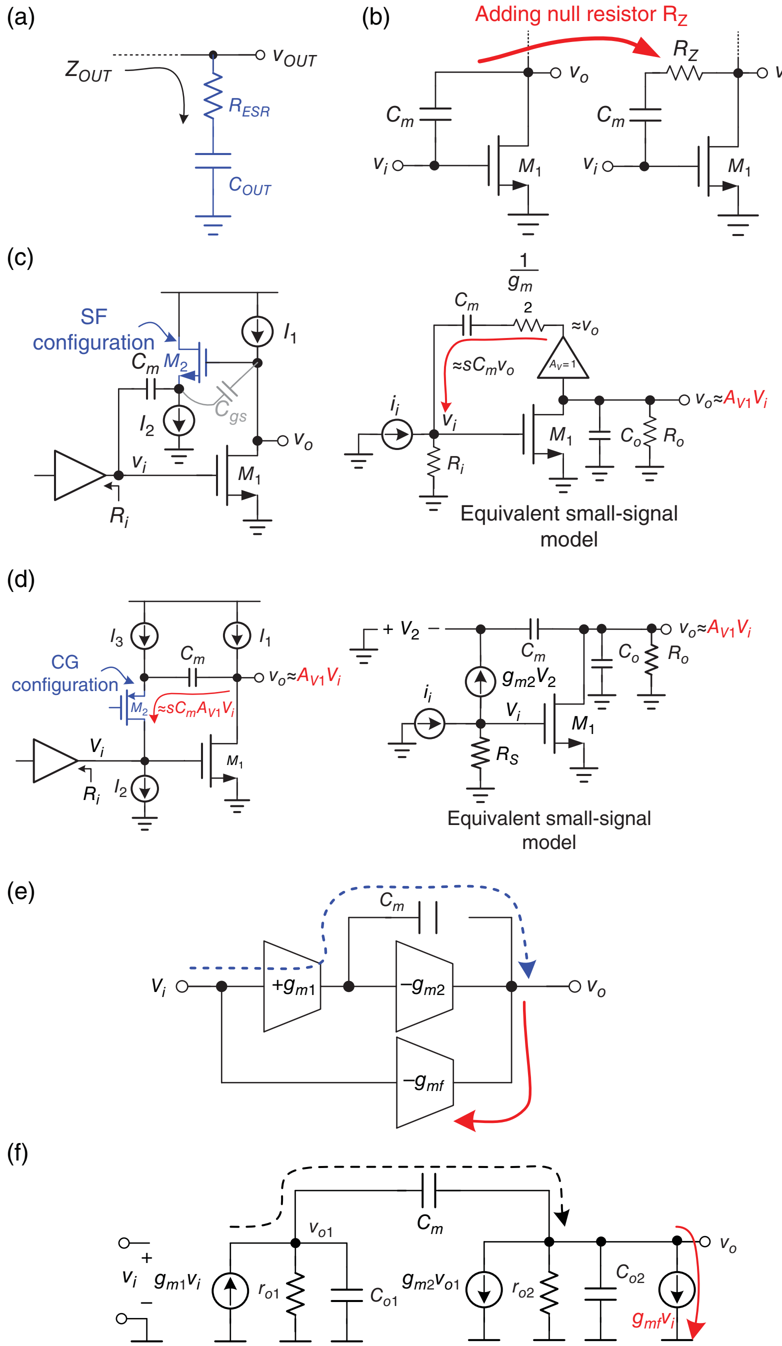

Figure 2.12 LHP zero generation methods. (a) ESR on the output capacitor. (b) Miller capacitor with a null resistor R_Z. (c) SF configuration used to remove RHP zero. (d) CG configuration used to remove RHP zero. (e) Feedforward path forms parallel architecture to obtain one LHP zero. (f) Small signal model of (e) showing the effect of the feedforward path

is, RHP zero moves toward infinity, and thus an LHP zero is generated. After calculation, the transfer function is derived as:

$$\frac{v_o}{i_i} = \frac{-g_{m1}R_oR_i(1 + C_m/g_{m2}s)}{R_oC_oC_m(1/g_{m2} + R_i)s^2 + [(1/g_{m2} + g_{m1}R_oR_i)C_C + R_oC_o]s + 1} \tag{2.5}$$

The generated LHP zero is expressed in Eq. (2.6), which is determined by controlling the transconductance value of M_2:

$$\omega_Z = -\frac{g_{m2}}{C_m} \tag{2.6}$$

The RHP zero disappears in the circuit, but a disadvantage must be pointed out in this case. The maximum allowable output voltage must be higher than $V_{OD(current\ source\ I2)} + V_{GS2}$, which indicates that the output voltage cannot be too low. Thus, another technique can be used to improve the solution to this problem. The circuit configuration is shown in Figure 2.12(d), where transistor M_2 works as a common gate (CG) configuration. The output voltage signal is converted into a current signal sC_mv_o, which is injected into the v_i node and summed with the input current i_i. The transfer function from i_i to v_o is expressed in Eq. (2.7) with one generated LHP zero, as shown in Eq. (2.8):

$$\frac{V_o}{i_i} = \frac{-g_{m1}R_iR_o\left(1 + {}^{C_m}/_{g_{m2}}s\right)}{{}^{R_o}/_{g_{m2}}C_oC_ms^2 + \left[(1 + g_{m1}R_i)R_oC_m + {}^{C_m}/_{g_{m2}} + R_oC_o\right]s + 1} \tag{2.7}$$

$$\omega_Z = -\frac{g_{m2}}{C_m} \tag{2.8}$$

Moreover, the feedforward path can generate an LHP zero via an active signal path, because it can alleviate the effect of RHP zero through C_m. Figure 2.12(e) shows a two-stage amplifier with the Miller compensation capacitor C_m. The feedforward amplifier with transconductance g_{mf} can provide a feedforward signal path that is used to move RHP zero toward infinity, thus transforming it into LHP. That is, if g_{mf} is equal to g_{m1}, then no net current signal out of the output exists, and thus the output voltage is zero when ω approaches infinity, as shown in Figure 2.12(f). After the transfer function is derived, setting $g_{mf} = g_{m1}$ can move zero to LHP and equal to $-g_{m2}/C_{o1}$, which is above the UGF, where C_{o1} is the output capacitance of the first stage. RHP zero is effectively removed from the system, even if LHP zero is not well defined in this case.

An explanation of LHP and RHP zeros helps in the following LDO designs. Before providing the detailed designs, the specifications and design considerations of the LDO regulator are carefully addressed in the following section.

2.3 Design Consideration for LDO Regulators

Several specifications should be mentioned in LDO designs. Such specifications can be classified into two categories, namely steady-state and dynamic characteristics. Steady-state specifications include dropout voltage $V_{dropout}$, line regulation, load regulation, temperature coefficient (TC), and maximum/minimum load current, among others. Meanwhile, dynamic specifications include line/load transient voltage variations, transient recovery time, pole/zero doublets, PM, and adaptive quiescent operating current, among others. The following subsections explain each specification in the LDO designs.

2.3.1 Dropout Voltage

As shown in Figure 2.13, when a certain desired V_{OUT} is assumed under certain loading conditions, different V_{IN} refer to various operating regions, including the off region, dropout region, and regulation region. Different operating regions lead to varying V_{OUT} performances. In the regulation region, V_{OUT} can be regulated within the allowable dropout voltage $V_{dropout}$. This voltage can be estimated by the voltage difference between the input voltage and the output voltage when V_{OUT} deviates from its nominal regulated voltage of approximately 2%. That is, the voltage across the power transistor that is defined as $V_{dropout}$. When $V_{dropout}$ increases with decreasing V_{IN}, the power transistor eventually loses its gain and V_{OUT} gradually deviates from its desired value. As soon as the power transistor turns to function as a switch instead of an analogy gain stage, the system loop gain drastically decreases and consequently, the LDO regulator loses its capability to regulate V_{OUT}. The V_{OUT} value is determined according to V_{IN}, I_{Load}, and R_{on}, where R_{on} is the equivalent resistance of the power transistor. Under this situation, the operation enters the dropout region. The voltage across the power transistor can basically be expressed as:

$$V_{dropout} = I_o R_{on} \tag{2.9}$$

As V_{IN} decreases further, R_{on} increases continuously because the input voltage that drives the power transistor is too low. Under this situation, V_{OUT} is decreased to the ground and the operation enters the off region.

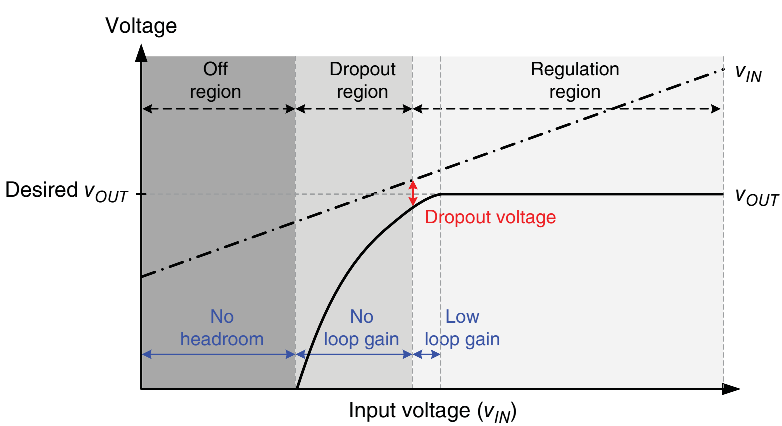

Figure 2.13 Characteristic curve of input voltage and output voltage that defines the dropout voltage $V_{dropout}$

Operation in the regulation region is necessary for LDO regulators to provide regulated performance and driving capability. When a small $V_{dropout}$ and a large I_{Load} are desired, LDO regulators should be out of the regulation region. The feedback controller can adjust the R_{on} that corresponds to $V_{dropout}$ to regulate V_{OUT} when operation occurs in the regulation region. The minimum $V_{dropout}$ is the worst-case scenario in the regulation region, when LDO regulators consider the largest I_{Load} and the minimum R_{on}. To ensure that LDO regulators are operating within the regulation region, increasing the size of the power transistors can reduce R_{on} and then ease the limitation of $V_{dropout}$. However, this process increases the complexity of the compensation network.

LDO regulators are typically utilized for two purposes. The first purpose is to function as the voltage converter to provide step-down supply voltage. In this case, the requirements are simple architecture, small silicon area, and low cost with minimal requirement for passive components such as inductors and capacitors. However, a large $V_{dropout}$ is inevitable because LDO regulators are used for step-down voltage conversion. The PCE of LDO regulators is sacrificed because of the considerable power loss across power transistors compared with SWRs.

The second purpose of LDO regulators is to work as post-regulators to provide a noise-suppression function for low-noise voltage in sensitive analogy circuits. $V_{dropout}$ is expected to be minimal to prevent power loss. In modern LDO regulator designs, the trade-off between efficiency and the silicon area occupied by power transistors is considered; thus, $V_{dropout}$ is designed within the range of 200 mV.

2.3.2 *Efficiency*

PCE is defined as the ratio of the output power P_{OUT} to the input power P_{IN}, as expressed in Eq. (2.10). The output power P_{OUT} is equal to the product of the output voltage and the output loading current. The input power is equal to the product of the input voltage and the input current I_{IN}:

$$\eta = \frac{P_{OUT}}{P_{IN}} = \frac{V_{OUT} I_{Load}}{V_{IN} I_{IN}} \tag{2.10}$$

In Eq. (2.11), I_{IN} includes the loading current I_{Load} and the quiescent current $\underline{I_q}$ that consists of all the biasing currents in the EA, the bandgap reference, feedback resistance network, and so on:

$$I_{IN} = I_{Load} + I_q \tag{2.11}$$

Overall, the PCE in Eq. (2.10) can be modified to Eq. (2.12), which is a product of voltage conversion efficiency and current efficiency as respectively defined in Eqs. (2.13) and (2.14):

$$\eta = \frac{I_{Load} V_{OUT}}{\left(I_{Load} + I_q\right) V_{IN}} = \eta_V \cdot \eta_I \tag{2.12}$$

where

$$\eta_V = \frac{V_{OUT}}{V_{IN}} \times 100\% \tag{2.13}$$

and

$$\eta_I = \frac{I_{Load}}{I_{Load} + I_q} \times 100\% \tag{2.14}$$

Under heavy loads, the PCE in Eq. (2.12) is approximately equal to the ratio of V_{OUT} to V_{IN}, because I_{Load} is considerably larger than I_q. This finding indicates that the dropout voltage $V_{dropout}$ limits the efficiency performance. Meanwhile, the current efficiency η_I cannot be ignored in expressing the overall PCE under light loads. The standby and light-load modes occupy most of the time in portable devices. That is, a high η_I becomes an important factor in determining battery usage time. Thus, to obtain high current efficiency, the quiescent current I_q should be adaptive to the variation in loading current I_{Load} to obtain a high η_I over a wide load range. Under light loads, a low quiescent current requirement is typically set to maintain a high η_I. Having a low I_q if the system loading current is low in standby mode has become a challenge. Under heavy loads, however, a high η_I can still be maintained if I_q is increased to correspond to the loading current. Even if I_q can be adjusted adaptively by the loading current, maintaining the performance of the LDO regulator remains a challenge because of the limited bandwidth and slew rate of the operational amplifier (OPA). The required performance of LDO designs is first reviewed. Then, possible techniques to satisfy the specifications are discussed later in this book.

2.3.3 Line/Load Regulation

Line and load regulations are two important steady-state performances of LDO regulators because they indicate the percentage of output voltage variation in case a perturbation occurs from the input voltage and the output loading current, respectively. Line regulation is defined as the output voltage variation caused by an input line voltage variation, as shown in Figure 2.14. Line regulation can be expressed as Eq. (2.15), where L_o is the loop gain of the LDO regulator; g_m and r_o are the transconductance and the output resistance of the power MOSFET, respectively; β is the feedback factor. To obtain good line regulation, a high loop gain is required but

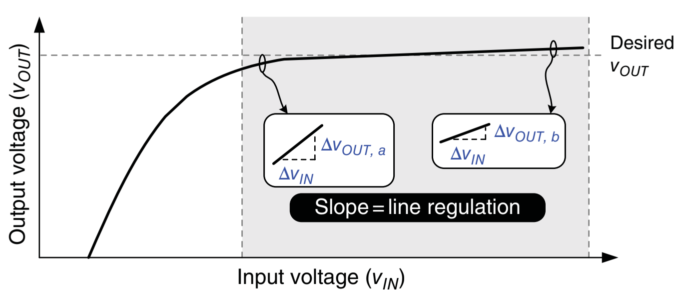

Figure 2.14 Definition of line regulation

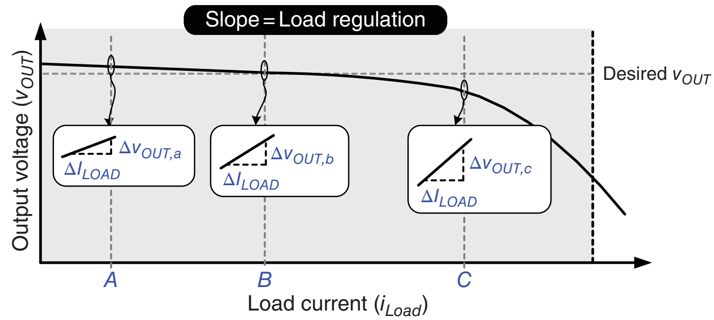

Figure 2.15 Definition of load regulation

stability may deteriorate. Thus, a trade-off occurs between regulation performance/accuracy and stability:

$$\text{Line regulation} = \frac{\Delta V_{OUT}}{\Delta V_{IN}} \approx \frac{g_m r_o}{L_o} + \frac{1}{\beta} \cdot \left(\frac{\Delta V_{REF}}{\Delta V_{IN}} \right) \tag{2.15}$$

When V_{IN} decreases, the dropout voltage also decreases. In Figure 2.14, the line regulation is drastically degraded when V_{IN} is reduced to a certain level. Given the constraint of dropout voltage, the loop gain of the LDO regulator is degraded because of the reduced V_{IN}. That is, $\Delta V_{OUT,a} > \Delta V_{OUT,b}$, assuming that the input variation ΔV_{IN} is the same as that in Figure 2.14.

The load regulation is defined by the output voltage variation in case of a change in loading current, as shown in Figure 2.15. To obtain good load regulation, a high loop gain is required, but the stability worsens and the compensation becomes complex. The expression of load regulation is given in Eq. (2.16), where L_o is the loop gain of the LDO regulator and r_o is the output resistance of the power MOSFET:

$$\text{Load regulation} = \frac{\Delta V_{OUT}}{\Delta I_{OUT}} = -\frac{r_o}{1 + L_o} \tag{2.16}$$

If a P-MOSFET is selected as the power transistor, then the loop gain of the LDO regulator depends strongly on the load conditions. A heavy load condition typically reduces gain, and thus the load regulation is degraded gradually as shown in Figure 2.15. When the load current change (ΔI_{Load}) is the same, the change in output voltage is large under heavy load conditions. That is, $\Delta V_{OUT,c} > \Delta V_{OUT,b} > \Delta V_{OUT,a}$ (Figure 2.15), because the loading current at point C is significantly higher than that at point A.

2.3.4 Transient Output Voltage Variation Caused by Sudden Load Current Change

The transient response can be used to test the dynamic performance of LDO regulators. This response can be categorized into two types. The first type, called load transient response, is caused by load current variation. The second type, called line transient response, is caused by line voltage variation.

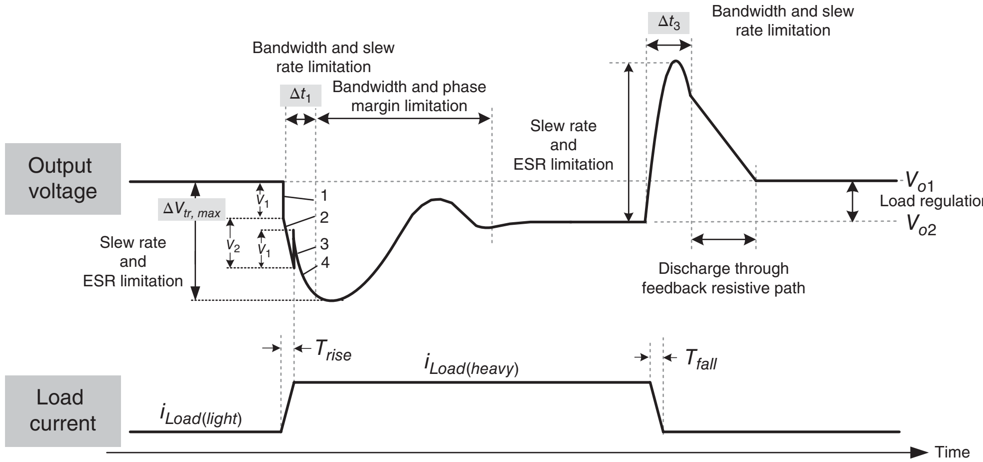

Figure 2.16 Load transient response in case of a load current change

As shown in Figure 2.16, when the LDO regulator experiences a load current step change, the output voltage requires time to be regulated by the negative feedback control. During a light-to-heavy load current change, the transient voltage variation can be divided into four parts. The first segment of the output voltage variation is contributed by the equivalent series inductance (ESL) in the output capacitor, because the voltage variation is equal to $V_1 = ESL^* di/dt$, which can also be expressed as $V_1 = ESL^*(I_{Load(heavy)} - I_{Load(light)})/T_{rise}$. Similarly, the ESR in the output capacitor also expresses the voltage drop V_2 as segment 2. When the load current is set to $I_{Load(heavy)}$, the ESL expresses the upward effect as segment 3. Finally, the output voltage experiences a transient response that is determined by the dynamic behavior of the converter and by the applied control method in segment 4. The ESR can sometimes be regarded as the current sensor to let the output ripple as the pulse width modulation (PWM) ramp in ripple-based control. The utilization of ESR as current sensor will be explained in detail in the following sections.

In case of a load current change, the pass device cannot immediately supply sufficient load current because this device works similarly to a faucet and requires time to be turned on to obtain a large current. Therefore, the output voltage experiences a voltage drop. The drop period Δt_1 depends on the closed-loop bandwidth BW_{cl} and the slew rate at the power MOSFET gate terminal. The response time can be approximated using Eq. (2.17), where C_{par} is the parasitic capacitance at the power MOSFET gate terminal and I_{sr} is the biasing current of the EA:

$$\Delta t_1 \approx \frac{1}{BW_{cl}} + t_{sr} = \frac{1}{BW_{cl}} + C_{par}\frac{\Delta V}{I_{sr}} \tag{2.17}$$

Here, BW_{cl} is the bandwidth at light loads.

Similarly, Δt_3 can be expressed as in Eq. (2.18). However, the values of Δt_1 and Δt_3 are different, because the bandwidths under light and heavy load conditions vary:

$$\Delta t_3 \approx \frac{1}{BW_{cl}} + t_{sr} = \frac{1}{BW_{cl}} + C_{par}\frac{\Delta V}{I_{sr}} \tag{2.18}$$

where BW_{cl} is the bandwidth at heavy loads.

In this case, we can derive the maximum transient voltage variation $\Delta V_{tr,max}$ during transient response. To simplify the expression, the ESL effect is first ignored. Thus, in case of light-to-heavy and heavy-to-light load changes, as shown in Figures 2.17 and 2.18, respectively, the power MOSFET delivers current that is either too low or too high, respectively, to the output load. Considering the energy difference between the input voltage source and the output loading, the output capacitor works as the buffer to keep the output voltage well regulated before a dynamic equilibrium is established. That is, the output capacitor can deliver energy to the output first before the gate-to-source voltage power MOSFET is adjusted to an adequate voltage (Figure 2.17). By contrast, additional energy is injected into the output capacitor before the power MOSFET is turned off to a suitable operating position (Figure 2.18). That is, the maximum transient output voltage variation $\Delta V_{tr,max}$ can be derived using Eq. (2.19) to represent the undershoot/overshoot voltage:

$$\Delta V_{tr,\max} = \frac{I_{Load(heavy)} - I_{Load(light)}}{C_{OUT}} \cdot \Delta t_1 \,(\text{or } \Delta t_3) + \Delta V_{ESR} \tag{2.19}$$

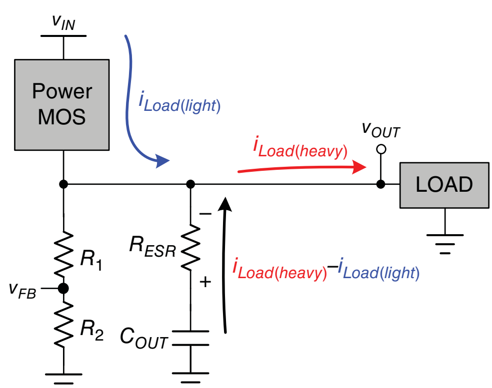

Figure 2.17 Output voltage drops in case of a light-to-heavy load change

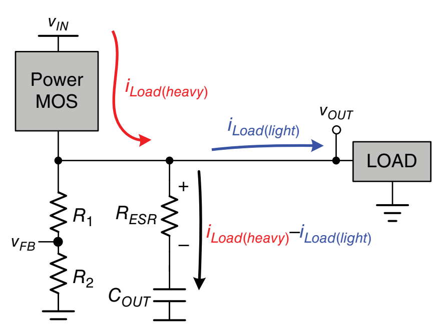

Figure 2.18 Output voltage overshoots in case of a heavy-to-light load change

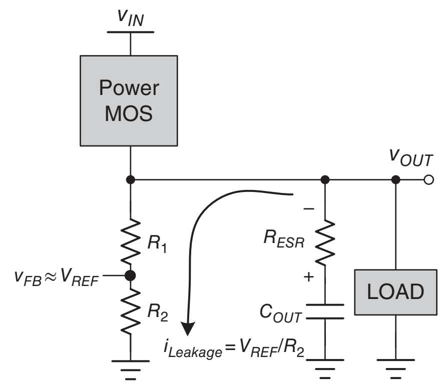

Figure 2.19 Output overshoot voltage is dissipated by the feedback divider resistors

The first term in Eq. (2.19) indicates that the output capacitor C_{OUT} functions as the buffer before the power MOSFET can work properly after the period Δt_1 or Δt_3, which is the time during which the power MOSFET starts to react to a light-to-heavy or a heavy-to-light load change, respectively. Simultaneously, current flows into or out of the output capacitor. The IR-drop voltage caused by the ESR can be represented by the second term in Eq. (2.19). Under both light-to-heavy and heavy-to-light load changes, the IR-drop voltage caused by the ESR will result in significant output transient voltage variation. Thus, ESR is undesirable in LDO regulators. In currently available commercial products, capacitors with low ESR are selected. For example, a multi-layer ceramic capacitor (MLCC) is frequently used in LDO regulators to ensure low output voltage ripples.

The overshoot voltage shown on the right of Figure 2.16 can be dissipated by the leakage current at the feedback divider resistors, as shown in Figure 2.19. Thus, the decreasing slope of the output voltage depends on the value of the divider resistors. However, a large overshoot voltage may damage the next Soc stage, which is implemented in an advanced nanometer process with low voltage characteristics. Thus, in conventional LDO regulators, a dummy resistive load can be connected to the output to dissipate additional energy. However, this process is applied at the cost of efficiency.

The other dynamic performance is line transient response. When the input voltage steps down to a smaller value, the power MOSFET will deliver a reduced load current, and thus cause a voltage dip. This response is similar to the load transient response in case of a light-to-heavy load current change. By contrast, when the input voltage steps up to a larger value, the power MOSFET will deliver a higher load current than the required output load, and thus produce an overshoot voltage. This response is similar to the load transient response in case of a heavy-to-light load current change.

Based on the preceding discussion, a conclusion can be drawn on which general methods can be used to improve the dynamic transient response. These methods include applying a wide bandwidth for a closed loop, implementing a fast slew rate at the power MOSFET gate terminal, using a large output capacitor, and reducing the ESR. These methods are similar to those presented in the discussion of the emulated reservoir and basin system. A large output capacitor can sustain the voltage variation in terms of temperature, but limits the bandwidth. With a large output capacitor, the compensation becomes too complex to be able to extend the bandwidth.

By contrast, when a large output capacitor is discarded, a wider bandwidth is required and the complexity increases. Moreover, a fast slew rate is required to compensate for the degraded transient response caused by the absence of a large output capacitor. This condition is the reason why the quiescent current is typically higher when a *C-free* structure is applied compared with when dominated pole compensation is implemented. In conclusion, a trade-off between performance and cost occurs in these methods.

2.4 Analog-LDO Regulators

As discussed in Sections 2.2 and 2.3, the design strategy in A-LDO regulators can be classified into two basic categories according to compensation skill. That is, the location of the dominant pole is first determined. In the first category, the dominant pole is located at the output because a large output capacitor is used. In the second category, called a *C-free* or capacitor-less LDO regulator, the dominant pole is generated by the Miller capacitance without using a large output capacitor.

Conventional LDO regulators use a large output capacitor with several microfarads to set the dominant pole at the output node. Considering that a large physical capacitor is used, the compensation method is called the dominant-pole compensation skill. An obvious disadvantage of this method is the use of a large off-chip capacitor, which increases the PCB area.

By contrast, the dominant pole can be generated by an amplified capacitance through the Miller effect. Using a large physical off-chip capacitor to form the dominant pole is unnecessary. The *C-free* LDO regulator got its name from the removal of the large off-chip capacitor, which was replaced with a small on-chip Miller capacitor to form the dominant pole and increase the system stability. When considering highly integrated LDO regulators for Soc applications, the *C-free* LDO regulator exhibits the advantages of reducing the effects of bonding wires and decreasing the silicon area occupied by the input/output (I/O) pads to connect off-chip passive components.

According to the following subsections, the specifications of LDO regulators for portable electronics should include three basic requirements: low quiescent current, wide input range operation, and improved regulation performance. Based on these requirements, we examine the advantages and disadvantages of the two categories of LDO regulators to identify the differences between these designs. The LDO regulators with the dominant-pole compensation skill are examined first.

2.4.1 Characteristics of Dominant-Pole Compensation

Figure 2.20 shows the LDO regulator with the dominant-pole compensation skill. In general, the dominant-pole compensation skill is utilized when the equivalent output capacitance of the next stage circuit is undetermined. Inserting a large output capacitor C_{OUT} can ensure that the dominant pole is located at the output node even if the input capacitance of the next stage circuit is unknown. Moreover, it can deal with the large load step occurring at the next stage circuit.

A basic LDO regulator includes an EA, feedback divider resistors R_1 and R_2, and a power P-MOSFET. In this regulator, a large gate-to-drain capacitance C_{gd} can be expected across the gate and drain terminals of the P-MOSFET. That is, the total parasitic capacitance C_{par} at the gate of the P-MOSFET is determined by the Miller capacitance $C_{gd}{}^{*}(1 + A_{v(Mp)})$, where $A_{v(Mp)}$ is the voltage gain of the P-MOSFET in the CS configuration. Moreover, the output resistance

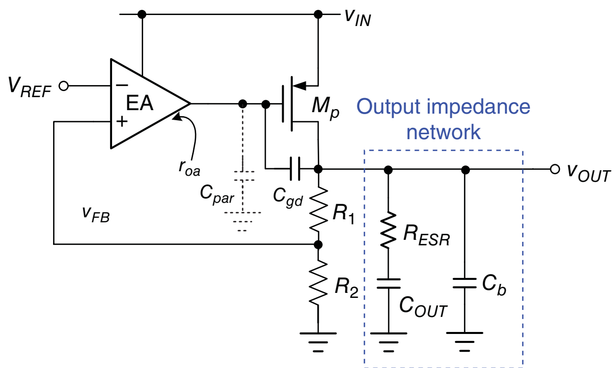

Figure 2.20 Conventional LDO regulator

r_{oa} of the EA is large because of its low quiescent current in portable electronics. The pole at the gate of the P-MOSFET, which is called the first non-dominant pole P_{1st_non}, will be at low frequencies and expressed through the following equation:

$$P_{1st\_non} = \frac{1}{r_{oa}C_{par}} \tag{2.20}$$

The dominant-pole compensation skill connects a suitable output impedance network to the output node. The constitution of the output impedance network includes two capacitors: the large output capacitor C_{OUT} and the bypass capacitor C_b, which has a smaller value than C_{OUT}. C_b can generate a pole at high frequencies to promote decade loop gain and reduce high-frequency noise. Moreover, the ESR resistance R_{ESR} of C_{OUT} should be considered in stability analysis, whereas the ESR resistance of C_b can be disregarded because C_{OUT} is larger than C_b.

In this case, all poles and zeros are examined if the output impedance network is used. The equivalent output impedance Z_{OUT} can be expressed as in Eq. (2.21), where r_{out} is the equivalent output resistance:

$$\begin{aligned} Z_{OUT} &= r_{out}||(R_1+R_2)||\frac{(1+sR_{ESR}C_{OUT})}{sC_{OUT}}||\frac{1}{sC_b} \\ &= \frac{\left(r_{op}||(R_1+R_2)\right)\cdot(1+sR_{ESR}C_{OUT})}{s^2\left(r_{op}||(R_1+R_2)\right)R_{ESR}C_{OUT}C_b + s\left[\left(r_{op}||(R_1+R_2)\right)+R_{ESR}\right]C_{OUT} + s\left[r_{op}||(R_1+R_2)\right]C_b + 1} \end{aligned} \tag{2.21}$$

The denominator in Eq. (2.21) is a second-order polynomial that indicates the two poles contributed by C_{OUT} and C_b, as expressed in Eqs. (2.22) and (2.23), respectively:

$$P_0 = \frac{1}{r_{out}C_{OUT}}, \text{ if } (R_1+R_2) >> r_{op} >> R_{ESR} \tag{2.22}$$

$$P_{2ndhskip-1pt\_non} = \frac{1}{R_{ESR}C_b} \tag{2.23}$$

Meanwhile, the numerator in Eq. (2.12) is a first-order polynomial. A zero, which is contributed by R_{ESR}, is located at low frequencies:

$$Z_{ESR} = \frac{1}{R_{ESR}C_{OUT}} \tag{2.24}$$

In this case, the relationship of three poles and one zero is illustrated in Figure 2.21(a). Two low-frequency poles are located below the UGF. Thus, ESR zero can alleviate the effect of the first non-dominant pole P_{1st_non} while ensuring that the second non-dominant pole P_{2nd_non} is above the UGF to satisfy the stability requirement. That is, the stability requirement sets an ESR stable region in the design of LDO regulators. If ESR zero is found at an extremely low frequency, a high gain will cause P_{2nd_non} to be located below the UGF, which reduces the system stability. By contrast, if ESR zero is located at high frequencies, then the effect of P_{2nd_non} will not be alleviated before the UGF, and thus the PM is insufficient. As shown in Figure 2.21(b), the curve of load current versus ESR value shows the ESR stable region set by the stability requirement.

Furthermore, different load currents will change the shape of the frequency response, as shown in Figure 2.21(c). According to Eq. (2.25), the dominant pole P_0 will move toward high frequencies if the load current increases. P_0 is regarded as a load-dependent pole:

$$P_0 = \frac{1}{r_{out}C_{OUT}} \propto I_{Load}\text{, where } r_{out} \propto \frac{1}{I_{Load}} \tag{2.25}$$

Similarly, the loop gain L_O and the UGF can respectively be derived from Eqs. (2.26) and (2.27):

$$L_O = \beta A_{EA} A_{v(Mp)} \propto g_{mp} r_{OUT} \propto \sqrt{I_{Load}} \times \frac{1}{\lambda I_{Load}} = \frac{1}{\sqrt{I_{Load}}} \tag{2.26}$$

where $\beta = \frac{R_2}{R_1 + R_2}$, A_{EA} is the error amplifier gain, and $g_{mp}(\propto \sqrt{I_{Load}})$ is the transconductance of the power MOSFET;

$$\text{UGF} = L_O \cdot P_0 \propto \sqrt{I_{Load}} \tag{2.27}$$

Although the loop gain L_O decreases when the load current increases, the UGF still moves toward high frequencies because P_0 moves toward such frequencies more drastically than the decrease in L_O. However, a further decrease in L_O will degrade the regulation performance without benefiting the system stability.

Figure 2.22 shows the output voltage waveforms in the LDO regulator with dominant-pole compensation. The PM degrades as the load increases, because the UGF moves toward high frequencies, and thus high-frequency poles appear below the UGF. When the PM degrades to a value that is smaller than 45°, the transient performance suffers from an obvious damping voltage variation (ΔV_{OUT}) and long settling time (t_s). When the PM degrades to a value that is smaller than 0° under an ultra-heavy load, the feedback loop becomes a positive feedback and the output voltage experiences oscillation.

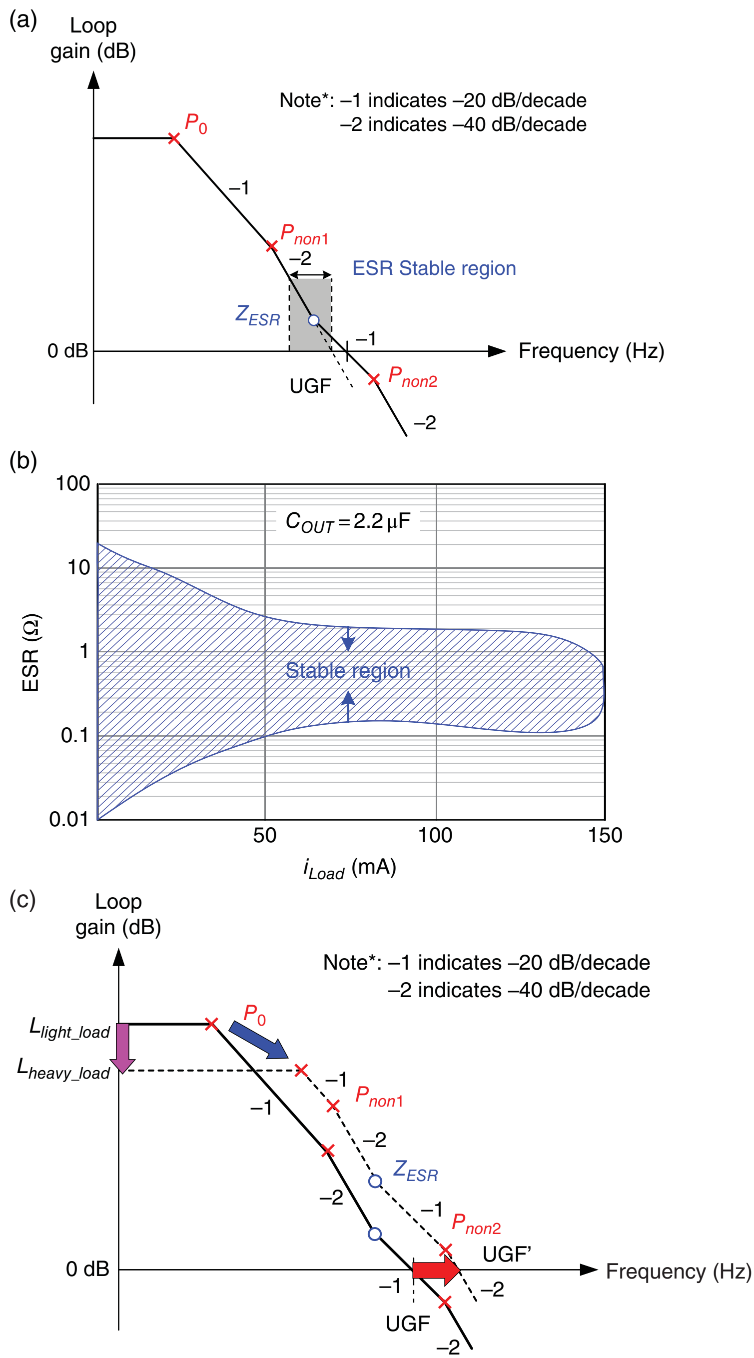

Figure 2.21 Analysis of the frequency response of the LDO regulator with dominant-pole compensation. (a) ESR stable region. (b) Relationship of load current vs. ESR. (c) Frequency response under different load current conditions

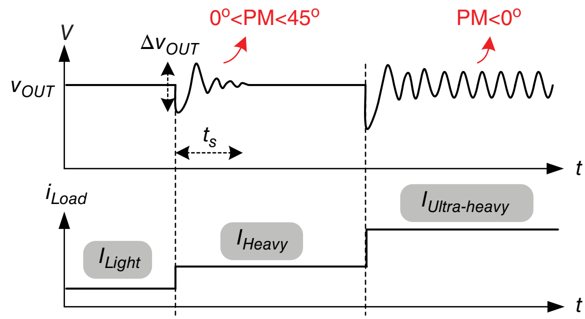

Figure 2.22 In dominant-pole compensation, the unstable phenomenon of the LDO regulator is caused by load current limitation as the load gradually changes from light to heavy

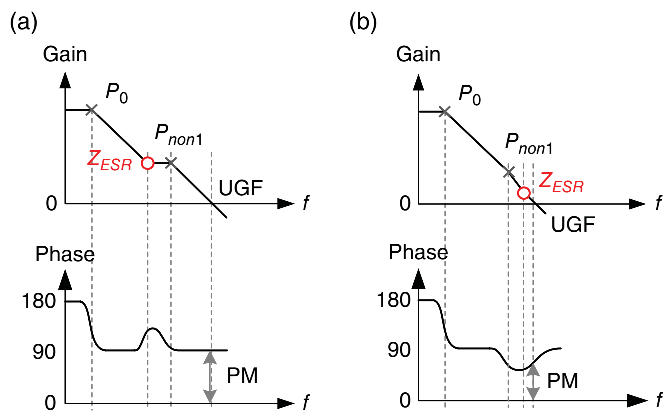

Figure 2.23 If the second non-dominant pole can be disregarded, then (a) a large ESR results in a robust and stable system, whereas (b) a small ESR can achieve a small recovery transient time

Moreover, when ESR zero is utilized to compensate for the first non-dominant pole, it can be designed at a frequency lower or higher than P_{non1}, as shown in Figure 2.23(a), (b), respectively. When P_{non2} is considerably higher than the UGF and can be disregarded, the case presented in Figure 2.23(a) is more robust because the frequency of Z_{ESR} is lower than P_{non1} and can increase the PM to approximately 90°. However, the LDO regulator with 90° PM exhibits a stable transient response but a long recovery time. Moreover, a low-frequency Z_{ESR} implies a large R_{ESR} value, which seriously deteriorates the dip voltage of the transient response. By contrast, the case shown in Figure 2.23(b) can achieve an improved transient response with an adequate PM of approximately 60°.

Furthermore, the variation in R_{ESR} caused by the temperature change leads to a difficulty in achieving system stability. A variable R_{ESR} causes different Z_{ESR} frequencies, which in turn result in a varying UGF, even if the load condition does not change. The PM can be deteriorated by P_{non2} when the UGF draws closer to P_{non2}.

Consequently, the ESR compensation skill can hardly ensure stability under different load conditions because of the non-constant unit gain frequency. That is, no simple rule can define the ESR compensation. The best means to ensure stability is to use a large ESR in the worst-case scenario. However, an undesirably large R_{ESR} may cause an unintended voltage dip when the load changes. As such, other techniques to generate an extra zero without using a large ESR for the dominant-pole compensation skill can be utilized. Moreover, inserting a buffer stage can cause the non-dominant pole that is moving toward higher frequencies to alleviate the UGF constraint, as discussed in Section 2.2.

2.4.1.1 Considering the Power MOSFET at the Triode Region

In the preceding discussion, the characteristic of frequency response is based on the power MOSFET that is operating in the saturation region. In general, when the voltage conversion ratio of the LDO regulator is large, $V_{dropout}$ is also large, which enables the power MOSFET to operate easily in the saturation region with an acceptable silicon area. By contrast, when $V_{dropout}$ is small or when the load range is large, keeping the power MOSFET operating within the saturation region is difficult. That is, the power MOSFET will operate in the triode region. In Soc applications, for example, the battery voltage and the conversion that supplies voltage to core devices are decreased simultaneously. Consequently, $V_{dropout}$ is reduced and the power transistor operates in the triode region. The characteristics of frequency response are different from the results derived in Eq. (2.27). The characteristics of the dominant pole and loop gain can be derived from Eqs. (2.28) and (2.29), respectively. In addition, the UGF expressed in Eq. (2.30) is constant and independent of load current conditions:

$$P_{don} = P_0 = \frac{1}{r_{out} C_{OUT}} \propto \sqrt{I_{Load}}\text{, where } r_{out} \propto \frac{1}{\sqrt{I_{Load}}} \tag{2.28}$$

$$L_o = \beta A_{oa} A_p = \beta g_{ma} r_{oa} g_{mp} r_{out} \propto \frac{1}{\sqrt{I_{Load}}} \tag{2.29}$$

$$\text{with} \begin{cases} \beta,\ g_{ma}, \text{ and } r_{oa} \text{ constant} \\ g_{mp} = \text{constant, when } V_{DS} \text{ is constant} \\ r_{out} \propto \dfrac{1}{\sqrt{I_{Load}}} \end{cases}$$

$$\text{UGF} = L_o \cdot P_{don} \propto \frac{1}{\sqrt{I_{Load}}} \cdot \sqrt{I_{Load}} = \text{constant} \tag{2.30}$$

Furthermore, for the power MOSFET that is operating in the triode region, the gain contributed by the power transistor degrades considerably. To maintain good output voltage regulation, enhancing the EA gain is necessary. Therefore, designing a gain-enhanced EA and an adequate pole/zero distribution is an important concern in the following discussion.

2.4.2 *Characteristics of* C-free *Structure*

An A-LDO regulator with Miller compensation can be used to supply energy to the Soc if the output capacitance from the next stage is known. No extra output capacitor is required, and thus this regulator is called the *C-free* LDO regulator. The advantage of this regulator is its high integration in the entire Soc, with an embedded power management system, because of the absence of an output capacitor. Given that the output capacitance consists of parasitic capacitance, the transient response of *C-free* LDO regulators will be considerably different from that of dominant-pole compensation, even if only one small bypass capacitor C_b is used to prevent a large dip voltage from occurring in case of load current change.

As shown in the emulated reservoir and basin system illustrated in Figure 2.3, the basin is too small to experience any sudden load current change. Thus, the only means to solve the sudden change in load current is to turn on the faucet rapidly to provide sufficient water to the output. Consequently, the *C-free* LDO regulator should be designed as a multi-stage architecture to enable it to attain a high DC gain, which can extend the bandwidth. Furthermore, a small Miller capacitor is typically used to guarantee system stability even when the DC gain is increased by a multi-stage architecture.

However, removing the output capacitor also results in several disadvantages in the regulated output voltage. Without a large output capacitor, the output voltage easily suffers from any noise disturbance or load transient variation. A noise disturbance at the output node is difficult to suppress because of the absence of a suitable bypass path. Such a path is formed by a large output capacitor in the dominant-pole compensation technique to generate high-frequency noise signals. When a heavy-to-light/light-to-heavy load transient occurs, redundant or insufficient energy will cause the voltage to overshoot or undershoot, respectively. To suppress the transient voltage variation, a large UGF is necessary to control the power transistor immediately and provide adequate power delivery. When a designer aims to obtain a large UGF for fast transient response, all non-dominant poles should be moved to frequencies that are considerably higher than the UGF. A multi-stage EA should be carefully designed to suppress any noise disturbance effectively.

Figure 2.24 shows the simplest LDO regulator designed using a three-stage OPA compensated with single Miller compensation (SMC) with a Miller capacitor C_m [1, 2]. By considering a large gate-to-drain capacitance C_{gd}, the LDO regulator has an inherent nested Miller compensation (NMC) structure [3–6]. The first stage consists of the transconductance g_{m1} of differential pairs to provide high gain. The second stage comprises positive g_{m2} as a single positive gain-boosting stage. The third stage involves the power P-MOSFET with its output impedance. Given its three stages, this structure has at least three poles. The dominant pole P_{non1} at the first stage output, which is contributed by a Miller capacitor C_m, is located at low frequencies. The first non-dominant pole P_{non2} appears at the gate of the power P-MOSFET because of its large size. The second non-dominant pole P_0, which is contributed by the output impedance because of its parasitic capacitance, is carried by the power line of the sub-supplied circuit and the equivalent output resistance.

Figure 2.25 shows a transistor-level example of the LDO regulator compensated by a single Miller capacitor C_m. The first stage, which is implemented by a differential amplifier, consists of transistors M_{P1}–M_{P3} and M_{N4}, M_{N5}. g_{m1} is the transconductance of the input differential pairs M_{P2} and M_{P3}. Transistors M_{N6} and M_{P7}, M_{P8} form the second stage to provide a positive gain-boosting effect, where g_{m2} is the transconductance of M_{N6}. The power

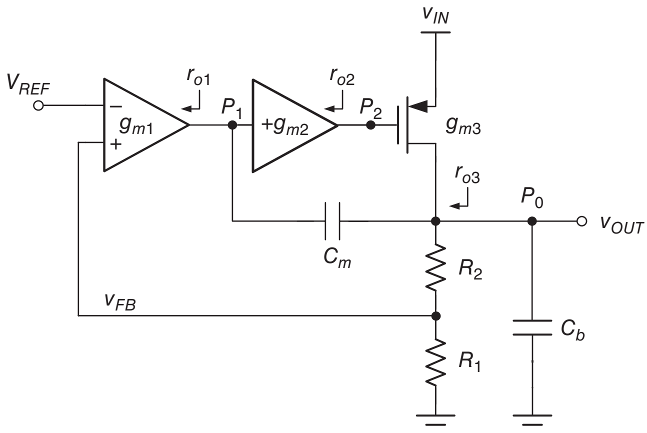

Figure 2.24 SMC for the three-stage LDO regulator

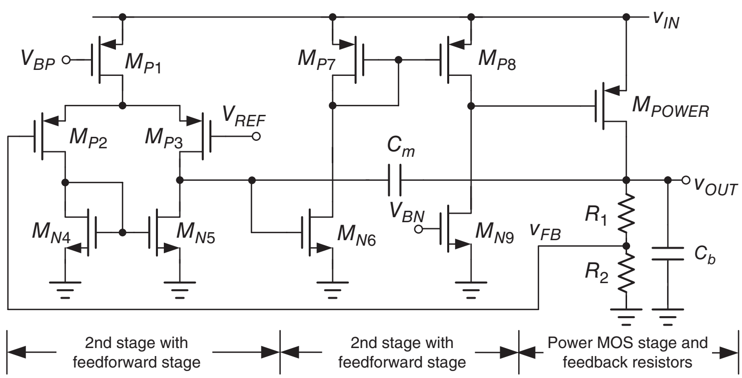

Figure 2.25 SMC capacitor for the three-stage LDO regulator

MOSFET and its output impedance form the third stage, where g_{m3} is the transconductance of M_{POWER}.

Figure 2.26 shows the simplified open-loop structure when considering all equivalent parasitic capacitances to analyze the frequency response, where g_{m1}, g_{m2}, and g_{m3} are the transconductances of the first, second, and third stages, respectively. r_{o1}, r_{o2}, and r_{o3} are the equivalent output resistance of each stage. C_1, C_2, and C_b are the equivalent output capacitance of each stage. In particular, C_{gd} is included because of the large capacitance contributed by the power P-MOSFET.

The small signal model shown in Figure 2.26 is illustrated further in Figure 2.27. Based on Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL) theorems, the transfer function from input to output can be given as in Eq. (2.31), where A_0 is the open-loop

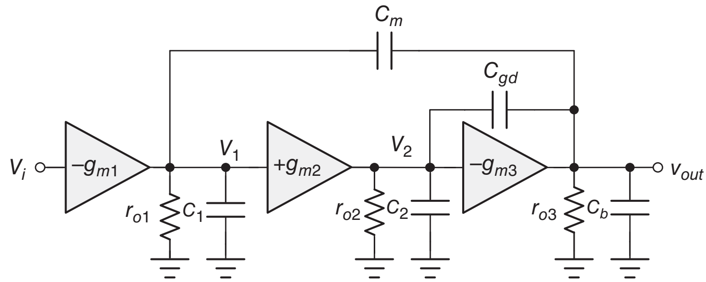

Figure 2.26 Analysis of the LDO regulator with SMC skill

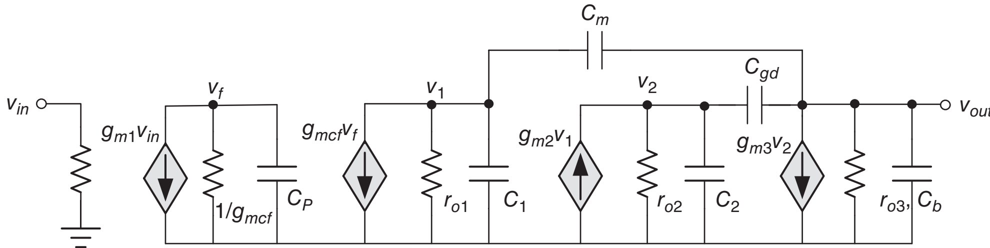

Figure 2.27 Equivalent small signal model for the basic structure of the three-stage *C-free* LDO regulator

DC gain and P_{dom} (the dominant pole of the system) are expressed in Eqs. (2.32) and (2.33), respectively:

$$\frac{v_{out}}{v_i} = A_o \frac{\left(1 + s\frac{C_{gd}}{g_{m3}} - s^2\frac{C_m C_{gd}}{g_{m2}g_{m3}}\right)}{\left(\frac{s}{P_{dom}} + 1\right)\left[s^2\frac{C_{gd}C_b}{g_{m2}g_{m3}} + s\frac{C_{gd}(g_{m3} - g_{m2})}{g_{m2}g_{m3}} + 1\right]} \tag{2.31}$$

$$A_0 = g_{m1}r_{o1}g_{m2}r_{o2}g_{m3}r_{o3} \tag{2.32}$$

$$P_{dom} = \frac{1}{r_{o1}[C_m(g_{m2}r_{o2}g_{m3}r_{o3})]} \tag{2.33}$$

Based on the numerator of Eq. (2.31), the second-order polynomial contains two zeros, which are separately located in RHP and LHP. The two zeros can be disregarded because they are located at high frequencies compared with the UGF. As mentioned in Section 2.2.2, if one LHP zero is required, then one null resistance can be in series with the Miller capacitor to convert RHP zero to LHP zero. One interesting result that can be observed is the UGF given in Eq. (2.34), which shows that its value is independent of the output capacitance because all non-dominant poles are pushed above the UGF:

$$\text{UGF} = A_0 \cdot P_{dom} = \frac{g_{m1}}{C_m} \tag{2.34}$$

The root locus of the *C-free* LDO regulator is shown in Figure 2.29 when the load current decreases. Under heavy loads, two non-dominant poles are separated at LHP. Two high-frequency non-dominant poles, namely P_{non1} and P_{non2}, can be expressed in Eqs. (2.35) and (2.36), respectively. These poles are contributed by the gate-to-drain capacitance C_{gd} of the power MOSFET and the output capacitance C_3, respectively:

$$P_{non1} \approx \frac{g_{m2}}{C_{gd}} \tag{2.35}$$

$$P_{non2} \approx \frac{g_{m3} C_{gd}}{C_2 C_b} \tag{2.36}$$

P_{non2} is load dependent because it is located at the output node. That is, P_{non2} moves toward high and low frequencies at heavy and light loads. In case of decreasing load current, two separated poles will run into each other to become complex poles, as shown at point B in Figure 2.28. In this case, these poles remain located at LHP, and thus the system is stable but the PM decreases continuously. Once the load current decreases further to light or ultra-light load conditions, these two poles may move from LHP to RHP. The system becomes unstable, as shown at point C in Figure 2.28.

The natural frequency ω_0 and the damping factor Q are derived from Eqs. (2.37) and (2.38), respectively:

$$\omega_0 = \sqrt{\frac{g_{m2} g_{m3}}{C_2 C_3}} \tag{2.37}$$

$$Q = \sqrt{\frac{C_2 C_b}{g_{m2} g_{m3}}} \frac{g_{m2} g_{m3}}{(g_{m3} - g_{m2}) C_{gd}} \tag{2.38}$$

Similarly, we can explain the scenario of complex poles under light loads at different Q values. Given the decrease in g_{m3} under light loads, the denominator will gradually decrease and even become negative, which indicates that a high Q effect becomes serious under light loads.

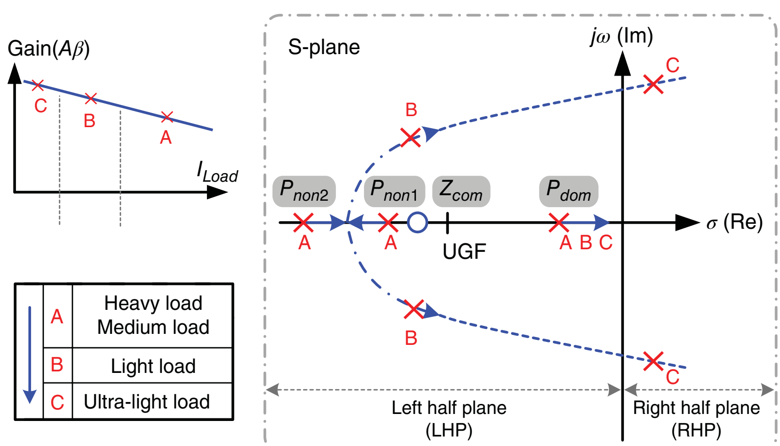

Figure 2.28 Root locus of the *C-free* LDO regulator when the load current decreases

As soon as g_{m3} becomes smaller than g_{m2}, the complex poles move from LHP to RHP. Then, the system becomes unstable. Figure 2.29(a), (b) shows the positions of the two non-dominant poles under heavy and light loads, respectively. Given the frequency peaking that occurs in Figure 2.29(b) caused by the high Q effect, the gain margin and PM are insufficient to guarantee system stability.

As illustrated in Figure 2.30, the output voltage gradually becomes unstable because of the variation in complex poles and the increase in Q, which is an expected phenomenon when the load current decreases continuously to ultra-light load conditions. Output voltage ringing and long settling time (t_s) during the transient period are caused by insufficient gain margin and PM.

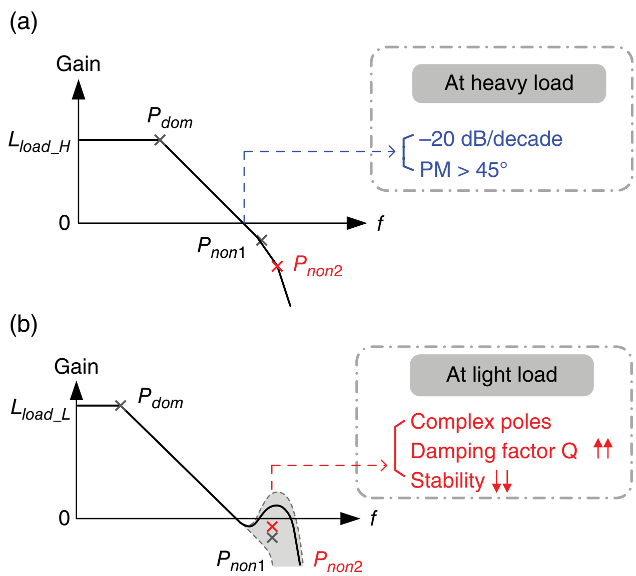

Figure 2.29 Frequency response of the *C-free* LDO regulator at (a) heavy loads and (b) light loads

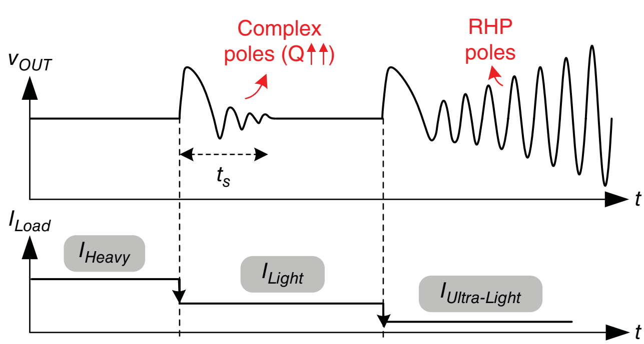

Figure 2.30 Load transient response in case of different load current changes in the *C-free* LDO regulator

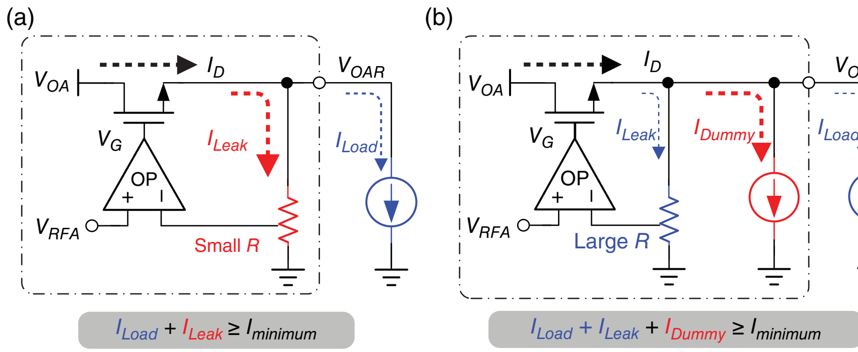

Figure 2.31 Conventional solution to alleviate minimum load limitation in the *C-free* LDO regulator with (a) an increased quiescent current or (b) a dummy loading current

Moreover, the slewing problem affects the transient response time of the *C-free* LDO regulator. Without the assistance of an output capacitor during the load transient period, the transient response can be improved by a slightly higher quiescent current to achieve a low impedance and a high slew rate (SR). However, the current efficiency is poor under light loads. Moreover, the biasing current of the second stage should be sufficiently large to drive the power MOSFET and satisfy the transient requirements. Consequently, a high quiescent current increases the values of g_{m2} and Q. That is, the minimum load current constraint will become larger than the expected value because increasing g_{m2} will cause the denominator to decrease according to Eq. (2.45). Compared with the corresponding LDO design with dominant-pole compensation, the *C-free* LDO regulator exhibits poor transient performance if the quiescent current is limited to the same value in both compensation techniques.

In designing *C-free* LDO regulators, the minimum load current requirement constraint seriously degrades the PCE under light-load or no-load conditions. An extra minimum loading current at V_{OUT} of tens or hundreds of microamperes is necessary. That is, to avoid an unstable light-load scenario, the load current should be larger than the minimum load current constraint. The *C-free* LDO regulator exhibits the disadvantage of minimum load current constraint, which may cause the efficiency to degrade under considerably light loads. In battery-powered electronics, such a regulator will shorten the battery usage time. The basic minimum load current can be established by the leakage current generated by the feedback divider resistors. A dummy loading current is sometimes consumed at the output. Consequently, efficiency is restricted because of the increased leakage current (I_{Leak}) or dummy load current (I_{Dummy}) payment to satisfy the minimum load current requirements, as shown in Figure 2.31(a), (b), respectively. Although an ultra-light load is required when a Soc system enters standby mode, the current wasted through the power MOSFET remains large and results in substantial power loss. Thus, the battery usage time of portable electronics is shortened.

2.4.2.1 Comparison between *C-out* LDO and *C-free* LDO

The preceding discussion presents the characteristics of dominant-pole compensation and the *C-free* structure. Table 2.2 provides a comparison of the LDO regulator design with different compensation methods. Considering Soc applications, the *C-free* LDO regulator is widely

Table 2.2 Comparison between dominant-pole compensation LDO and *C-free* LDO

	Dominant-pole compensation LDO	*C-free* LDO
Dominant pole	Output capacitor	Miller capacitor
UGB to load condition	1. Saturation-region power MOSFET → Dependent 2. Linear-region power MOSFET → Independent	Independent
Load range	Limitation of maximum load	Limitation of minimum load
Quiescent current	1. Increase for good SR and transient response	1. Increase for good SR 2. Increase for moving non-dominant pole to higher frequency 3. Increase for stability at ultra-light load
Defense of voltage variation	Excellent	Poor
DVS speed	Slow	Fast

utilized compared with the dominant-pole compensation LDO regulator. However, based on the table, the *C-free* LDO controls the degraded performance because of the discarded large output capacitor. To improve performance and achieve competitive specifications, the following subsections introduce the design methodologies for the *C-free* LDO regulator.

2.4.3 Design of Low-Voltage C-free *LDO Regulator*

For the Soc applications fabricated in an advanced process, the voltage stress that can be sustained by core devices is drastically scaled down. Implementing a high gain in conventional designs via a cascode structure is difficult, because of the decreasing voltage headroom caused by the decreasing input voltage. Given the decreasing voltage headroom, designing a multi-stage *C-free* LDO regulator under a low-input-voltage operation is useful. Under low-voltage operation, a conventional LDO regulator with dominant-pole compensation suffers from low DC gain, and thus the regulation performance deteriorates drastically. In the currently available nanometer CMOS process, the Soc requires the local LDO regulator to provide regulation performance and high power supply rejection (PSR). That is, a multi-stage *C-free* LDO regulator with a capacitor-less structure and a high DC gain is one suitable design for low-input-voltage operation. In this study, we present several design techniques for *C-free* LDO regulators to improve the regulation performance.

Figure 2.32 shows the schematic of a multi-stage *C-free* LDO with a compensation-enhancement multi-stage amplifier (CEMA) [7]. To obtain high gain, the structure must consist of three gain stages. Transistors M_7 and M_8 constitute the first gain stage. The second stage is composed of transistor M_{11} with a current mirror to obtain a positive gain. Meanwhile, the

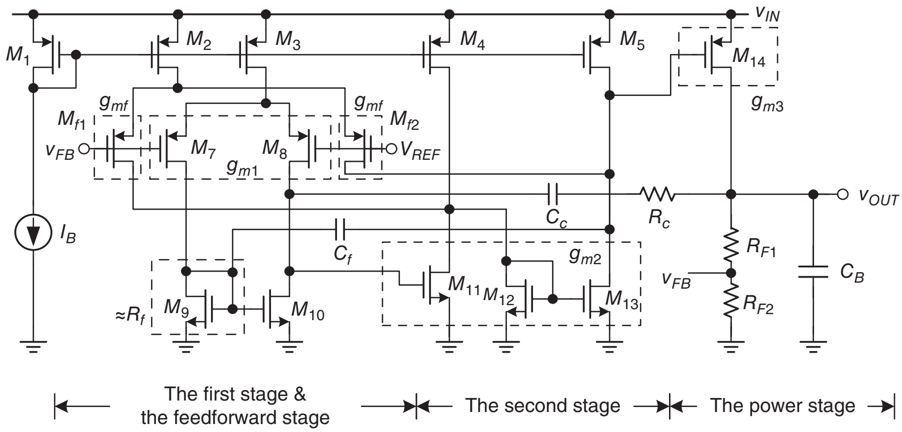

Figure 2.32 Schematic of the multi-stage *C-free* LDO regulator with CEMA

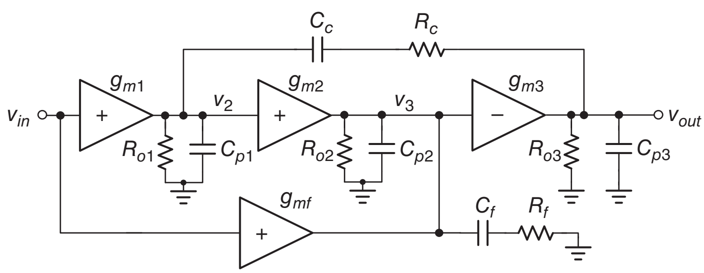

Figure 2.33 Small signal model of the CEMA structure in an open loop

power MOSFET M_{14} constitutes the third gain stage. The feedforward stage consists of transistors M_{f1} and M_{f2} to generate one compensation zero. The equivalent resistance R_f is composed of the diode-connected transistor M_9, and thus has an equivalent resistance of $1/g_{m9}$. This structure can be supplied by sub-1 V to overcome the small voltage headroom in the analog design and achieve on-chip compensation by simultaneously utilizing two small compensation capacitors. Given the cascaded structure, the voltage gain of the multi-stage amplifier can be increased even under low-voltage operation. A high gain and a large UGF can still be maintained by the multi-stage amplifier to achieve good regulation and a fast transient response, respectively.

Figure 2.33 shows the equivalent circuit of the multi-stage *C-free* LDO design, where C_c, C_f, R_c, and R_f are the compensation components. $L_O(s)$ expressed in Eq. (2.39) is the transfer function of the open loop. After simplifying Eq. (2.39), the DC gain and the positions of poles and zeros are shown in Eq. (2.40), which includes three poles (ω_{pL2}, ω_{ph1L2}, and ω_{ph2L2}) and two zeros (ω_{zL2} and ω_{zhL2}):

$$L_O(s) \approx \frac{K[1 + a_1 s + a_2 s^2]}{(1 + sC_c g_{m2} g_{m3} R_{o1} R_{o2} R_{o3})[1 + b_1 s + b_2 s^2]} \tag{2.39}$$

where

$$K = g_{m1}g_{m2}g_{m3}R_{o1}R_{o2}R_{o3}$$
$$a_2 = \frac{g_{m1}g_{m2}R_{o1}R_cR_fC_cC_f}{g_{m1}g_{m2}R_{o1} + g_{mf}},\ a_1 = R_cC_c + \frac{g_{mf}R_{o1}C_c}{g_{m1}g_{m2}R_{o1} + g_{mf}} + R_fC_f$$
$$b_1 = \frac{\left(C_f + C_{p2}\right)\left(C_c(R_{o1} + R_{o3} + R_c) + C_{p3}R_{o3}\right)}{C_cg_{m2}g_{m3}R_{o1}R_{o3}} + \frac{C_{p3}(R_{o1} + R_c)}{g_{m2}g_{m3}R_{o1}R_{o2}}$$
$$b_2 = \frac{C_{p3}\left(C_f + C_{p2}\right)(R_{o1} + R_c)}{g_{m2}g_{m3}R_{o1}}$$

$$L_O(s) \approx \frac{K\left(1 + \frac{s}{\omega_{zL2}}\right)\left(1 + \frac{s}{\omega_{zhL2}}\right)}{\left(1 + \frac{s}{\omega_{pL2}}\right)\left(1 + \frac{s}{\omega_{ph1L2}}\right)\left(1 + \frac{s}{\omega_{ph2L2}}\right)} \tag{2.40}$$

According to Miller's theorem, pole ω_{pL2}, which is provided by Eq. (2.41) using a small on-chip capacitor C_c within the range of several pico-farads, is regarded as the system dominant pole:

$$\omega_{pL2} = \frac{1}{C_cg_{m2}g_{m3}R_{o1}R_{o2}R_{o3}} \tag{2.41}$$

The compensation zero ω_{zL2}, as expressed in Eq. (2.42), is generated by the feedforward gain stage in CEMA. Moreover, the on-chip compensation resistor R_c can further push ω_{zL2} toward low frequencies to achieve adequate PM:

$$\omega_{zL2} \approx \frac{g_{m1}g_{m2}}{C_c\left(g_{mf} + R_cg_{m1}g_{m2}\right)} \tag{2.42}$$

The generated low-frequency pole/zero pair, namely ω_{pL2} and ω_{zL2}, can guarantee system stability. Considering the other two non-dominant poles contributed by the second-order polynomial of the denominator of Eq. (2.39), the stability will deteriorate when complex poles are formed. To avoid complex poles, the criteria presented through Eq. (2.43) should be held:

$$b_1^2 \geq 4b_2$$
$$\Rightarrow \left(\frac{\left(C_f + C_{p2}\right)\left(C_c(R_{o1} + R_{o3} + R_c) + C_{p3}R_{o3}\right)}{C_cg_{m2}g_{m3}R_{o1}R_{o3}} + \frac{C_{p3}(R_{o1} + R_c)}{g_{m2}g_{m3}R_{o1}R_{o2}}\right)^2 \geq 4 \cdot \frac{C_{p3}\left(C_f + C_{p2}\right)(R_{o1} + R_c)}{g_{m2}g_{m3}R_{o1}} \tag{2.43}$$

In particular, satisfying the aforementioned inequality becomes difficult if the value of C_{p2} is small when the LDO regulator with a small power MOSFET requires working under low-power operation. Consequently, a compensation capacitor C_f is utilized to satisfy Eq. (2.43) easily. The pole-splitting technique is achieved for two non-dominant poles by providing the low impedance of $1/g_{mf}$ through a short path with C_f at high frequencies. That is, the pole

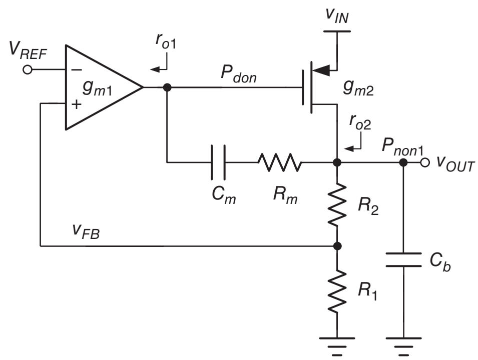

Figure 2.34 Two-stage *C-free* LDO regulator

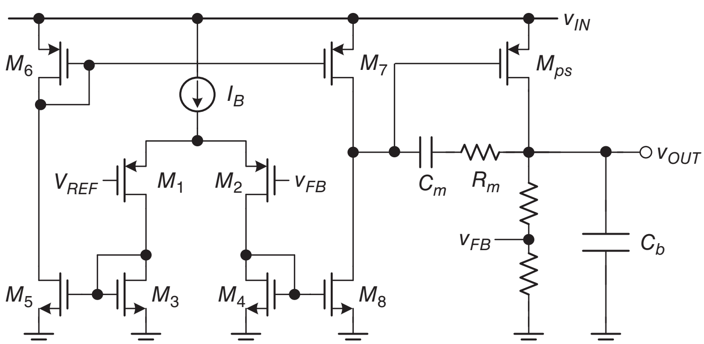

Figure 2.35 Circuit implementation of the two-stage *C-free* LDO regulator

at the second stage output can be moved toward higher frequencies. To understand the advantage of the multi-stage *C-free* LDO further, Figure 2.34 illustrates a simplified *C-free* LDO with two-stage structure for detailed comparison.

Figure 2.35 provides an example to illustrate the two-stage *C-free* LDO regulator.

The transfer function can be derived from Eq. (2.44) according to the small signal model shown in Figure 2.36:

$$L_O'(s) \approx A_0 \frac{\left(1+\frac{s}{z_1}\right)}{\left(\frac{s}{P_{dom}}+1\right)\left(\frac{s}{P_{non1}}+1\right)} \tag{2.44}$$

where

$$A_0 = g_{m1} r_{o1} g_{m2} r_{o2}$$

$$P_{dom} \approx \frac{1}{r_{o1}\left[\left(C_m + C_{gd}\right) g_{m2} r_{o2}\right]}$$

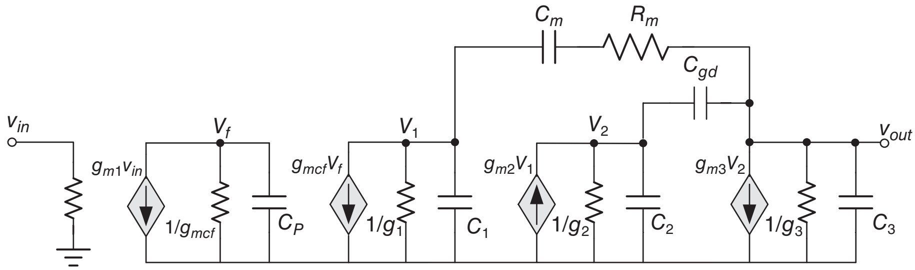

Figure 2.36 Equivalent small signal model of the two-stage *C-free* LDO regulator

$$P_{non1} = \frac{1}{r_{o2}C_b}, \ Z_1 = \frac{1}{R_m C_m}$$

In Eq. (2.44), the dominant pole P_{dom} is contributed by the Miller capacitor and C_{gd} at the node of the gate of the power P-MOSFET. Given that the power MOSFET is sufficiently large, C_{gd} behaves as the Miller capacitance without requiring an extra physical capacitor. By contrast, the only non-dominant pole P_{non1} is generated at the output node. To extend the UGF further, zero Z_1, which is provided by C_m and R_m, can be used to cancel the effect of P_{non1}. However, the two-stage *C-free* LDO cannot achieve a high gain. Moreover, the UGF is limited by P_{non1} under light loads because Z_1 cannot compensate for the PM over a wide load range.

Figure 2.37 presents a comparison of the frequency responses of the LDO regulators with a two-stage structure, a three-stage structure, and a three-stage structure with CEMA. The $L_O'(s)$ from the two-stage *C-free* LDO has a DC voltage gain that is significantly smaller than 40 dB, which cannot guarantee a regulated output driving voltage in the power management module. Meanwhile, the $L_O(s)$ from the three-stage structure with CEMA can effectively provide a high DC gain even under a low supply voltage operation of 1 V. That is, the DC voltage gain can be raised higher than 80 dB to achieve good regulation performance. In addition, the compensation zero enhancement and non-dominant pole splitting in CEMA are indicated. The location of the output filter pole of the DC/DC converter is also provided in Figure 2.37. The system PM varies from 55° to 80° under different load conditions.

2.4.4 *Alleviating Minimum Load Current Constraint through the Current Feedback Compensation (CFC) Technique in the Multi-stage* C-free *LDO Regulator*

In general, the LDO regulator should guarantee high PSR and fast transient response. Hence, these two factors should be simultaneously considered along with minimum load current. In this case, the multi-stage LDO with the current feedback compensation (CFC) technique can be used to alleviate the minimum load constraint for the *C-free* LDO regulator while achieving high PSR and fast transient response.

The CFC structure consists of three stages, as shown in Figure 2.38. The first stage, which involves M_1–M_6, converts differential in-signals to single-end out-signals and achieves high gain. The second stage involves M_7–M_{12} and R_B. During this stage, transistors M_7–M_{10} form

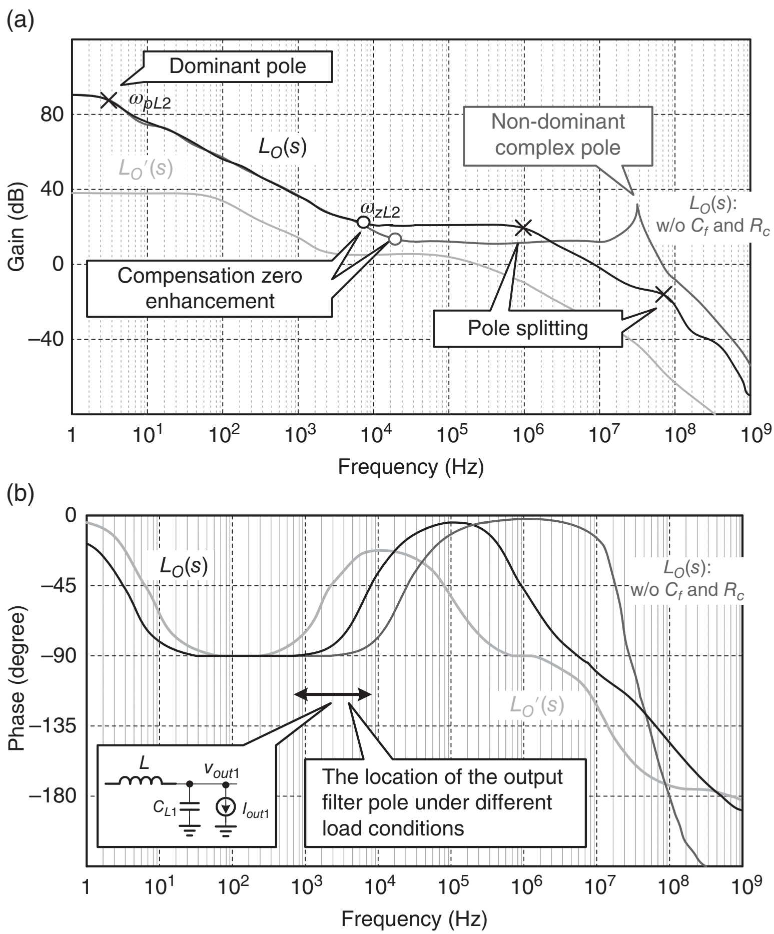

Figure 2.37 Comparison of three different EAs in frequency response. (a) Gain. (b) Phase

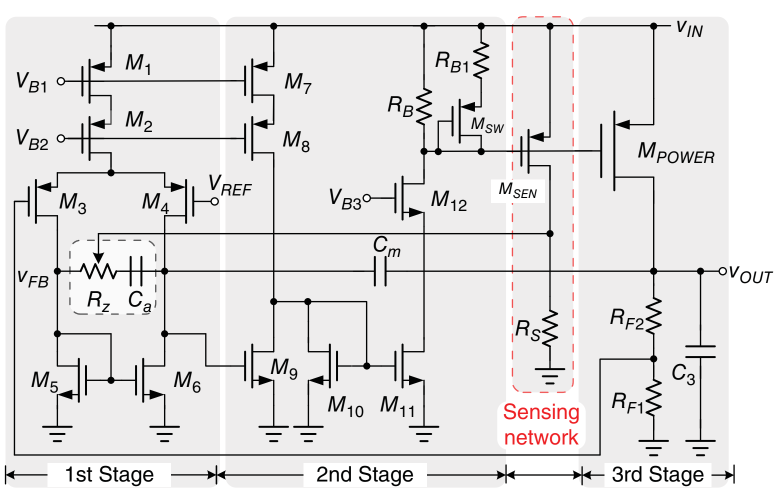

Figure 2.38 Architecture of the CFC *C-free* LDO regulator

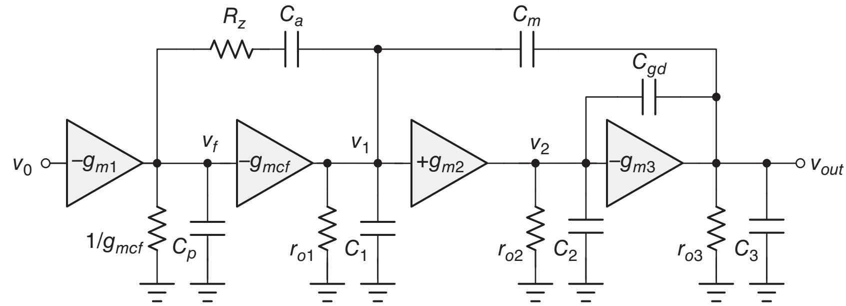

Figure 2.39 Equivalent small signal circuit of the CFC *C-free* LDO regulator

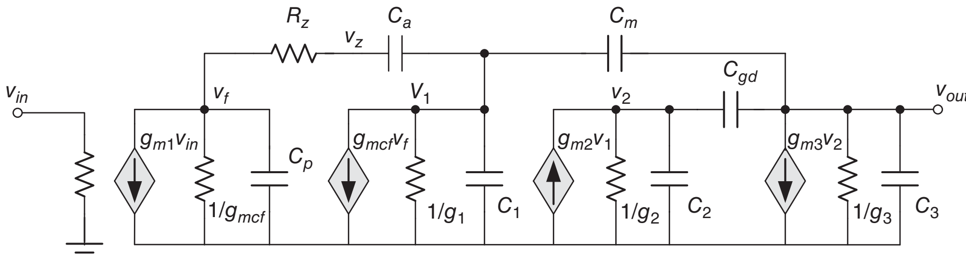

Figure 2.40 Complete small signal model of the CFC *C-free* LDO regulator

a wideband stage and create a ground reference. Meanwhile, transistors M_{11} and M_{12} form a CS stage with resistive load R_B to achieve high PSR performance. The third stage is structured by the power P-MOSFET with CS configuration. The feedback divider resistors R_{F1} and R_{F2} form the shunt feedback to regulate the output voltage. The Miller capacitor is connected to both the ground reference node, which is the output of the first stage, and the output node to avoid noise that is directly passing from the supply source to the output. The compensation network C_a and the resistance R_Z are used to control the gain and the phase dynamically under different loads during the first stage. The value of the resistance R_Z can be adjusted by the current sensing network, which is composed of M_{SEN} and R_S.

The analysis structure of the CFC *C-free* LDO regulator is illustrated in Figure 2.39. g_{m1}, g_{m2}, and g_{m3} are the equivalent transconductances of each stage. r_{o1}, r_{o2}, and r_{o3} are the equivalent output reactances of each stage. C_1, C_2, and C_3 are the lumped parasitic capacitors of each stage. Given that the value of the gate-to-drain capacitor of the power P-MOSFET is large, the parasitic capacitor should be considered and is represented as C_{gd}.

Based on the KVL and KCL theorems, the transfer function of Figure 2.40 from the input to the output can be derived from Eq. (2.45), where A_o is the open-loop DC gain as expressed in Eq. (2.46). The dominant pole of this system is P_{dom}:

$$\frac{v_{out}}{v_0} = -A_{vo}\frac{(sC_aR_z+1)(a_2s^2+a_1s-1)}{\left(1+\dfrac{s}{P_{dom}}\right)\left(1+\dfrac{s}{P_{non1}}\right)}\cdot\frac{1}{(b_2s^2+b_1s+1)} \tag{2.45}$$

where

$$a_2 = \frac{C_m C_2}{g_{m2} g_{m3}},\ a_1 = \frac{g_{m2} C_{gd}}{g_{m2} g_{m3}},\ b_2 = \frac{C_2 C_3}{g_{m2} g_{m3}}$$

$$b_1 = \frac{\left(\frac{1}{g_{mcf}} + R_z\right)\left({}^{C_3}\!/_{r_{o2}} + {}^{C_2}\!/_{r_{o3}} + (g_{m3} - g_{m2})C_{gd}\right)C_a C_m + (C_m + 2C_a)C_2 C_3}{\left({}^{C_3}\!/_{r_{o2}} + {}^{C_2}\!/_{r_{o3}} + (g_{m3} - g_{m2})C_{gd}\right)(C_m + 2C_a) + 2g_{m2} C_{gd} C_a + \left(\frac{1}{g_{mcf}} + R_z\right) g_{m2} g_{m3} C_a C_m}$$

$$A_{vo} = g_{m1} r_{o1} g_{m2} r_{o2} g_{m3} r_{o3} \tag{2.46}$$

The system consists of four poles and three zeros. One LHP zero Z_{LHP} for dynamic compensation is contributed by the capacitor C_a and the resistor R_Z:

$$Z_{LHP} = \frac{1}{C_a R_Z} \tag{2.47}$$

The two zeros provided by the second-order polynomial are each located in RHP and LHP. These zeros can be disregarded because they are considerably higher than the UGF. The dominant pole in Eq. (2.48), which is proportional to the root of I_{Load}, is decided by C_m and the output resistance in the first stage:

$$P_{dom} = \frac{g_1 g_2 g_3}{g_{m2} g_{m3} C_m} \propto \frac{1}{\sqrt{I_{Load}} \times \frac{1}{I_{Load}}} = \sqrt{I_{Load}} \tag{2.48}$$

The second and third non-dominant poles, P_{non2} and P_{non3}, are respectively generated by the gate-to-drain capacitance C_{gd} at the gate of the power P-MOSFET and the capacitor C_3 at the output node. Given that P_{non3} depends strongly on I_{Load}, the two poles from the second-order polynomial in the denominator of Eq. (2.45) can be expressed in different forms under various load conditions for a clear analysis. P_{non3} moves toward high frequencies under heavy loads, and thus P_{non2} and P_{non3}, which are roughly expressed as in Eq. (2.49), are far from each other. By contrast, P_{non2} and P_{non3} are close to each other when P_{non3} moves toward low frequencies under light loads.

$$P_{non2} = \frac{g_{m2}}{C_{gd}},\ P_{non3} = \frac{g_{m3} C_{gd}}{C_2 C_3} \text{ under heavy loads} \tag{2.49}$$

To discuss the effect of complex poles, the natural frequency ω_0 and a damping factor featured in P_{non2} and P_{non3} can be derived as follows:

$$\omega_o = \sqrt{\frac{g_{m2} g_{m3}}{C_2 C_3}} \text{ and}$$

$$Q = \sqrt{\frac{C_2 C_3}{g_{m2} g_{m3}}} \frac{\left(g_2 C_3 + g_3 C_2 + (g_{m3} - g_{m2})C_{gd}\right)(C_m + 2C_a) + 2g_{m2} C_{gd} C_a + \left(\frac{1}{g_{mcf}} + R_Z\right) g_{m2} g_{m3} C_a C_m}{\left(\frac{1}{g_{mcf}} + R_Z\right)\left(g_2 C_3 + g_3 C_2 + (g_{m3} - g_{m2})C_{gd}\right) C_a C_m + (C_m + 2C_a) C_2 C_3} \tag{2.50}$$

under light loads

To alleviate the influence of a high Q value, which results from the two non-dominant poles under light loads, an extra pole P_{non1}, expressed in Eq. (2.51), is inserted into the frequency around the UGF:

$$P_{non1} = \frac{g_{m2}g_{m3}C_m}{\left(\frac{1}{g_{mcf}} + R_Z\right)g_{m2}g_{m3}C_aC_m + \left({}^{C_2}/_{r_{o3}} + g_{m3}C_{gd}\right)(C_m + 2C_a)} \propto \frac{1}{k_1 + k_2\sqrt{I_{Load}}} \quad (2.51)$$

where

$$k_1 = C_a\left(\frac{1}{g_{mcf}} + R_Z\right) \text{ and } k_2 = \frac{C_2(C_m + 2C_a)}{g_{m2}C_m}$$

Figure 2.41 illustrates the root locus of the poles and zeros when the load current increases from light to heavy. The UGF is nearly constant in the aforementioned *C-free* structure. According to Eqs. (2.48) and (2.51), P_{dom} moves to a high frequency, whereas P_{non1} moves to a low frequency. The lower frequency of P_{non1} deteriorates the PM, and thus the compensated zero with a large resistor R_Z can dynamically move to a lower frequency under heavy loads to guarantee stability. Moreover, the complex poles will separate into two poles as the load current increases. This phenomenon implies that the effects of P_{non2} and P_{non3} can be disregarded under heavy loads.

Compared with the Q value of the basic *C-free* LDO regulator shown in Eq. (2.38), the CFC technique uses the C_a, R_Z, and dynamic output impedance of the second stage to introduce other factors and maintain a low Q value over a wide range of load conditions, as indicated in Eq. (2.50). Moreover, this technique realizes the minimum load constraint by an existing LHP pole even under ultra-light loads.

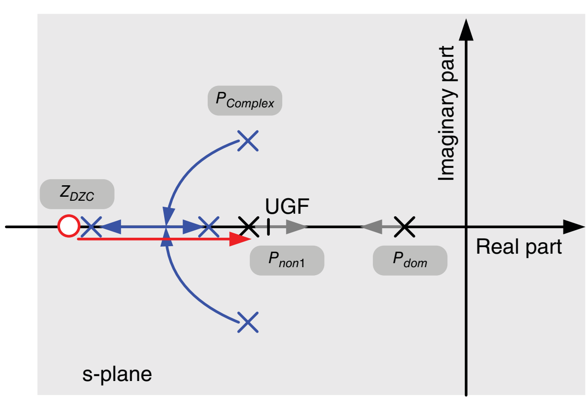

Figure 2.41 Poles and zero locations of the CFC LDO as the load increases

2.4.4.1 Ultra-Light Load Condition with Small R_Z

Under light loads, P_{non2} and P_{non3} form complex poles with a three-stage LDO design. In this case, Eq. (2.50) can be simplified to Eq. (2.52) and Q can be reduced by the second and third terms of the denominator, that is $C_2C_3(C_m+2C_a)/((1/g_{mcf})+R_Z)$ and $g_2C_aC_mC_3$, to decrease its value:

$$Q=\sqrt{\frac{C_2C_3}{g_{m2}g_{m3}}}\frac{g_{m2}g_{m3}C_aC_m}{(g_{m3}-g_{m2})C_aC_mC_{gd}+g_2C_aC_mC_3+\dfrac{C_2C_3(C_m+2C_a)}{\left(\dfrac{1}{g_{mcf}}+R_Z\right)}} \tag{2.52}$$

The compensated resistor R_Z should be small to achieve a low Q value. Based on direct observation, R_Z increases the equivalent resistance at the output node through the path of C_m at high frequencies. That is, the equivalent resistance is R_Z in series with $1/g_{mcf}$. The sensing network can adjust R_Z to have a small value under ultra-light loads. Q and P_{non1} can be derived from Eqs. (2.53) and (2.54), respectively, as follows:

$$P_{non1}=\frac{g_{mcf}}{C_a} \tag{2.53}$$

$$Q=\sqrt{\frac{C_2C_3}{g_{m2}g_{m3}}}\cdot\frac{\frac{1}{g_{mcf}}g_{m2}g_{m3}C_aC_m}{(C_m+2C_a)C_2C_3}=\sqrt{\frac{C_2C_3}{g_{m2}g_{m3}}}\cdot\frac{\frac{1}{g_{mcf}}g_{m2}g_{m3}C_m}{\left(\frac{C_m}{C_a}+2\right)C_2C_3} \tag{2.54}$$

Although R_Z is helpful in reducing Q, the peaking of Q still affects stability. To improve the constraint of minimum load current further, pole P_{non1} is set at a frequency that is near the UGF. Figure 2.42 illustrates the contribution of P_{non1}, which accelerates the decay of the gain after surpassing the frequency of the UGF.

To maintain the PM at approximately 60°, P_{non1} must be placed twice higher than the UGF, as shown in Eq. (2.55). This setup results in a low Q of 5.

$$P_{non1}>2\cdot\text{UGF}=2\frac{g_{m1}}{C_m} \tag{2.55}$$

To avoid complex poles and an unstable system, the first non-dominant pole must be set to at least half the natural frequency, as shown in Eq. (2.56). Given that the magnitude rolls off with a slope of −20 dB/decade after the UGF and a slope of −40 dB/decade after the first non-dominant pole, a gain margin of at least 18 dB exists for complex poles and low Q approximations.

$$P_{non1}<\frac{1}{2}\omega_o \tag{2.56}$$

The compensated zero is located at a relatively high frequency compared with the UGF, because R_Z is small. Consequently, the PM of the overall system can be determined by Eq. (2.57), which is near 60°:

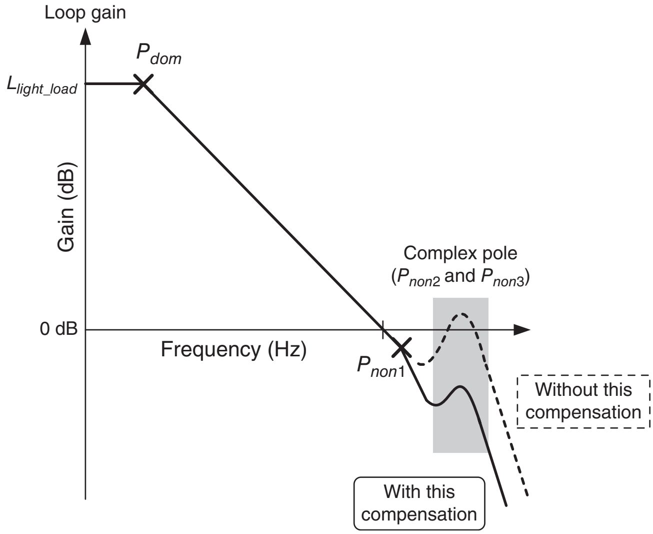

Figure 2.42 CFC capacitor-free LDO under a light load condition

$$\begin{aligned}\mathrm{PM} &= 180^\circ - \tan^{-1}\left(\frac{\mathrm{UGF}}{p_{don}}\right) - \tan^{-1}\left(\frac{\mathrm{UGF}}{p_{non1}}\right) - \tan^{-1}\left(\frac{\frac{\mathrm{UGF}}{\omega_o}}{Q\left[1-\left(\frac{\mathrm{UGF}}{\omega_o}\right)^2\right]}\right) \\ &= 90^\circ - \tan^{-1}\left(\frac{\mathrm{UGF}}{p_{non1}}\right) - \tan^{-1}\left(\frac{\frac{\mathrm{UGF}}{\omega_o}}{Q\left[1-\left(\frac{\mathrm{UGF}}{\omega_o}\right)^2\right]}\right) = 90^\circ - 26.56^\circ - 2^\circ \approx 60^\circ\end{aligned} \tag{2.57}$$

Based on the preceding analysis, compensation capacitors C_m and C_a can be derived from Eqs. (2.58) and (2.59), respectively, as follows:

$$C_m = 2\frac{g_{m1}}{g_{mcf}}C_a = 4g_{m1}\sqrt{\frac{C_2C_3}{g_{m2}g_{m3}}} \tag{2.58}$$

$$C_a = 2g_{mcf}\sqrt{\frac{C_2C_3}{g_{m2}g_{m3}}} \tag{2.59}$$

2.4.4.2 Light-to-Medium Load Condition

As the load increases from light to medium, that is 1–10 mA, P_{non1} moves toward lower frequencies because R_Z increases. As the R_Z value increases, P_{non1}, ω_0, and Q can be derived from

Eqs. (2.60) to (2.62), respectively. In this case, a low Q can also be maintained, and Eq. (2.63) still holds true because of the slight probability of R_Z:

$$P_{non1} = \frac{1}{\left(\frac{1}{g_{mcf}} + R_Z\right)C_a} \tag{2.60}$$

$$\omega_o = \sqrt{\frac{g_{m2}g_{m3}}{C_2C_3}} \tag{2.61}$$

$$Q = \sqrt{\frac{C_2C_3}{g_{m2}g_{m3}}} \cdot \frac{g_{m2}}{C_{gd}} \tag{2.62}$$

$$P_{non1} > 2 \cdot \text{UGF} \tag{2.63}$$

The PM of the overall system can be determined by P_{dom} and P_{non1}, as shown in Eq. (2.64):

$$\begin{aligned} \text{PM} &= 180^\circ - \tan^{-1}\left(\frac{\text{UGF}}{P_{dom}}\right) - \tan^{-1}\left(\frac{\text{UGF}}{P_{non1}}\right) \\ &= 90^\circ - \tan^{-1}\left(\frac{\text{UGF}}{P_{non1}}\right) \\ &\approx 60^\circ \end{aligned} \tag{2.64}$$

2.4.4.3 Heavy Load Condition with Large R_Z

As the load increases further from 10 to 100 mA, P_{non1}, as shown in Eq. (2.65), will move to a lower frequency, because both the output reactance g_3 and R_Z increase. The low-frequency zero Z_{DZC}, expressed in Eq. (2.66), will alleviate the P_{non1} effect:

$$P_{non1} = \frac{g_{m2}g_{m3}C_m}{\left(\frac{1}{g_{mcf}} + R_Z\right)g_{m2}g_{m3}C_aC_m + g_3C_2(C_m + 2C_a)} \tag{2.65}$$

$$Z_1 = \frac{1}{R_ZC_a} \tag{2.66}$$

The second-order polynomial in Eq. (2.45) can be simplified to Eq. (2.67) under heavy loads. If the discriminant function of the second-order polynomial in Eq. (2.67) is smaller than zero, then a pair of complex poles exists in the system. However, these complex poles are located at high frequencies and have no effect on the system stability. If the discriminant function of the second-order polynomial in Eq. (2.67) is larger than zero, then two separate poles exist in the system. Finally, if a second non-dominant pole exists, then the locations of the poles are shown in Eq. (2.68), which are found at high frequencies and have negligible effects on the system stability:

$$s^2\frac{C_2C_3}{g_{m2}g_{m3}}+s\frac{g_3C_2+g_{m3}C_{gd}}{g_{m2}g_{m3}}+1\approx s^2\frac{C_2C_3}{g_{m2}g_{m3}}+s\frac{g_{m3}C_{gd}}{g_{m2}g_{m3}}+1 \tag{2.67}$$

$$\begin{aligned}&P_{non2}=\frac{g_{m2}g_{m3}}{g_3C_2+g_{m3}C_{gd}} \quad \text{and} \quad P_{non3}=\frac{g_3C_2+g_{m3}C_{gd}}{C_2C_3}\\ &\Rightarrow P_{non2}=\frac{g_{m2}}{C_{gd}} \quad \text{and} \quad P_{non3}=\frac{g_{m3}C_{gd}}{C_2C_3}\end{aligned} \tag{2.68}$$

The overall system contains two low poles and one low zero. The system stability can be determined as follows:

$$\begin{aligned}\text{PM}&=180^\circ-\tan^{-1}\left(\frac{\text{UGF}}{p_{don}}\right)-\tan^{-1}\left(\frac{\text{UGF}}{p_{1st\text{-}non}}\right)+\tan^{-1}\left(\frac{\text{UGF}}{z_{LHP}}\right)\\ &=90^\circ-\tan^{-1}\left(\frac{\text{UGF}}{p_{1st\text{-}non}}\right)+\tan^{-1}\left(\frac{UGF}{z_{LHP}}\right)\\ &\approx 60^\circ\end{aligned} \tag{2.69}$$

2.4.4.4 Summary of CFC Capacitor-Free LDO Regulators

The frequency responses of CFC capacitor-free LDO regulators under different load conditions are summarized in Figure 2.43. Under ultra-light load conditions smaller than 1 mA, the dominant pole contributes a 90° phase shift. The first non-dominant pole contributes a near 30° phase shift, and the complex poles contribute minimal phase shift. The overall system stability can be maintained at 60°. Under light to medium load conditions, that is approximately 1–10 mA, the dominant pole contributes the same phase shift. The first non-dominant pole moves to lower frequencies and contributes a 30° phase shift. The complex poles move to higher frequencies and do not contribute any phase shift. Thus, the system stability is maintained at 60°. Finally, under medium to heavy load conditions, approximately 10–100 mA, the dominant pole contributes the same phase shift. The first non-dominant pole moves further

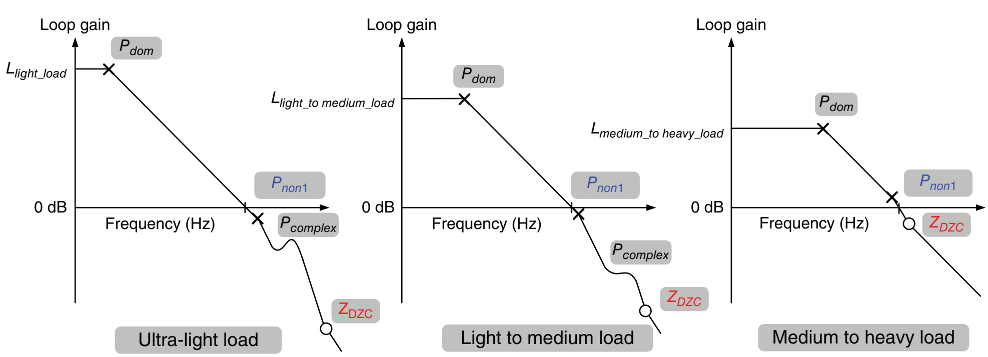

Figure 2.43 Variations of poles and zero locations in the CFC *C-free* LDO regulator over a wide load range

toward even lower frequencies, and the dynamic zero also moves toward lower frequencies to alleviate the pole effect. Therefore, the PM of the CFC *C-free* LDO regulator can be maintained at 60° within the entire load range.

2.4.5 *Multi-stage LDO Regulator with Feedforward Path and Dynamic Gain Adjustment (DGA)*

With the development of advanced nanometer processes in recent years, the supply voltage has been continuously scaled down. Consequently, LDO regulators are required to operate under sub-1 V. To understand the design in detail, an example implemented in a 65 nm CMOS technology is presented here. The design specifications are described as follows. The input voltage V_{IN} can range from 0.9 to 1.2 V; the latter is the voltage stress of the 65 nm core device. The dropout voltage is approximately 200 mV, with a maximum load current of 100 mA. To achieve stable operation, a minimum load of 50 μA is required.

Figure 2.44 shows a schematic of the four-stage LDO with feedforward path and dynamic gain adjustment (DGA). Moreover, the feedforward gain stage contributed by M_{13} is included to accelerate the transient response at the gate of the power P-MOSFET M_P. As mentioned earlier, the feedforward path contributes LHP zero in the frequency domain to improve stability. The DGA mechanism, which consists of M_{14} and R_S, is used to sense the load current and adjust the resistance value of R_p to guarantee stability even at ultra-light loads.

Figure 2.45 shows the structure of the sub-1 V multi-stage LDO design. Considering its cascade structure, the voltage gain of the multi-stage amplifier can be increased under low-voltage operation. A high-voltage gain is basically composed of three gain stages: g_{m1}, which includes g_{md}, g_{m2}, and g_{mL}. Moreover, a feedforward gain stage g_{mf} accelerates the slew rate at the gate of the pass element and generates LHP zero to improve LDO stability. In addition, the compensation capacitor C_c is amplified by Miller's theorem to determine the dominant pole in the system. C_f is inserted to split high-frequency complex poles under light loads and prevent

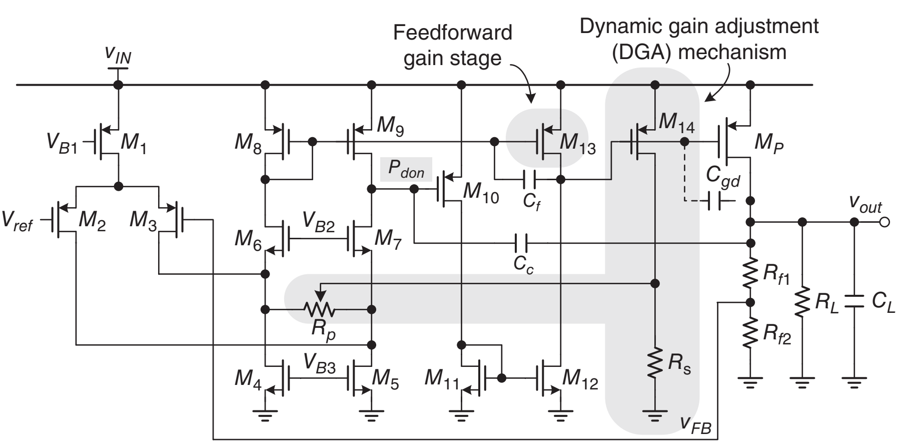

Figure 2.44 Schematic of the sub-1 V four-stage LDO with feedforward gain stage and DGA mechanism

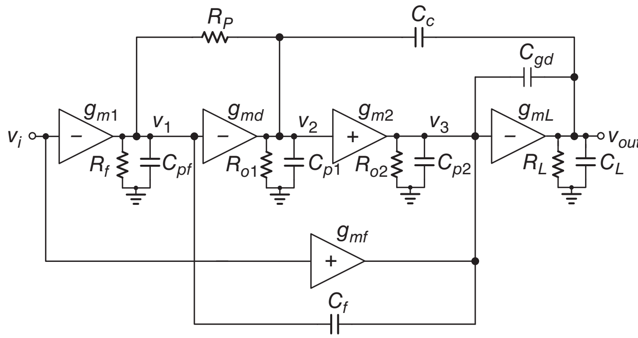

Figure 2.45 Structure of the sub-1 V multi-stage LDO regulator

non-dominant complex poles from moving toward RHP to ensure stability. Furthermore, resistance R_p is proportionally adjusted to I_{Load} by the DGA mechanism. Hence, R_p can be used to adjust the DC voltage gain and the UGF to avoid phase deterioration under ultra-light loads. The transfer function is given in Eq. (2.70) according to Figure 2.45. The exact analysis of the multi-stage LDO can be illustrated through three different output load conditions, as discussed in the following subsections.

$$\frac{v_{out}}{v_{in}} \approx \frac{g_{m1}g_{m2}g_{mL}R_{o1}R_{o2}R_L \frac{R_f\left(g_{md}R_p-1\right)}{\left(R_p+g_{md}R_{o1}R_f\right)}\left[1+s\left(R_fC_f+\frac{g_{mf}R_{o1}C_c}{g_{m1}g_{m2}R_{o1}+g_{mf}}\right)\right]}{\left(1+sC_cg_{m2}g_{mL}R_{o1}R_{o2}R_L\right)\left[1+s\frac{\left(R_p+R_f\right)\left(g_{mL}\left(C_{gd}+C_f\right)-g_{m2}\left(C_c+C_L\right)+g_{mf}\left(C_f+C_2+C_c\right)\right)}{g_{m2}g_{mL}}+s^2\frac{C_L\left(C_{gd}+C_f+C_2\right)}{g_{m2}g_{mL}}\right]} \tag{2.70}$$

2.4.5.1 Medium to Heavy Load Currents (g_{mL} Significantly Larger than g_{m1} and g_{m2})

As the load current increases from 10 to 100 mA, the transfer function in Eq. (2.70) is approximated as:

$$\frac{v_{out}}{v_{in}} \approx \frac{g_{m1}g_{m2}g_{mL}R_{o1}R_{o2}R_L\left(1+s\frac{g_{mf1}C_c}{g_{m1}g_{m2}}\right)}{\left(1+sC_cg_{m2}g_{mL}R_{o1}R_{o2}R_L\right)\left[\begin{array}{l}1+s\dfrac{\left(C_{gd}+C_f+C_2\right)\left(R_p+R_f\right)}{g_{m2}}\\ +s^2\dfrac{C_L\left(C_{gd}+C_f+C_2\right)}{g_{m2}g_{mL}}\end{array}\right]} \tag{2.71}$$

The dominant pole ω_{don}, a feedforward LHP zero ω_{zf}, and a system UGF are given respectively as:

$$\omega_{don} = \frac{1}{C_cg_{m2}g_{mL}R_{o1}R_{o2}R_L} \tag{2.72}$$

$$\omega_{zf} = \frac{g_{m1}g_{m2}}{C_c g_{mf1}} \tag{2.73}$$

$$\text{UGF} = \frac{g_{m1}}{C_c} \tag{2.74}$$

The first non-dominant pole at the gate of the power MOSFET is close to the feedforward zero ω_{zf}. Moreover, the second non-dominant pole at the output node is located at high frequencies because of the large g_{mL}. Consequently, a one-pole system is achieved with a theoretical PM of 90°. Thus, the stable operation of the multi-stage LDO is derived under medium to heavy load conditions because of the split high-frequency LHP non-dominant poles.

2.4.5.2 Light to Medium Output Load Currents (g_{mL} Slightly Larger than g_{m1} and g_{m2})

As the load current increases from a light to a medium load range, that is approximately 1–10 mA, the non-dominant pole at the output node moves toward the origin, which results in the occurrence of complex poles, with the first non-dominant pole at the gate of the power MOSFET. Based on the transfer function of Eq. (2.70), the relationship among non-dominant complex poles can be exhibited as follows:

$$\text{UGF} = \frac{g_{m1}}{C_c} \tag{2.75}$$

Under this condition, complex poles are located at high frequencies, but will have an influence on the phase shift. The complex pole frequency ω_o and Q are respectively indicated by:

$$\omega_o = \sqrt{\frac{g_{m2}g_{mL}}{C_L\left(C_{gd} + C_f + C_2\right)}} \tag{2.76}$$

$$Q = \sqrt{\frac{C_L\left(C_{gd} + C_f + C_2\right)}{g_{m2}g_{mL}}} \times \left[\frac{g_{m2}g_{mL}}{\left(R_p + R_f\right)\left(g_{mL}\left(C_{gd} + C_f\right) - g_{m2}\left(C_c + C_L\right) + g_{mf}\left(C_f + C_2 + C_c\right)\right)}\right] \tag{2.77}$$

Increasing g_{mL} and C_f can push complex poles toward higher frequencies, and thus decrease the Q value to improve the system stability. Thus, implementing C_f can minimize the unintended phase shift caused by complex poles under light to medium load conditions.

2.4.5.3 Ultra-Light Load Current (g_{mL} Smaller than g_{m1} and g_{m2})

Under ultra-light loads (i.e., below 1 mA) a small g_{mL} will lead to an unstable LDO system. As mentioned in the discussion on light to medium load conditions, non-dominant complex poles with a high Q factor reduce the PM. Consequently, a higher Q factor is derived with decreasing

g_{mL}. Non-dominant complex poles will move toward the RHP to deteriorate system stability. The boundary condition between stable and unstable operations can be expressed as:

$$g_{mL}\left(C_{gd}+C_f\right)-g_{m2}(C_c+C_L)+g_{mf}\left(C_f+C_2+C_c\right)>0$$
$$\Rightarrow g_{mL}>\frac{g_{m2}(C_c+C_L)-g_{mf}\left(C_f+C_2+C_c\right)}{\left(C_{gd}+C_f\right)} \tag{2.78}$$

RHP poles can be eliminated when the s term of the second-order function in Eq. (2.70) is positive. Thus, a minimum value of g_{mL} must be contained to ensure stability. A minimum load current of the multi-stage LDO can be obtained to maintain a minimum g_{mL} value.

Nevertheless, non-dominant complex poles under ultra-light loads still affect the phase shift around the UGF. A load-dependent resistance R_p, proportional to the output loading current, is used to adjust the DC voltage gain in the DGA mechanism. Given the high voltage gain at ultra-light loads, decreasing the DC voltage gain and the UGF does not obviously influence the transient response. The DGA mechanism can minimize the effect caused by non-dominant complex poles, and thus ensure LDO stability.

Figure 2.46 shows the frequency response under an ultra-light load condition of 50 μA with and without a flying capacitor C_f and the DGA mechanism. In the case without both C_f and the DGA mechanism, non-dominant complex poles appear near the UGF and result in a poor PM. With the implementation of C_f, the Q factor decreases but an inadequate PM remains. The DGA mechanism can slightly decrease the DC voltage gain and the UGF of the LDO to enhance the system PM. Thus, operation under an ultra-light load can be realized.

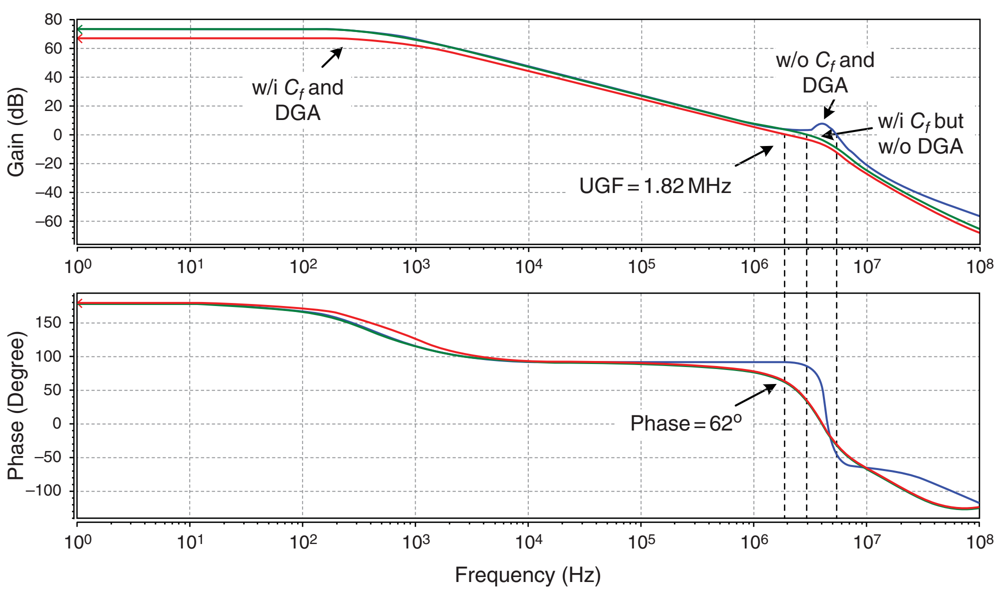

Figure 2.46 Comparison of frequency responses under a 50 μA load condition at different implementations

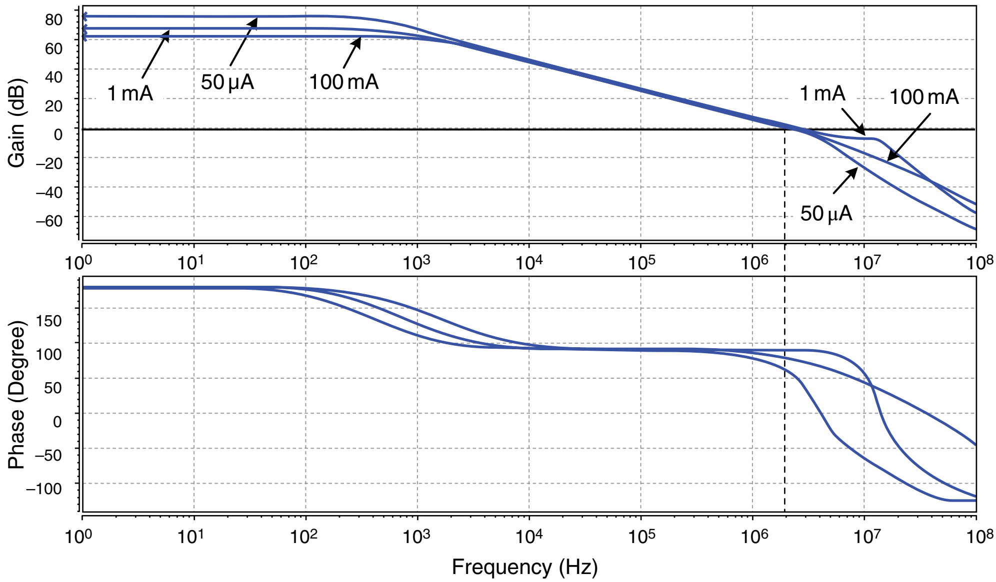

Figure 2.47 Frequency responses under different output load conditions

Figure 2.47 shows the frequency responses under different load conditions, ranging from 50 μA to 1 mA, and then to 100 mA. With the DGA mechanism, which can dynamically adjust the DC voltage gain through load detection, the correct operating function is achieved under all load conditions.

Figure 2.48 shows the load transient response under an input voltage of 1 V. In Figure 2.48(a), the output voltage is regulated to 0.8 V with undershoot and overshoot voltages of approximately 52 and 85 mV, respectively, when the load changes from 50 μA to 100 mA, and vice versa, within a load step of 1 μs. The transient recovery time is approximately 0.6 μs. By contrast, the transient response under an output voltage of 0.6 V is presented in Figure 2.48(b). The undershoot and overshoot voltages are 43 and 70 mV, respectively, with a load step ranging from 50 μA to 100 mA, and vice versa. The transient recovery time is approximately 0.6 μs.

2.5 Design Guidelines for LDO Regulators

In this section of the book we provide typical guidelines for designing LDO regulators. The crucial design parameters that should be considered carefully before implementing a circuit are as follows:

- reading the specifications carefully
- load considerations
- output capacitor considerations
- on-chip capacitor considerations
- MOSFET-cap considerations
- system compensation considerations

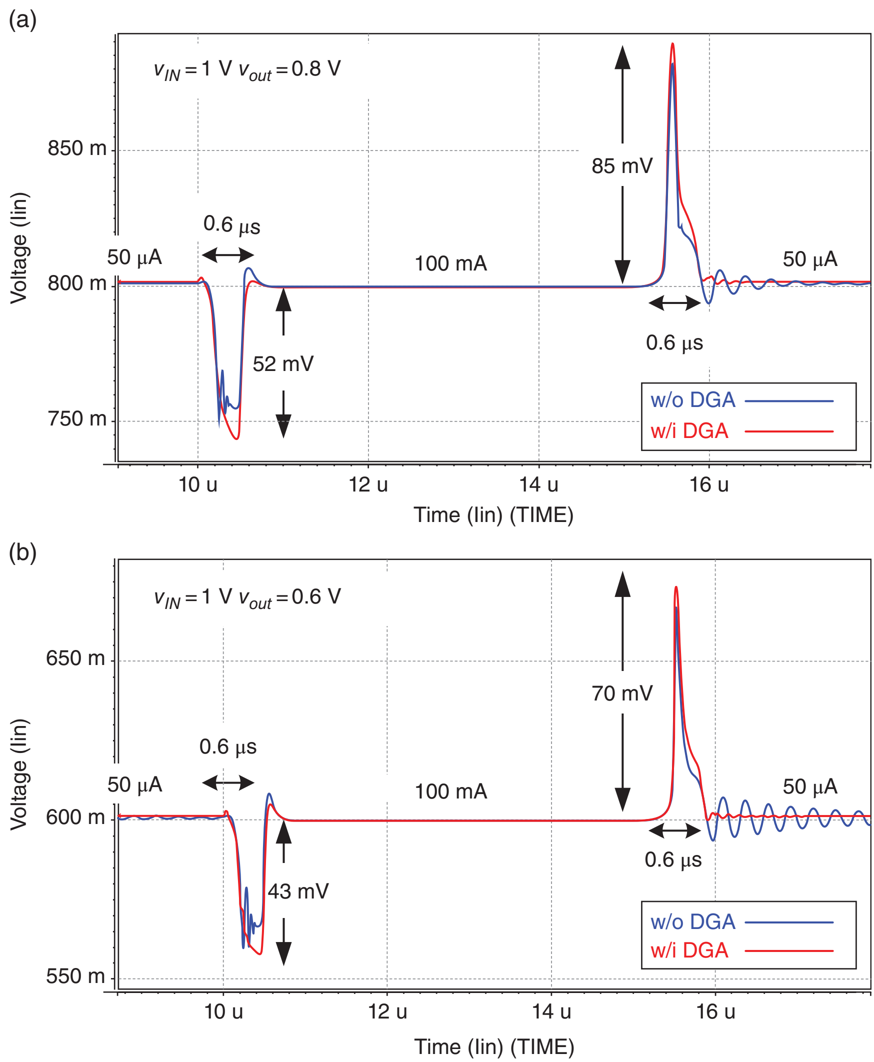

Figure 2.48 Load transient responses under 50 µA to 100 mA, and vice versa. (a) $V_{IN} = 1$ V, $V_{OUT} = 0.8$ V. (b) $V_{IN} = 1$ V, $V_{OUT} = 0.6$ V

- voltage range
- power MOSFET size estimation.

Before starting, the specifications should be read carefully. Compensation methods and techniques have both advantages and limitations, that is, the actual design exhibits trade-off with performance. Consequently, some important conditions provided in the specifications can help designers determine an adequate structure.

For example, load considerations include the desired driving capability and the load range. The driving capability suggests that designers should provide a wide bandwidth, a large SR, or a large output capacitance to sustain V_{OUT} regulation during instances of changing loads. To satisfy the load range requirement, a designer should be careful with the maximum/minimum load limitation when using dominant-pole compensation or a *C-free* structure. In some applications, such as power for PA, the required large capacitance with nano- to micro-farads at the node of the supply voltage is unavoidable. Such a large capacitance can provide low frequency. For the desired gain, PM, and UGF, a pole can obstruct frequency response compensation regardless of whether dominant-pole compensation or a *C-free* structure is applied. By contrast, on-chip capacitances are utilized for the desired poles or zeros to provide adequate frequency response. In this case, the capacitance value should be determined carefully when considering the cost of the silicon area. On-chip capacitance can be realized using different types of device, such as poly–insulator–poly (PIP), metal–insulator–metal (MIM), or MOSFET-capacitance. In particular, the cross voltage can directly vary the capacitance when MOSFET-capacitance is used. Thus, the designer should ensure that cross voltages are sufficiently large under all conditions with varying loads, input voltages, and output voltages. In general, the specifications of load consideration and output capacitance loading provide sufficient information for the designer to plan the compensation strategy. Meanwhile, the operating ranges of the input or output voltage are related to the frequency response and stability because different cross voltages can vary the gain, small signal resistance, and operating region. Once a power transistor operates in the linear and saturation regions at different input/output voltages, the gain and small signal resistance of the transistor significantly change and seriously influence the frequency response. In this case, compensation should be adjusted adequately for different situations. Finally, the size of the power transistor is estimated. The desired dropout voltage and driving capability can basically determine the minimum size of a power transistor. Hence, the designed size should be larger than the minimum size. By contrast, the maximum size of a power transistor should be limited not only because of concerns about using unnecessary silicon area, but also because an extremely large power transistor can degrade the DC operation of the EA under light loads. For example, to drive a large p-type power transistor under a light load, the driving voltage level should be extremely high. This voltage may be the upper voltage headroom of the last EA stage, and thus the gain is decreased drastically. Consequently, a power transistor with adequate size should be designed.

2.5.1 Simulation Tips and Analyses

After circuit implementation, performance can be checked through the following simulations. AC analysis is used to analyze the frequency response, including gain, PM, and UGF. Moreover, checking the slew rate in case of a large signal operation is also important. DC analysis is employed to check the operating region of each MOSFET, and DC specifications such as line and load regulations. Transient analysis is utilized to check dynamic performance during the transient period and power-on sequence. Planning the power-on sequence with an adequate soft start is important to ensure safe operation, without any damage caused by overcurrent, overvoltage, and so on, before quiescent operation is established.

Finally, full-chip simulation should be carefully executed. In this step, designers should consider possible parasitic components existing on the silicon and the PCB. Such components include the capacitance of the back-end circuit; on-resistance through the power delivery path; and R, L, and C in bond wires, among others. Moreover, designers should consider process, voltage, and temperature (PVT) variations. However, considering numerous factors draws from the simulation time. As previously stated, appropriate design and simulation considerations can effectively guarantee the yield rate after fabrication:

- AC analysis
- DC analysis
- transient analysis
- power-on sequence and soft-start design
- full-chip simulation
- consideration of PVT variations.

Designers should know how to utilize simulation tools to analyze circuit performance, as discussed in the preceding paragraphs. These tools can also be used to verify the original functions and yield rate after fabrication. Therefore, the authors strongly encourage designers to anticipate all possible situations that chips may encounter in practice. Designers should first learn to establish simulation environments. Understanding design issues and practical system operating environments is helpful for this purpose. However, checking all possible considerations is impossible because such a process not only increases the design complexity but also requires considerable simulation time. Therefore, the trade-off between robust simulation and efficient design procedure should be balanced carefully.

2.5.2 Technique for Breaking the Loop in AC Analysis Simulation

An open-loop structure without a feedback path is applied to reduce complexity or alleviate the BW constraint, while the regulated voltage is varied significantly because of the loading effect. To guarantee the quality of the regulated voltage, a closed-loop structure is used as the basis to design power regulators. The stability of the closed-loop structure should be designed carefully; such stability is generally confirmed by the gain margin and PM. However, a closed-loop structure experiences difficulties in AC simulation because the processes of injecting the stimulus and obtaining the output signal should be determined. The characteristics of a closed-loop system can be obtained from an open-loop analysis whose bode plot reflects its corresponding gain margin and PM. That is, the analysis is simplified by conducting an open-loop analysis, which is achieved by breaking the loop. Thus, determining how to break the loop adequately is important to maintain simulation accuracy without being affected by the loading effect. Two methods are recommended for this purpose, and these are introduced as follows.

Figure 2.49 illustrates a typical circuit block that represents a power converter with a feedback loop. Block A provides the open-loop gain, whereas block β provides the path to feed the output voltage V_{OUT}, which enables the reference voltage V_{REF} to regulate the output voltage.

When breaking the loop, the input and output impedances should be considered at the broken node to build an accurate model. In Figure 2.50, the output impedance $Z_{o,\beta}$ of block β is added to the front end of the breaking node while the input impedance $Z_{i,A}$ of block A is added to the back end of the breaking node. In Figure 2.51, the test AC signal source v_{ac}, a test inductor L, and a test capacitor C are used to extract the transfer function during simulations. The test inductor L conducts DC information to maintain the DC operating voltage in a steady state. Meanwhile, L blocks AC information, and thus the circuit forms as an open loop in view of the small signal aspect. By contrast, the test capacitor C conducts only the test AC signal

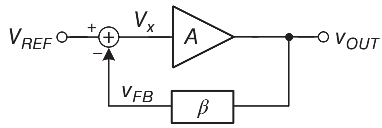

Figure 2.49 Model of a typical converter circuit

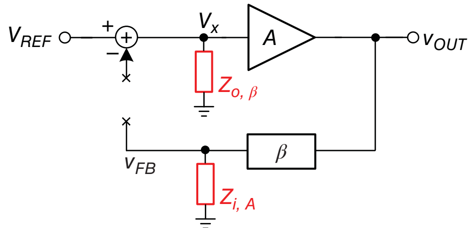

Figure 2.50 Basic principle for breaking a feedback loop

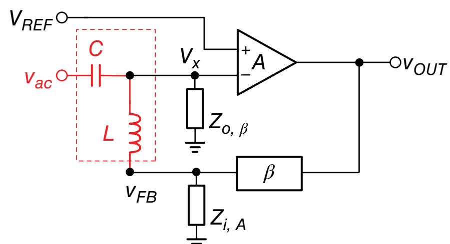

Figure 2.51 General method for breaking a feedback loop

v_{ac}, but blocks any DC information. Consequently, the DC voltage of V_X will not be influenced by the AC signal source. Given that L and C are expected to be ideal components with infinite values, their values should be high, on the order of several megas. For example, L can be 100 MH and C can be 100 MF. The transfer function can be obtained by observing the value of v_{FB}/v_{ac}.

However, deriving accurate input/output impedance values is not easy or convenient. As such, one possible method to break a loop is to select a high impedance node as the breaking node to reduce the complexity of the analysis. For example, a MOSFET gate is typically chosen as the breaking node because of its high impedance. Consequently, the input impedance $Z_{i,A}$ can be disregarded. Moreover, the output impedance $Z_{o,\beta}$ can also be ignored. The AC and DC components at V_x can be determined directly by v_{ac} and v_{FB} through C and L, respectively, to derive the transfer function. This procedure does not provide an exact solution to the transfer function, because the assumption is high impedance at the gate of the power MOSFET. Furthermore, designers may experience a situation in which the high impedance node is difficult to find in the loop. When the breaking node is not a high impedance node, the corresponding impedance $Z_{i,A}$ should be considered; however, this action increases the complexity of analysis.

Figure 2.52 presents another approach to establish an accurate model related to the original circuit. Test capacitors, C_1 and C_2, and test inductors, L_1 and L_2, provide adequate AC and DC paths for the same purpose mentioned earlier. The values of these test passive components are set on the order of several megas. When considering the DC domain, the DC operating voltage

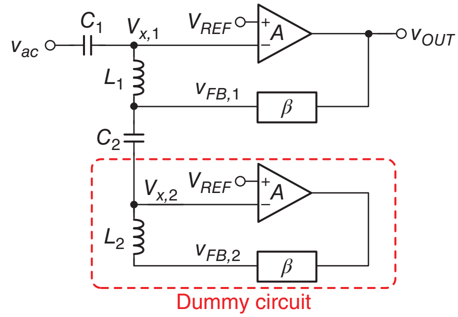

Figure 2.52 Accurate model for breaking a feedback loop

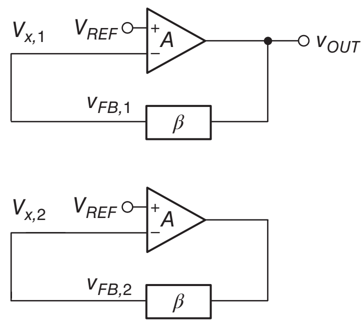

Figure 2.53 DC operating points correctly established

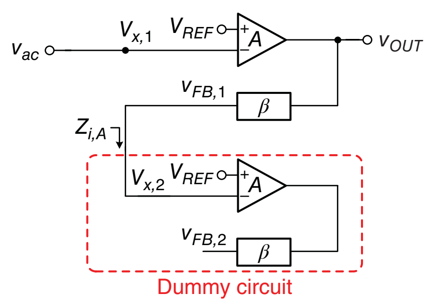

Figure 2.54 In the AC domain, the circuit topology reflects an exact solution to the transfer function without being affected by the selection of the breaking node

of each circuit block can be established as shown in Figure 2.53, because of the short paths provided by the inductors L_1 and L_2, and the open paths provided by C_1 and C_2. By contrast, when considering the AC domain, Figure 2.54 shows that the capacitors C_1 and C_2 are shorter and the inductors L_1 and L_2 are opened, and thus the circuit block exhibits one open loop and the dummy circuit is utilized to provide $Z_{i,A}$. Consequently, the AC and DC characteristics are confirmed, and thus this approach can reflect an accurate model without being affected by the selection of the breaking node. Similarly, the transfer function can be obtained from the value of $v_{FB,1}/v_{ac}$.

2.5.3 Example of the Simulation Results of the LDO Regulator with Dominant-Pole Compensation

In this subsection, we provide basic simulation methods to verify the performance and function of the LDO regulator whose specifications are listed in Table 2.3. These will guide the reader to perform the simulations step by step. The verified performances include stability, dropout voltage, line regulation, load regulation, overshoot/undershoot voltage, and settling time during transient response.

In AC analysis, Figure 2.55 illustrates the closed loop in the LDO regulator with dominant-pole compensation. The first step is to identify the possible location of the breaking node in the entire LDO regulator. In general, the input impedance of the EA is assumed to be infinitely large because the first stage is implemented by a CS amplifier. Thus, Figure 2.56(a) shows that the breaking node is selected on the path from V_{FB} to the non-inverting terminal of the EA. Figure 2.56(b) shows that L_{ideal} and C_{ideal} are added to provide the paths for conducting DC and AC signals, respectively, to determine the quiescent operating points and the transfer function, respectively. The conducting path for DC signals can maintain the function of DC operation with adequate values of V_{IN}, V_{OUT}, V_{REF}, and I_{Load}. Based on DC operation, the conducting path for AC signals can determine the frequency response. Readers may believe that this process is an alternative method to connect DC voltage sources directly to the breaking node as quiescent operating points. However, a high loop gain in the LDO regulator will result in a severe error if a slight error caused by the deviation of quiescent operating points occurs.

Table 2.3 Specifications of the LDO regulator

Parameter	Value	Unit
Compensation	Dominant-pole compensation	
Technology	0.25 μm CMOS (5 V device)	
Input voltage (V_{IN})	3.0–4.5	V
Output voltage (V_{OUT})	2.7	V
Reference voltage (V_{REF})	1.2	V
Load current (I_{Load})	1–100	mA
Output capacitor (C_{OUT})	4.7	μF
ESR of output capacitance (R_{ESR})	5	Ω

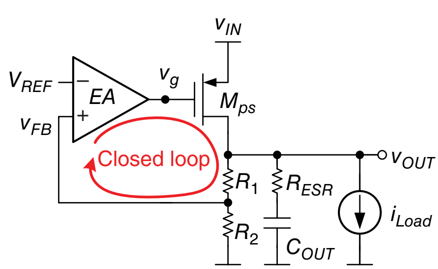

Figure 2.55 Illustration of the LDO regulator for AC analysis

Given the simple single-stage design, the poles and zeros of the EA, which are located at high frequencies, can be disregarded. Therefore, two poles, P_1 and P_2, which are located at V_{OUT} and v_g, respectively, are considered for stability analysis. Moreover, ESR zero Z_1, which is contributed by R_{SER} and C_{OUT} at V_{OUT}, can have a phase boost of 90°. If dominant-pole compensation is used, Figure 2.57 shows the frequency response when V_{IN}, V_{OUT}, and I_{Load} are 3 V, 2.8 V, and 100 mA, respectively. P_1 is located at approximately 1 kHz and zero Z_1 is designed at approximately 4 kHz to cancel the effect of P_2. Compared with those in Figure 2.57, the frequency responses in Figures 2.58 and 2.59 show the different locations of zero Z_1 and provide different performances. In Figure 2.58, the UGF and PM increase and decrease, respectively, because a large R_{ESR} (50 Ω) causes Z_1 to be located at lower frequencies compared with P_2. In Figure 2.59, the UGF and PM decrease and increase, respectively, because a smaller R_{ESR}

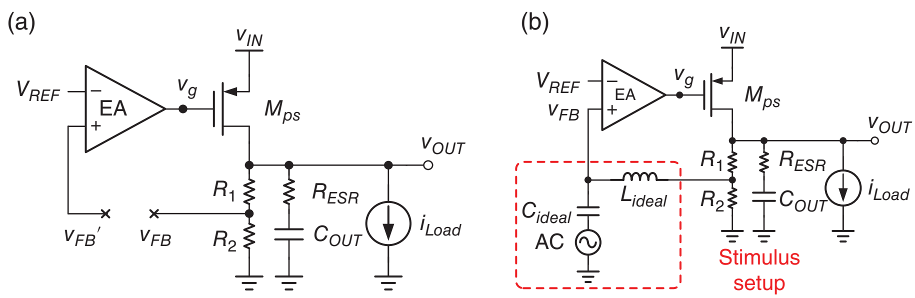

Figure 2.56 (a) Conceptual approach to break the loop at the high impedance node. (b) Setup of the stimulus

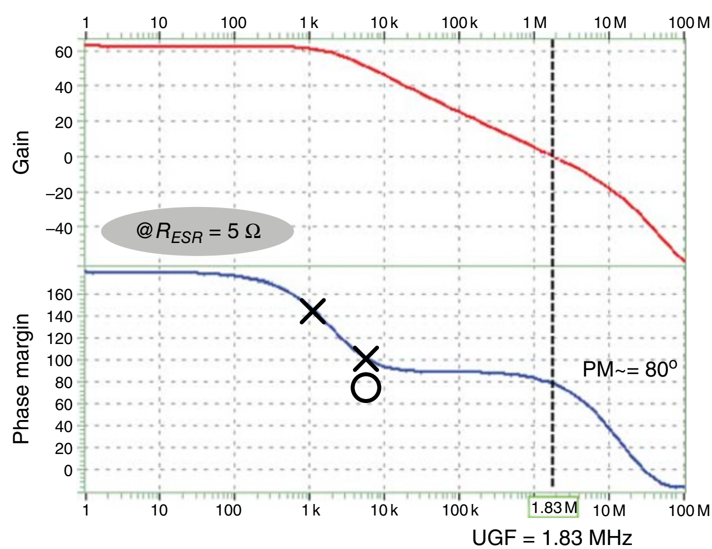

Figure 2.57 Frequency response with R_{ESR} = 5 Ω and I_{Load} = 100 mA

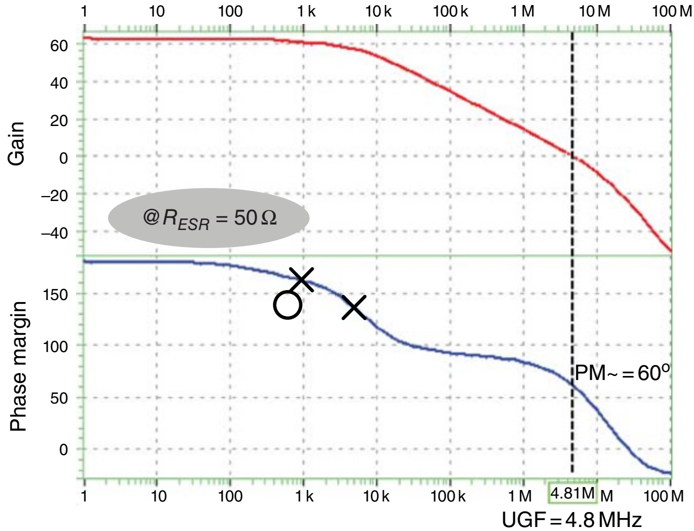

Figure 2.58 Frequency response with R_{ESR} = 50 Ω and I_{Load} = 100 mA

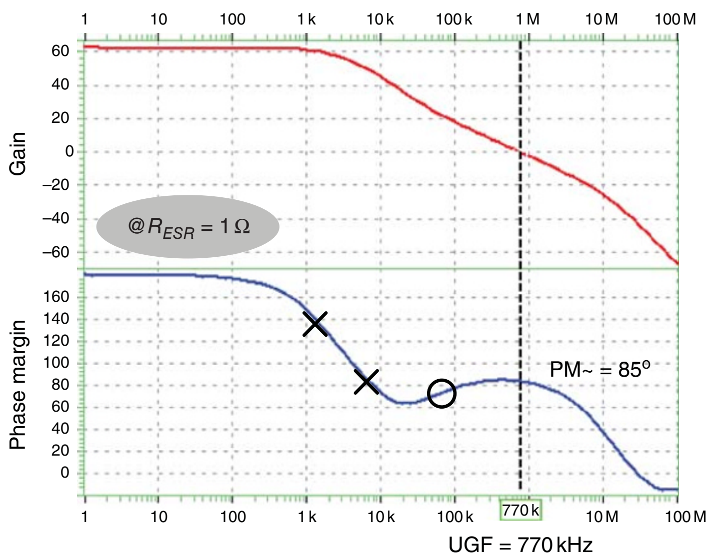

Figure 2.59 Frequency response with R_{ESR} = 1 Ω and I_{Load} = 100 mA

(1 Ω) causes zero Z_1 to be located at higher frequencies compared with P_2. Consequently, R_{ESR} = 1 Ω can provide adequate frequency compensation.

In Figure 2.60 the frequency response is verified under different loading conditions, including 1 and 100 mA. A decreasing loading condition leads to higher gain ($\text{Gain}_{(1\ \text{mA})} > \text{Gain}_{(100\ \text{mA})}$), lower frequency of P_1 ($P_{1(1\ \text{mA})} > P_{1(100\ \text{mA})}$), and smaller UGF ($\text{UGF}_{(1\ \text{mA})} < \text{UGF}_{(100\ \text{mA})}$). Moreover, when checking the simulation conditions, a complete verification should include

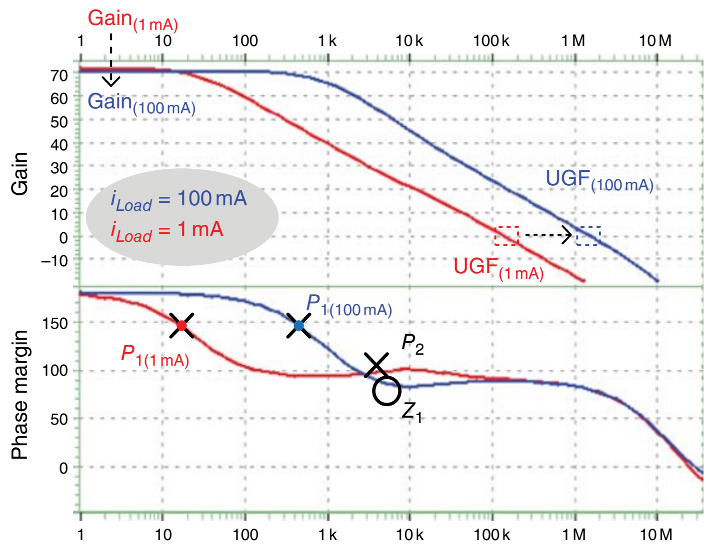

Figure 2.60 Comparison of frequency responses under I_{Load} = 1 mA and I_{Load} = 100 mA

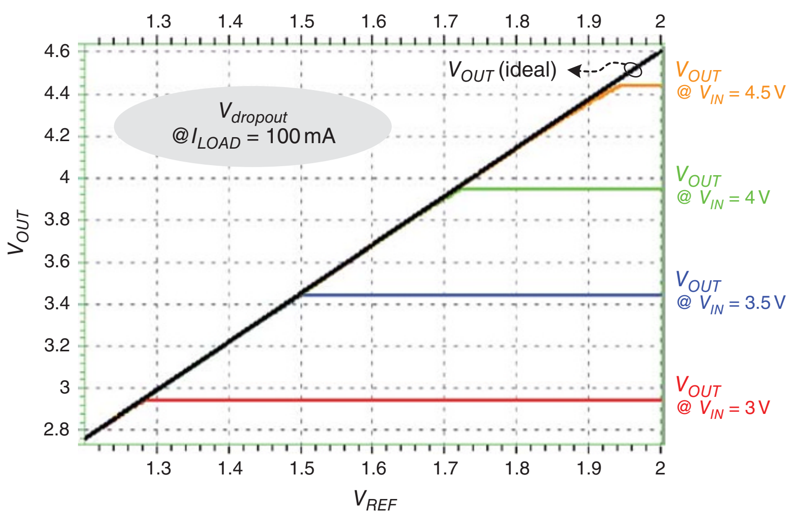

Figure 2.61 Dropout voltage with different V_{IN} values ranging from 3 to 4.5 V

all combinations of different V_{IN}, V_{OUT}, I_{Load}, temperature, and so on, to ensure system stability even under the worst operating conditions (Figure 2.60).

DC analysis can verify the dropout voltage, load regulation, and line regulation. First, the dropout voltage $V_{dropout}$ is observed if V_{IN} and V_{OUT} increase continuously, as shown in Figure 2.61. If I_{Load} is set to its maximum value of 100 mA, V_{OUT} is observed in case of increasing V_{IN} and V_{REF}. The ideal output value $V_{OUT(ideal)}$ can track the increasing trend of V_{REF}

because of the negative feedback control. When V_{IN} is set to 3 V, increasing V_{REF} causes V_{OUT} to also increase. As shown in Figure 2.62(a), where V_{IN} is set to 3 V, the loop gain becomes weak, and thus the dropout voltage can be determined as approximately 140 mV when the curve of V_{OUT} deviates from $V_{OUT(ideal)}$. Similarly, Figure 2.62(b)–(d) shows $V_{dropout}$ when V_{IN} is 3.5, 4, and 4.5 V, respectively.

Figures 2.63 and 2.64 show the load regulation and the line regulation, respectively. In Figure 2.63, the load regulation is verified when V_{IN} is 3 V, V_{OUT} is 2.7 V, and I_{Load} ranges from 1 to 100 mA. The variation at V_{OUT} is 2 mV, which indicates that the performance of load regulation is approximately 20.2 mV/A. In Figure 2.64, I_{Load} is set to its maximum condition of

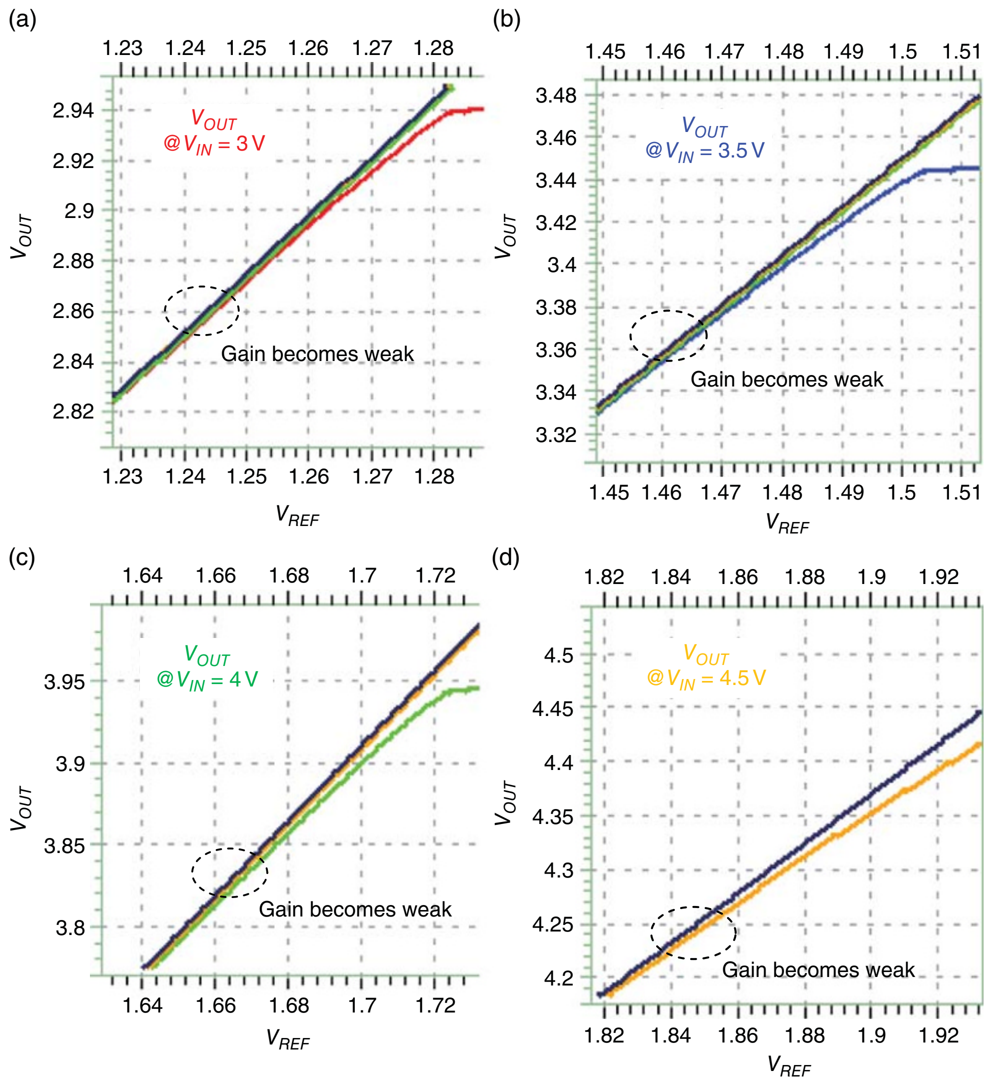

Figure 2.62 Dropout voltage with (a) V_{IN} = 3 V, (b) V_{IN} = 3.5 V, (c) V_{IN} = 4 V, and (d) V_{IN} = 4.5 V

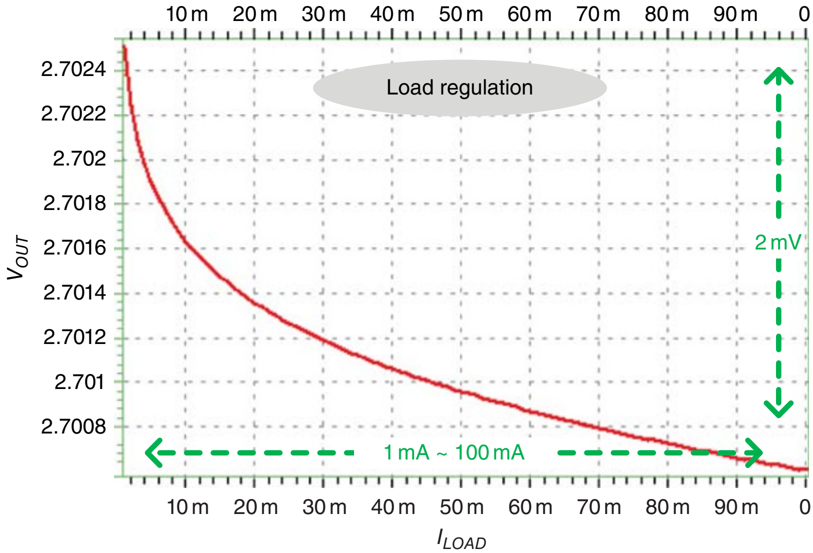

Figure 2.63 Load regulation when I_{Load} ranges from 1 to 100 mA

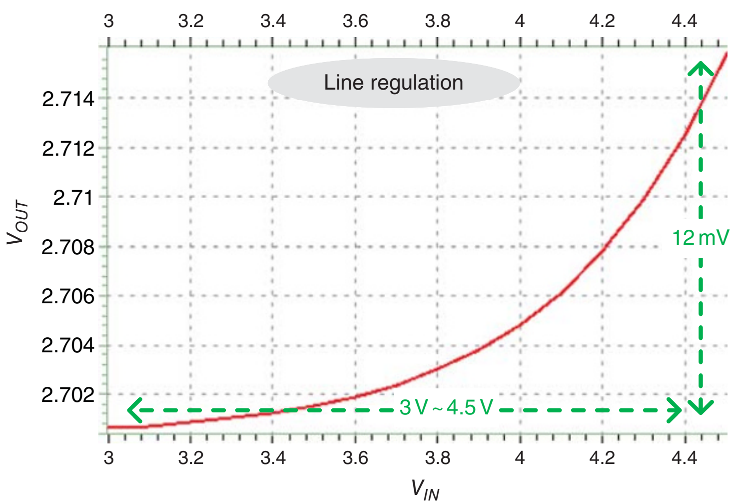

Figure 2.64 Line regulation when V_{IN} ranges from 3 to 4.5 V

100 mA, because this condition will lead to the worst loop-gain case. Line regulation is verified when V_{IN} ranges from 3 to 4.5 V, V_{OUT} is 2.7 V, and I_{Load} is 100 mA. The variation at V_{OUT} is 12 mV, which indicates that the performance of load regulation is approximately 8 mV/V.

Figure 2.65 shows the simulation results of transient response. In the beginning, V_{IN} is set to increase from 0 to 3 V within 50 μs, which emulates the power-on situation in real cases. After the power-on period of 50 μs, the system has not yet settled down because V_g and I_{MPS} still have varying values, although V_{OUT} is regulated at 2.7 V. V_{OUT} consists of the cross voltages of C_{OUT} and R_{ESR}. This variable indicates that C_{OUT} has not yet been charged to the desired voltage of

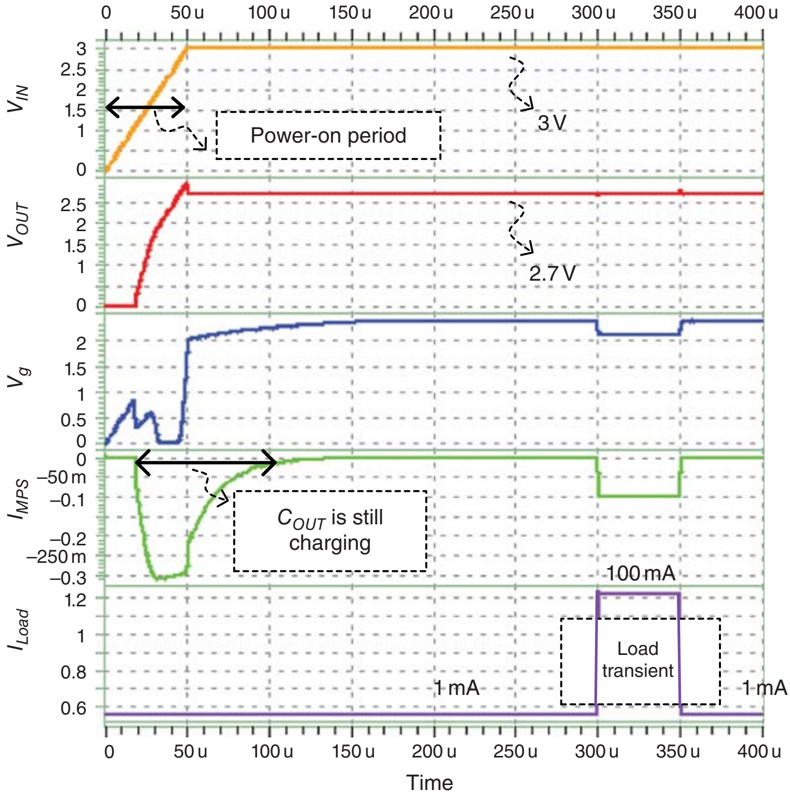

Figure 2.65 Transient response

2.7 V. To analyze the load transient response, designers should avoid setting any loading change during this unstable period.

During the transient response period, the loading current changes from 1 to 100 mA and from 100 to 1 mA at 300 and 350 μs, respectively. The slew rate of the loading current is 1 μs. Figure 2.66 provides detailed zoomed-in images of waveforms that occur during the load transient period. When the loading current changes from 1 to 100 mA, the EA controls V_g to adjust the corresponding driving capability of the power P-MOSFET. In particular, V_g experiences a sudden voltage change from 2.34 to 2 V, which incurs a large signal variation and indicates that the large-signal characteristics of the EA should be carefully designed. The slew rate of the EA always affects the transient response, even if the PM is adequate. Figures 2.67 and 2.68 show the voltage undershoot, voltage overshoot, and settling time. Moreover, the performance of load regulation can be verified from V_{OUT} in a steady state under different loading current conditions, as shown in Figure 2.69.

If the load transient performance cannot satisfy the specifications, designers can modify the design or the compensation to recheck the simulation results. Notably, regardless of whether AC analysis or transient response analysis is checked first, the other should always be checked after. The cross reference between AC analysis and transient response analysis should be consistent.

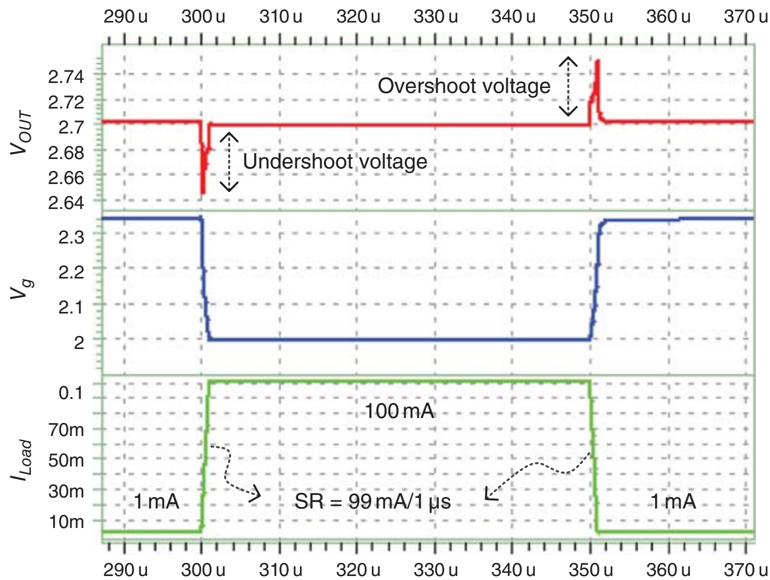

Figure 2.66 Transient response when I_{Load} changes from 1 to 100 mA, and vice versa

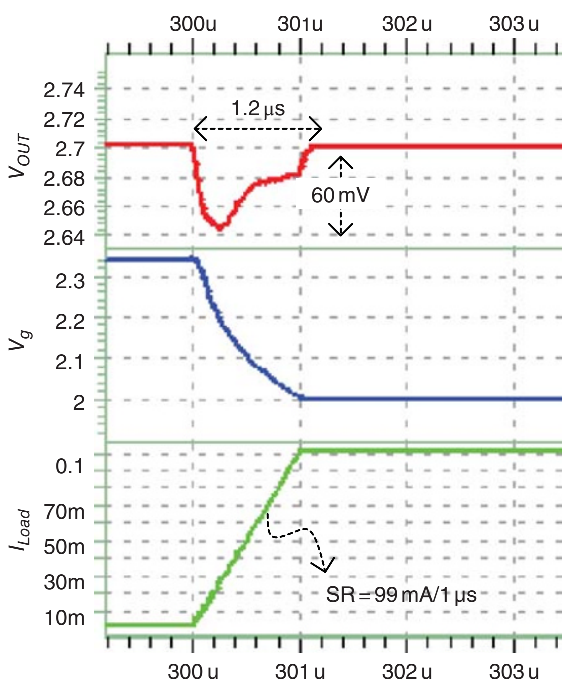

Figure 2.67 Load transient from light to heavy loads

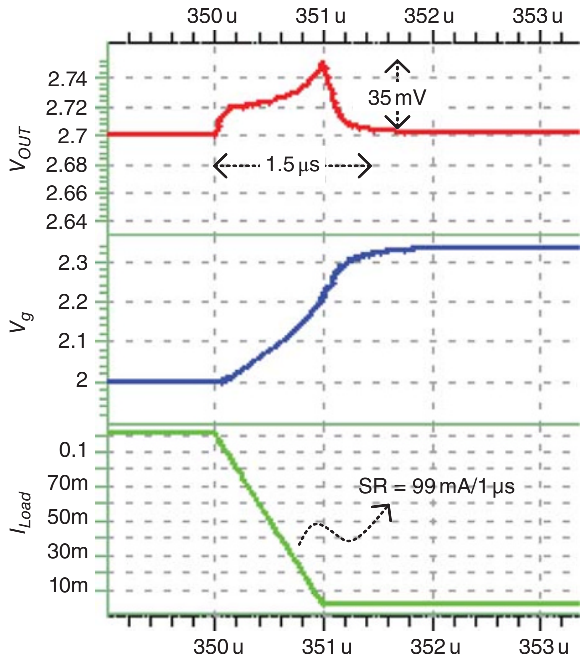

Figure 2.68 Load transient from heavy to light loads

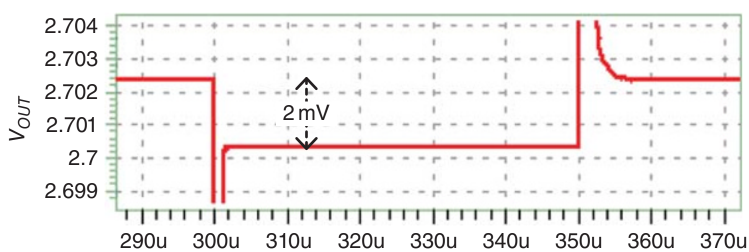

Figure 2.69 Load regulation when I_{Load} changes from 1 to 100 mA, and vice versa

2.6 Digital-LDO (D-LDO) Design

To obtain a high PCE, several approaches have been developed to maintain stability and consume a low quiescent current, even under no-load conditions. However, a low quiescent current induces difficulty in performance enhancement for most LDO regulators. Moreover, for adequate PM and transient response, a structure becomes more complex and the design cost increases correspondingly. Thus, D-LDO regulators have been proposed; the controllers of these regulators are implemented by digital circuits, to turn on/off the power MOSFET array fully [8–10]. Considering their digital implementation, D-LDO regulators can feature fast transit response, a simplified compensation network, and they can work under ultra-low supply voltages. We first introduce the concept of D-LDO regulators and then provide an advanced controller to improve the performance further.

2.6.1 Basic D-LDO

Figure 2.70 illustrates the basic structure of a D-LDO regulator, which consists of a power MOSFET array, a comparator, and a digital controller. In the design of the power MOSFET array, a large power MOSFET is divided into several sub-MOSFET units (SMUs). Each unit is driven by an n-bit digital control signal from the digital controller. Instead of a complex and noise-sensitive analog amplifier, a comparator is used to monitor the output voltage V_{OUT} and compare it with the reference voltage V_{REF}. That is, the logic high or low of the comparator output signal is fed into the digital controller to determine the number of turning-on SMUs. Different numbers of turning-on SMUs can be regarded as several on-resistances that are parallel to an equivalent on-resistance of the power MOSFET. Consequently, different driving capabilities that correspond to various loading current conditions can be adjusted. For example, under heavy loading current conditions, many SMUs are turned on to reduce the equivalent on-resistance and obtain high driving capability.

In general, a simple digital controller can be realized by an up/down counter or a shift register that is controlled by a one-bit compact or output. Moreover, an additional clock signal can be used to synchronize all control signals. The size of the SMUs in the MOSFET array can be designed in binary-code form. As shown in Figure 2.71 [8], the digital controller can be realized by a shift register, wherein the SMUs in the MOSFET array are designed with uniform size.

Compared with the power MOSFET that is controlled by an analog amplifier, a digital control method fully turns on/off each SMU in the power MOSFET array. From a digital perspective, fully turning a system on and off indicates high noise immunity. In a steady state, the number of turning-on power MOSFETs indicates the loading current condition. The shift register will have a dynamic equilibrium under a bistable condition that oscillates between two coding words. That is, the number of turning-on SMUs increases and decreases back and forth if equilibrium is established. However, the back-and-forth operation incurs voltage ripples, as shown in Figure 2.72; these ripples are similar to the switching voltage ripples in SWRs. Unlike

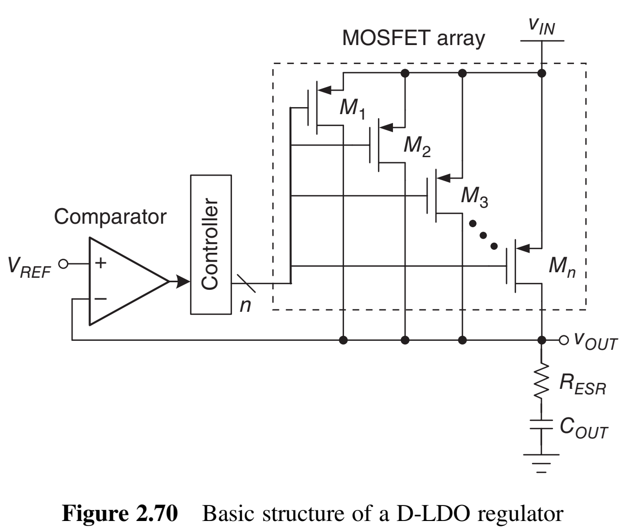

Figure 2.70 Basic structure of a D-LDO regulator

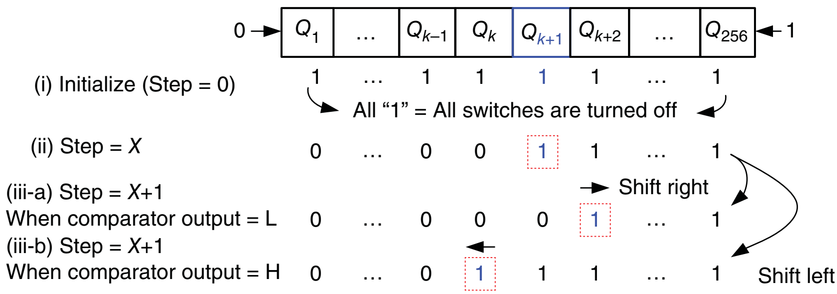

Figure 2.71 Shift register used to control on/off switching of the power MOSFET array

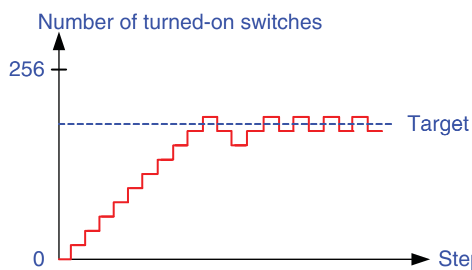

Figure 2.72 Up/down counting control method in D-LDO regulators causes undesired voltage ripples

A-LDO regulators, the undesired voltage ripples will degrade the advantages of the ripple-free characteristic of LDO regulators. The resolution of a switching MOSFET array determines how large a voltage ripple is. A high resolution results in a low output ripple. Furthermore, the transient response time is determined by the frequency of a clock signal and the resolution of the switching MOSFET array. A high frequency results in a fast response and a short recovery time. Nevertheless, the operating frequency cannot be increased without limitation because a high-frequency operation will result in insubstantial power consumption, thus worsening the PCE. That is, the bandwidth of D-LDO regulators is determined by the operating frequency and the resolution of the power MOSFET array when the digital control method is used.

A fast transient response can be achieved more easily in the digital control method than in the analog control method if the operating frequency is simply increased without considering the PCE. Moreover, the digital control method has another advantage, that is, only the pole at the output should be considered because full-swing signals can push poles and zeros to high frequencies. The system can be simplified into a single-pole system, alleviating the constraint of the loading current range, particularly for the minimum light load requirement in the *C-free* LDO regulator.

Given the advantages of fast transient and wide load range operation, D-LDO regulators are more suitable for DVS-based applications. Moreover, a wide operating voltage ranges from the device threshold voltage V_{th} to nearly the highest supply voltage (even if only the minimum biasing current is used to ensure voltage regulation). When considering an advanced nanometer process, D-LDO regulators are also among the candidates for point-of-load (POL) power converters. The following subsections introduce several advanced control methods for D-LDO regulators. For example, the lattice asynchronous self-timed control (LASC) technique is introduced to effectively reduce the power consumption required for synchronous clock control and output voltage caused by a bi-stable operation. Expanding the concept of the DVS technique from the task layer to the instruction layer, the instruction cycle-based dynamic voltage scaling (iDVS) technique is introduced to achieve the advantage of effective power saving. Considerable power can be saved at both the architectural and the circuit levels if the iDVS technique is used.

2.6.2 D-LDO with Lattice Asynchronous Self-Timed Control

A basic D-LDO regulator synchronously adjusts the driving capability with a predefined reference clock. Although a fast transient response and low-voltage operation can be obtained with D-LDO regulators compared with A-LDO regulators, the power consumption remains constrained by the synchronous clock. To improve the performance, a LASC control for *C-free* D-LDO regulators, as illustrated in Figure 2.73, has been developed without any clock signal because of its asynchronous control. The operation of the LASC is similar to that of a clock-free bidirectional shift register that determines the activity of each SMU. Without a synchronous clock, asynchronous control realizes the hand-shaking operation between adjacent SCUs. Therefore, the trade-off between the PCE and the transient response speed can be compromised. That is, in case of a load change, the equivalent clock

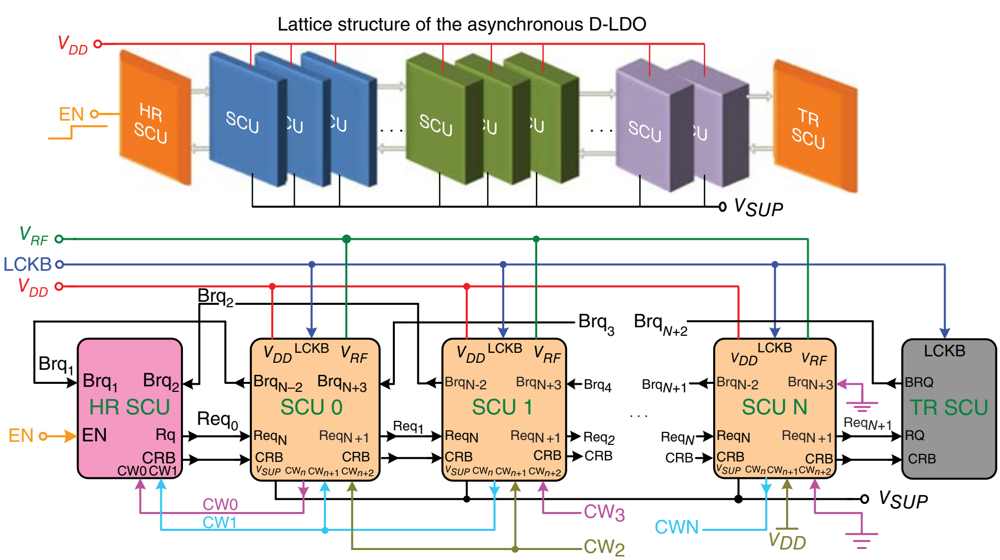

Figure 2.73 Implementation of an asynchronous D-LDO regulator with the LASC technique [11]

frequency is effectively increased to achieve a fast transient response. By contrast, the equivalent clock frequency is automatically decreased for energy/power saving in standby mode. This condition indicates that the D-LDO regulator with the asynchronous control technique can simultaneously exhibit the advantages of high PCE and fast transient response.

A driven source is an event, and thus the problems of clock skew and synchronous surging current never occur. Figure 2.74 shows that the LASC D-LDO regulator consists of an SCU, an SR-latch comparator, a heading reflector (HR), and a terminal reflector (TR). As shown in Figure 2.74(a), the SCU contains a Muller C-gate, an SR-latch comparator, a power switch, a path multiplexer, and control logics to modulate power switches to ensure the regulated output voltage V_{SUP}. In Figure 2.74(b), the SR-latch comparator is triggered by an active high enable signal EN, which is controlled by the forward request pulse of the previous stage. The dynamic comparator compares V_{SUP} with V_{RF} to generate signals CW_0–CW_N to turn on/off the corresponding power switch.

The path multiplexer determines the forward request signal Req_{n+1}, either from the prior stage or the later stage backward request signal Brq_{n+3} according to the results CW_n, CW_{n+1}, and CW_{n+2}. The table in Figure 2.74(a) shows the overall operating principle of the SCU-based Muller C-gate self-timed control. As shown in Figure 2.74(c), HR ensures that all SCUs in the LASC D-LDO regulator will return to their initial states. HR also guarantees that power switches are turned off by setting the signal current-ripple based (CRB) to low when the EN signal is forced to be low. The Muller C-gate shown in Figure 2.74(d) is a basic component of asynchronous circuits. The behavior of an n-input Muller C-gate changes the output state to 3high if all inputs are high and to low if all inputs are low. Otherwise, the n-input Muller C-gate maintains the same output as that in the previous state. As shown in Figure 2.74(e), the rising edge detector (RED) circuit generates a single pulse to trigger the HR circuit to pump the first request pulse, and thus activate the LASC D-LDO regulator when EN changes from low to high. To address the boundary condition, the TR circuit, as depicted in Figure 2.74(f), helps the forward request signal reflected from the termination when V_{SUP} cannot acquire a sufficient power supply during the final SCU stage. Furthermore, HR prevents the backward request signal from disappearing when V_{SUP} derives an overcharge load during the first SCU stage.

Figure 2.75(a), (b) shows the timing diagram of a single SCU stage operation under different conditions that correspond to the circuit in Figure 2.74(a). When V_{SUP} is smaller than the reference voltage V_{RF} in an SCU stage that is triggered by the signal Req_n from the prior stage, the level-active SR-latch comparator outputs low-signal CW_n to turn on the power switch. Thus, the voltage for V_{SUP} can be increased to track V_{RF}. The forward request signal Req_{n+1} is generated by the self-time control mechanism after a deterministic delay, that is "delay X + delay Y," when the next SCU stage performs a shift-right operation, which activates additional power switches to regulate V_{SUP}. If V_{SUP} is greater than V_{RF}, then the control signal CW_n will be pulled high to turn off the power switch at this stage. The backward request signal Brq_{n-2} will be triggered by the self-time mechanism after a deterministic delay, that is "delay X + delay Y," when the prior SCU stage performs a shift-left operation to reduce the driving capability of V_{SUP}. Figure 2.75(c) shows the operation timing diagram of the LASC D-LDO regulator. First, EN is pulled low and then, the LCKB signal is forced to high during the power-on reset state. All SCU stages are initialized to turn off all power switches. Once the power-on sequence of the

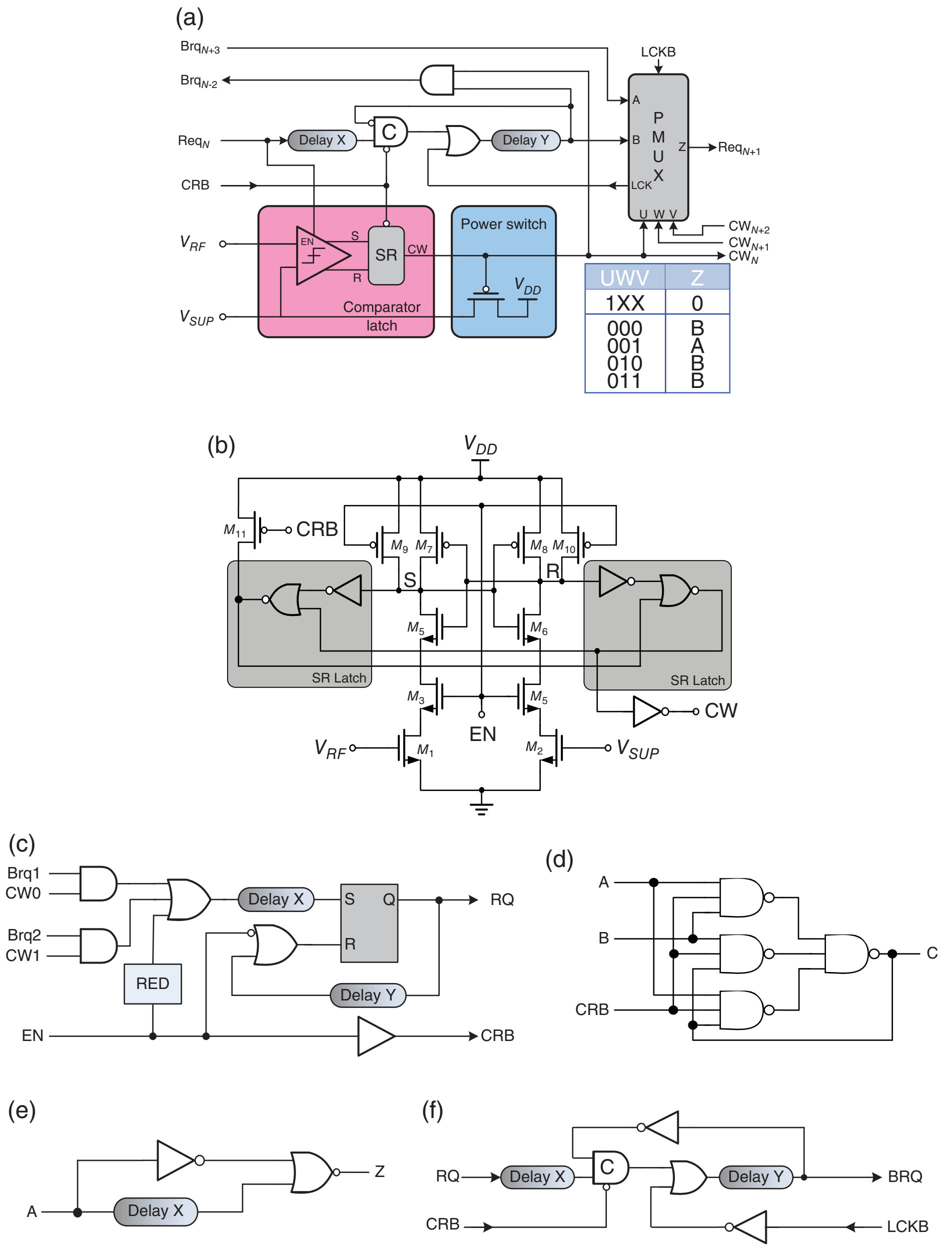

Figure 2.74 Implementations of (a) SCU, (b) SR-latch comparator, (c) HR, (d) Muller C-gate, (e) rising edge detector, and (f) TR

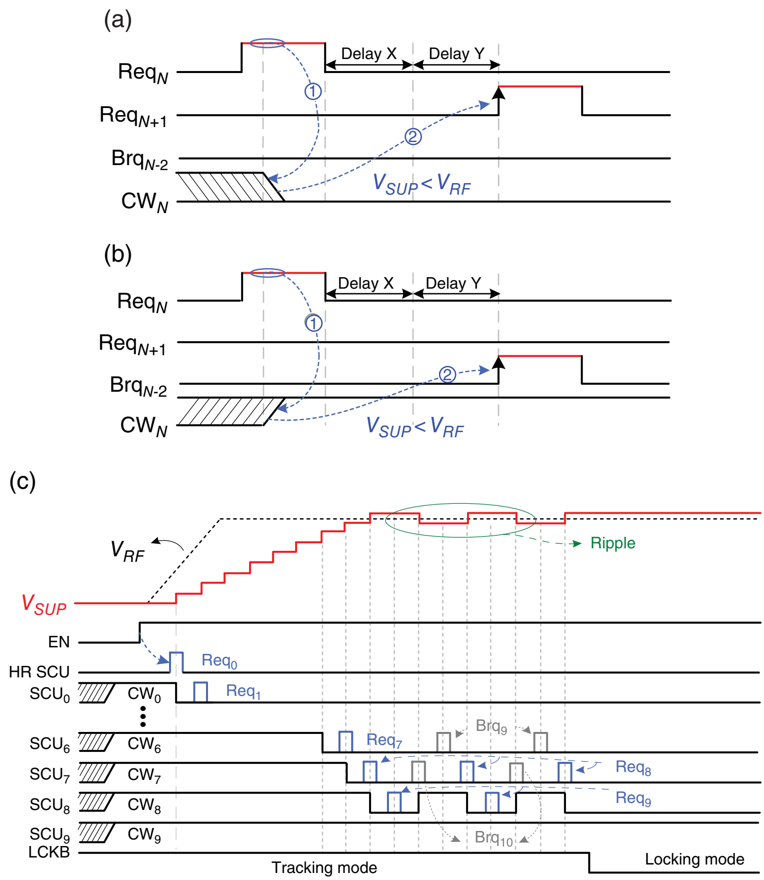

Figure 2.75 Timing diagrams. (a) Single SCU operation when V_{SUP} is smaller than V_{RF}. (b) Single SCU operation when V_{SUP} is larger than V_{RF}. (c) LASC operation when EN is activated

processor is complete and the EN signal is forced to high, the HR SCU pumps the first request signal Req_0 into the LASC controller, and thus the asynchronous D-LDO regulator output voltage V_{SUP} can start tracking the reference voltage V_{RF} according to the required power. During the up-tracking period, the LASC functions as the shift-right operation to increase the number of turning-on power switches by shifting the control signals from CW_0 to CW_N. When V_{SUP} reaches its target value of V_{RF}, backward request signals are generated to stop the delivery of supplementary power to V_{SUP}. When the LASC operation converges to the adjacent SCU stages, or when the present supply voltage is adequate for correct operation, the signal LCKB is cleared to change the operating state from the tracking mode and return to the locking mode. The operation of the LASC D-LDO regulator ends with the indication of the signal LCKB. Thus, output voltage ripples are eliminated in the LASC D-LDO regulator because all SCUs are in a steady state. Therefore, all devices are also in a steady state and the current consumption closely approaches the 0.18 μm process core device leakage current, which is approximately

80 nA because of the fully digital D-LDO regulator. The LASC D-LDO regulator achieves a fast response and an ultra-low static current consumption simultaneously.

2.6.3 Dynamic Voltage Scaling (DVS)

2.6.3.1 Introduction to Basic DVS

Portable electronics are essential products in our daily lives, covering a wide range of applications, including devices and gadgets used for entertainment, communication, and biomedicine. These electronic products contain processors, such as digital signal processors (DSPs), advanced reduced instruction set computing machines (ARMs), and microcontroller units (MCUs), as core components. Therefore, designing low-power processors to save as much power as possible, and extend the battery life of portable devices, is critical.

Figure 2.76 shows the hierarchical processor architecture. The figure also demonstrates that programs are executed from the high-level operating system (OS) layer to the lowest transistor component layer. The program stored in the memory is accessed by the OS to dispatch and schedule different priority tasks. The basic unit of a task is the individual instruction.

After the processor decodes the instructions, the logic gate circuits are activated to perform specific computations. The corresponding layer then accepts the logical control signals to enable or disable millions of transistors. Hence, various techniques have been presented to reduce the power consumption of processors according to the hierarchical processor architecture. In Figure 2.76, state-of-the-art multiple-threshold voltage and body-bias adjustment techniques are employed to reduce the power consumption down to the lowest level of the architecture, that is, the transistor component layer [12]. For simplicity, the clustered-voltage-scaling (CVS) technique at the logic gate layer is adopted to reduce power, because

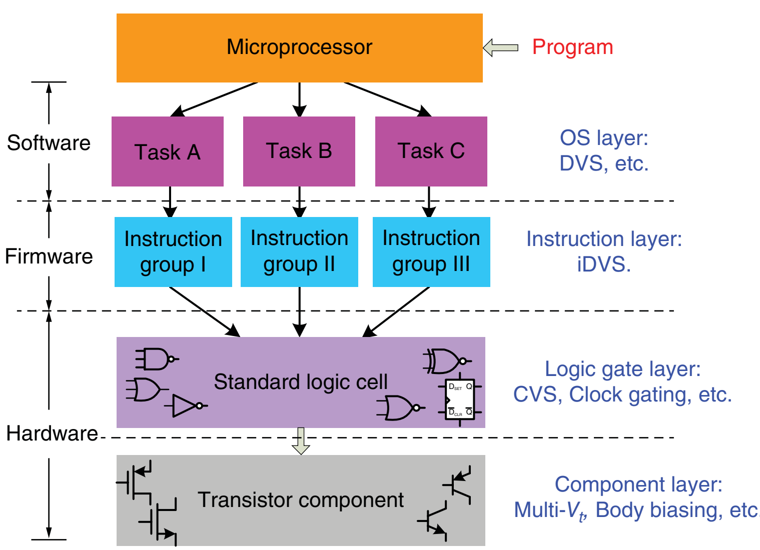

Figure 2.76 Low-power management strategy for processors

of the distinct CVS of the characteristics of each function block [13]. These techniques exhibit limited power-reduction capabilities and require specific processes supported by the foundries, or careful layout placement of a logical cell with multi-power grids. By contrast, the DVS technique in the OS layer is an effective means to reduce the power consumption, because dynamic power consumption depends on a quadratic function of the supply voltage V and the clock frequency f, as shown in Eq. (2.79), where C is the equivalent dynamic operating capacitance:

$$P \propto CV^2 f \tag{2.79}$$

The DVS technique is appropriate for low-power DSP designs fabricated using the standard CMOS process [14–16]. The conventional DVS task-based control circuit presented in [17] and illustrated in Figure 2.77 uses a closed loop to ensure that the clock frequency ($f_{DESIRED}$) satisfies the desired processor operating clock frequency, which is assigned by the OS and stored in the frequency register for a specific task execution. If peak performance is unnecessary, then the operating clock frequency can be decreased to save considerable power. To evaluate the operating clock frequency, the ring oscillator is provided with a real-time supply voltage V_{SUP} that is generated by an inductor-based SWR to determine the digital numerical clock frequency (f_{CLOCK}). Moreover, the supply-determined f_{CLOCK} is compared with $f_{DESIRED}$ to identify the digital frequency error signal (f_{ERROR}) and produce the control signal by using a digital filter. Finally, the drivers located after the digital loop filter turn the power MOSFETs on/off to modify the output voltage V_{SUP}. Given the closed loop, V_{SUP} can be adjusted to be sufficiently high to guarantee that f_{CLOCK} is close to $f_{DESIRED}$.

Therefore, the dynamic frequency scaling (DFS) technique is implemented by the rapid processor clock frequency according to the minimal and dynamic scaling supply voltage. If high efficiency is considered, then the suitable power supply regulator is an inductor-based converter with a low transient response. Consequently, the DVS tracking speed is restricted from a few microseconds to a few milliseconds. The delay caused by voltage tracking decreases efficiency and processor performance. Various fast voltage tracking methods for high-performance DVS response have been reported to satisfy the DVS requirement [18, 19].

The conventional task-based DVS technique allows all tasks in a scheduler to complete just-in-time operations. The OS depends on run-time workload to adjust the supply voltage dynamically, which leads to a substantial power saving [20, 21]. However, the supply voltage V_{DD} of

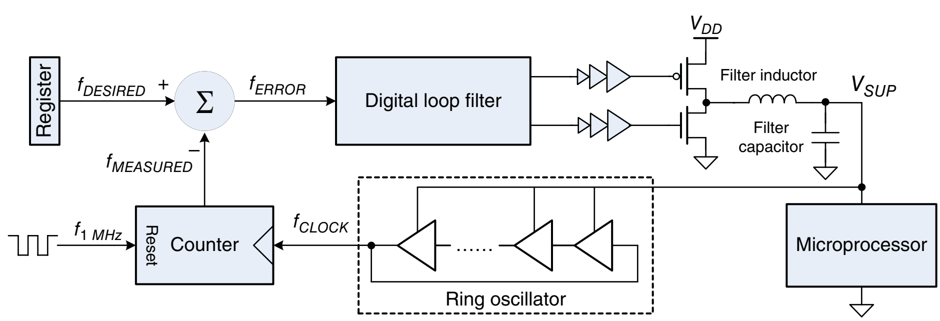

Figure 2.77 Conventional task-based DVS control circuit

the conventional task-based DVS is limited by the highest power instruction in a task operation, as illustrated in Figure 2.78(a). That is, V_{DD} is kept high without being scaled down to save power. Thus, the conventional task-based DVS with a conservative scheduler will fail to scale down V_{DD} because the processor has no slack time. Alternatively, the conventional task-based DVS technique can change the processor operating clock frequency to facilitate a voltage scaling operation; however, this process deteriorates its performance.

A rapidly changing processor clock frequency induces several problems when controlling peripheral modules. These problems include control of signal timing errors and a missing communication data latch in peripheral devices such as synchronous dynamic random access memory (SDRAM), inter-integrated circuit (I^2C), analog-to-digital converter (ADC), digital-to-analog converter (DAC), universal asynchronous receiver/transmitter (UART), and flash memory peripheral interface. The reason for such a problem is the dependence of peripheral devices in Soc on a constant clock and predictable control signals. An alternative iDVS technique [22] employs the DVS technique on the instruction layer to overcome the aforementioned drawbacks. One task consists of numerous instructions, as shown in Figure 2.78(b). If the DVS technique is applied on the instruction layer, then more slack time slots can be derived in the iDVS technique. This technique can reduce the power consumption more than the conventional task-based DVS technique.

To further understand this, the processor is regarded as a dynamic loading and emulated by an adjustable resistor that is controlled by instructions, as shown in Figure 2.79(a). The iDVS technique, which is based on different instructions, does not change or stall the processor operating clock frequency. Thus, the processor performs with a minimum power supply to save power. Through the task-based DVS technique, power consumption can be reduced if the operating tasks have low power consumption. Low-power DSP designs can have a powerful energy-saving capability if the DVS technique is applied on the instruction layer. Figure 2.79(b) shows an example of DSP with the iDVS technique to save power dissipation, wherein the LASC D-LDO regulator is used as the voltage buffer to improve the transient response. The SWR cannot satisfy the slew rate requirement of the DSP core. Thus, the LASC D-LDO regulator can exhibit fast reference tracking to satisfy the voltage slew rate requirement. Consequently, the DSP performance and efficiency can be maintained simultaneously. Moreover, at the DSP core side, the iDVS controller requires the adaptive instruction-cycle control (AIC) circuit to guarantee fast voltage tracking speed and high operating frequency. Moreover, an additional design flow for the iDVS technique, with the help of an automatic computer-aided design (CAD) tool, is also required to maximize power-saving performance.

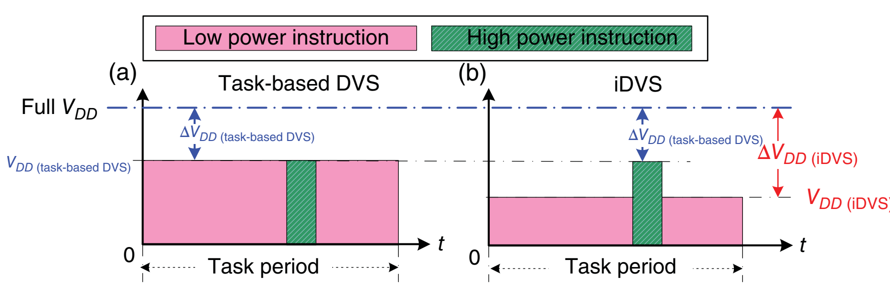

Figure 2.78 (a) Conventional task-based DVS limited by high power instruction. (b) iDVS effectively reduces power consumption

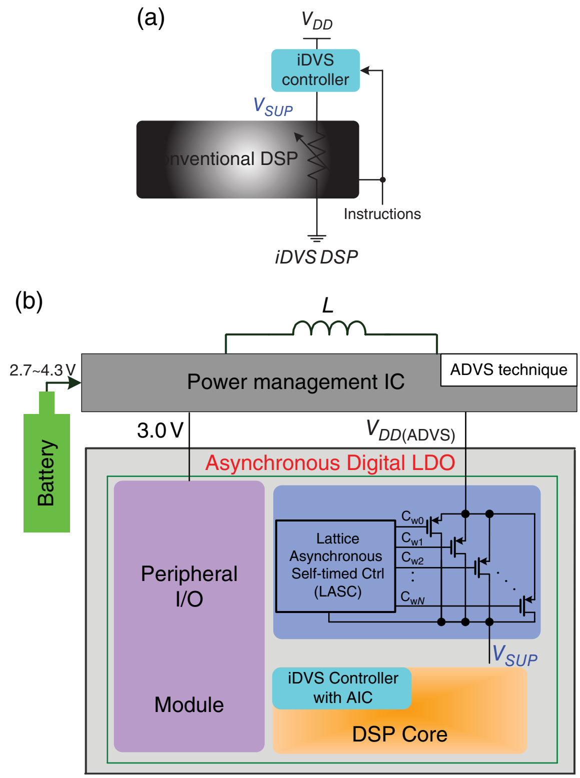

Figure 2.79 (a) Concept of iDVS operation. (b) DSP core with implementation of the iDVS technique

2.6.3.2 Instruction-Cycle-Based Dynamic Voltage Scaling (iDVS)

Processors are designed to provide versatile application programs. When listening to a Moving Picture Experts Group audio layer III (MP3) while simultaneously browsing a picture from a flash memory device, the OS dispatches file-system access and the MP3/Joint Photographic Experts Group (JPEG) decoding algorithm. Similarly, when taking a phone call, the OS of an embedded system launches the key-scan service and the speech compressing/decompressing code-excited linear prediction (CELP) algorithm once mobile phone buttons are pressed. Although these task programs have different characteristics, their fundamental unit is the instruction unit. The basic steps of program execution in a processor are instruction fetching, decoding, execution, and storing, as illustrated in Figure 2.80(a). The most complicated part is the execution unit, shown in Figure 2.80(b), which can provide all types of hardware circuit to support different complex instructions. Each instruction has a corresponding critical data path to complete execution. Critical paths occupy only a small fraction of the total number of paths within a chip. However, the clock speed of a synchronous processor is determined by the worst delay of the critical paths. These critical paths usually map high-power-consuming and long-data-path instructions that are subject to single-instruction multiple-data (SIMD) instructions, such as divide (DIV), normalize (NORM), and multiply-and-accumulate (MAC). Long slack time occurs in non-critical path instructions.

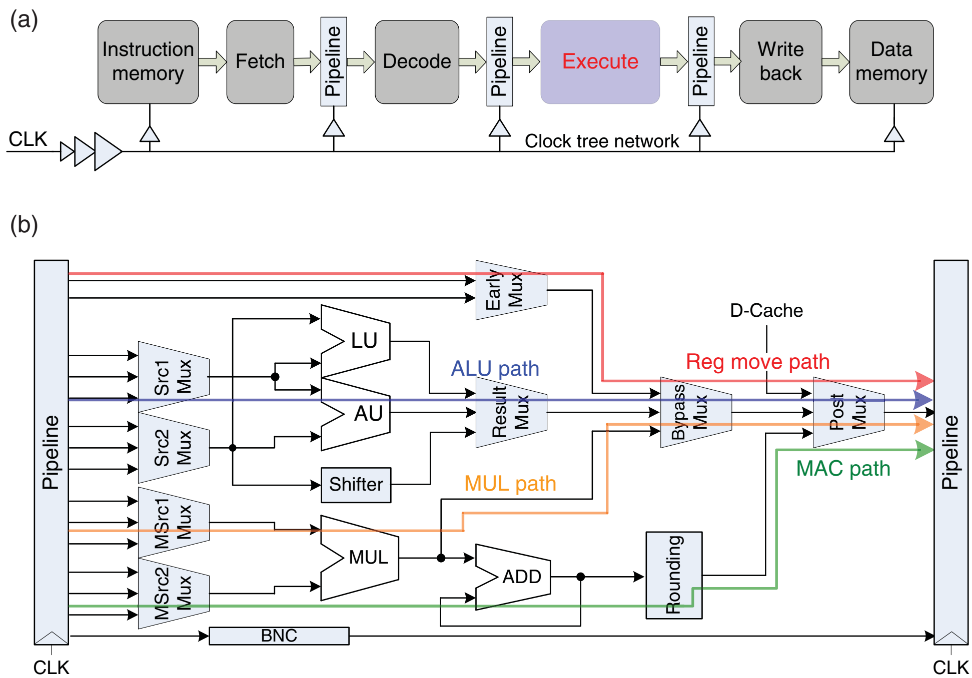

Figure 2.80 (a) Basic steps of program execution. (b) Critical paths during instruction execution

Different instructions present varying slack time, as shown in Figure 2.81(a). The measured slack time and its corresponding power consumption with different instructions are shown in Figure 2.81(b), with the supply voltage fixed at 1.8 V. In summary, a long slack time corresponds to a low power consumption because of the small supply voltage provided by the iDVS technique. Reducing the supply voltage in CMOS circuits affects the propagation delay T_d expressed in Eq. (2.80), which is inversely proportional to V_{DD} and the maximum operating frequency f_{max}:

$$T_d \propto \frac{V_{DD}}{(V_{DD}-V_t)^n} \propto \frac{1}{f_{\max}} \tag{2.80}$$

where V_t is the threshold voltage determined by foundry process parameters.

For simplicity, instructions can be classified into different power groups, namely Groups 1–4, for the iDVS technique to provide the corresponding V_{DD}.

The test chip of a 23-stage ring oscillator in a 0.18 μm CMOS process uses 1.8 V core devices. Figure 2.82 shows the maximum operating frequency f_{max} and the total power consumption with respect to the supply voltage variation. According to the data measured at the supply voltage of 1.8 V, the unit of the y-axis is the normalized operating frequency and the normalized power consumption. The linear relationship between circuit operating frequency and supply voltage V_{DD} ranges from 1.2 to 1.8 V. If V_{DD} is scaled down from 1.8 to 1.4 V, then f_{max} is still larger than half of the maximum operating frequency at 1.8 V. Thus, the scaling range of V_{DD} for

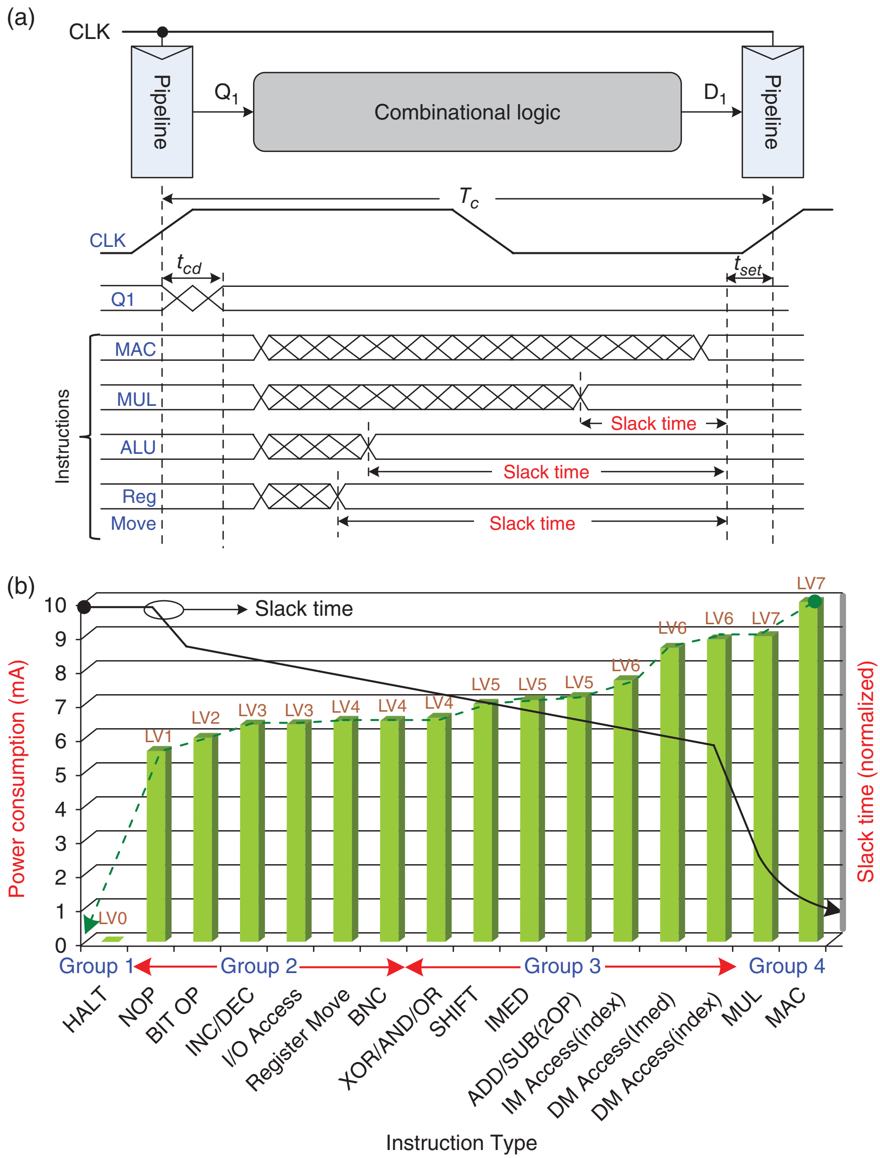

Figure 2.81 (a) Slack time in different instructions. (b) Measured slack time and power consumption in different instructions

normal operation in iDVS is from 1.2 to 1.8 V, excluding HALT and no operation (NOP) instructions. The minimum V_{DD} should be larger than 1 V; otherwise, the level-shift signal will experience serious delay when traveling from the level-shift circuit to peripheral I/O modules. Power reduction can be achieved if the iDVS technique reduces the V_{DD} of the instruction execution on the non-critical path while maintaining a high supply voltage on the critical paths to satisfy complex instruction timing requests. The iDVS dynamically adjusts V_{DD} based on the

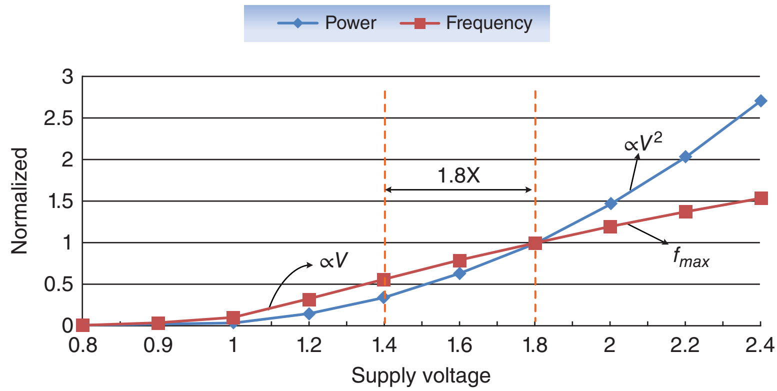

Figure 2.82 Normalized operating frequency and normalized power consumption vs. supply voltage in a 23-stage ring oscillator fabricated via the 0.18 μm CMOS process

instruction-cycle domain to guarantee sufficient power for the appropriate execution of instructions.

In conventional DVS systems, a voltage transition will stall the entire processor operation unless the required power for instructions is available, which results in serious degradation of the million instructions per second (MIPS) performance. By contrast, the iDVS technique uses an embedded all-digital LASC D-LDO regulator with low quiescent current to achieve high-speed voltage tracking capability and provide in-demand power for instruction execution. Moreover, the LASCD-LDO regulator receives assistance from the AIC circuit to avoid the aforementioned drawbacks. The AIC scheme can adaptively adjust the instruction execution cycle time to guarantee that each instruction is correctly executed during voltage tracking for high-performance iDVS operation. That is, the iDVS-based design processor does not change the processor clock frequency or stall the entire processor clock during DVS operation. Consequently, the clock frequency is kept constant, which is suitable for controlling peripheral I/O devices in the Soc.

2.6.3.3 Adaptive Instruction-Cycle Control

An instruction unit occupies one clock cycle in the reduced instruction set computing (RISC) design. A real-time adaptive instruction cycle should be performed in the AIC circuit to adapt to the scaling supply voltage level of the LASC D-LDO regulator.

Figure 2.83(a), (b) shows the topologies of the iDVS controller and the AIC circuit, respectively. The instruction critical path (ICP) shown in Figure 2.83(b) emulates the relative instruction group critical path delay, which is synthesized by the standard cell delay component after timing verification through the iDVS CAD design flow shown in Figure 2.84. Given that a processor has thousands of data paths and millions of logic gates, identifying the corresponding critical data path for each instruction and the relative operating voltage for the iDVS technique is an important issue. Manually analyzing the correlation between critical data paths and instructions is impractical. Circuit extraction tools obtain register-transfer-level (RTL) components and parasitic RC from the ICP to derive the parameters required by the ICP emulator. The extracted circuit netlist is used for simulation program with integrated circuit emphasis

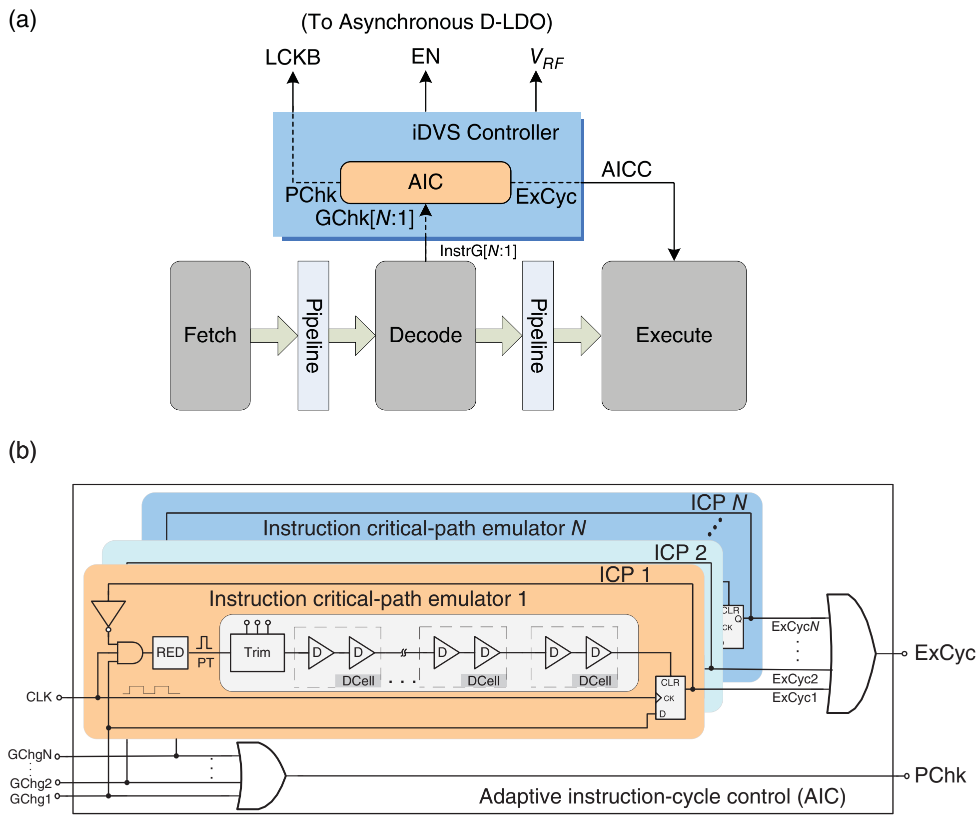

Figure 2.83 (a) Topology of the iDVS controller with the AIC circuit. (b) AIC circuit

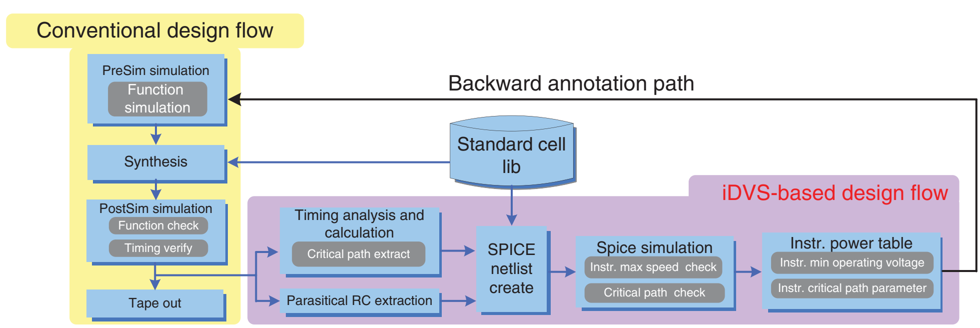

Figure 2.84 Standard cell library design flow with the iDVS design flow

(SPICE) simulations to obtain the minimum operating voltage for each instruction under PVT variations.

The design flow contains three steps, outlined as follows. First, hardware specifications are coded into the hardware description language (HDL) according to the traditional design flow to synthesize the cell-based circuit for post-simulation, which can check the function and verify timing. The critical path of each instruction at the post-stimulation stage can be extracted by the timing analysis tool. A SPICE simulator is used to establish the critical path table for correlating the minimum operating voltage to each corresponding instruction. The timing parameter of each ICP is also extracted to create the ICP in the AIC circuit. The final step is the backward annotation of each instruction power catalog and timing constraint to the iDVS controller in the HDL design. Considering the assistance of RC extraction and the timing analysis tools, the iDVS technique can fit any standard cell library provided by foundries.

Instructions with the same characteristics of data path or power consumption are grouped into one ICP emulator. Each ICP contains a RED, standard-cell delay components, a delay trimming module, and control logics. The delay trimming module is an option for minimizing mass-production deviation after a minor adjustment. Figure 2.85 shows the operating states of the iDVS controller with the timing diagrams, as presented in Figure 2.86. The DSP instruction cycle is synchronized with the edge-triggered clock signal CLK. In each cycle, different instructions as presented in Figure 2.81 are decoded to generate the instruction group signal InstrG[N:1]. When the DSP consecutively executes the instruction stream, the iDVS controller monitors the required power for each instruction according to the instruction power table generated by the CAD design flow.

Once iDVS detects that the required execution power of the next instruction is different from that of the current execution instruction group, the instruction group changes the signals GChg [N:1] to the AIC circuit.

In the next stage, the iDVS controller enters the tracking mode from its operating state to activate the LASC D-LDO regulator by setting the signal LCKB to high. As shown in Figure 2.86(a), given the characteristics of the DSP pipeline structure, the voltage transition command is issued before an instruction is executed prior to the one-clock cycle. Once RED detects the instruction group change signals, which are synchronized with CLK, RED will induce one pulse signal PT to the ICP emulator. The next operation of the AIC circuit is similar to the race condition, to test whether the instruction can finish execution within one instruction cycle at the present supply voltage V_{SUP}. If PT passes through the ICP emulator and simultaneously exceeds the rising edge of the CLK, then the AIC circuit will pull the signal ExCyc to low, which is synchronized by the iDVS controller to generate the signal AICC. Thus, the signal AICC is set to low to inform the DSP execution unit that an extra cycle is not required during the instruction cycle.

By contrast, the passing of PT through the ICP emulator, and the lag of the rising edge of the CLK, indicates that V_{SUP} provides insufficient power. DSP should insert an extra cycle to

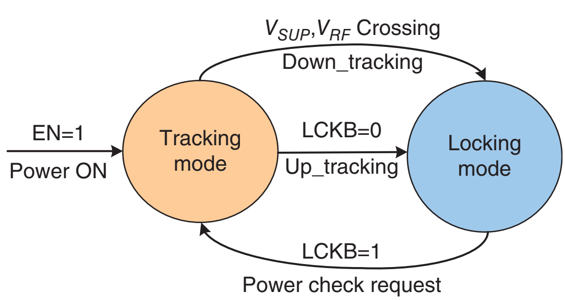

Figure 2.85 Operating states of the iDVS controller

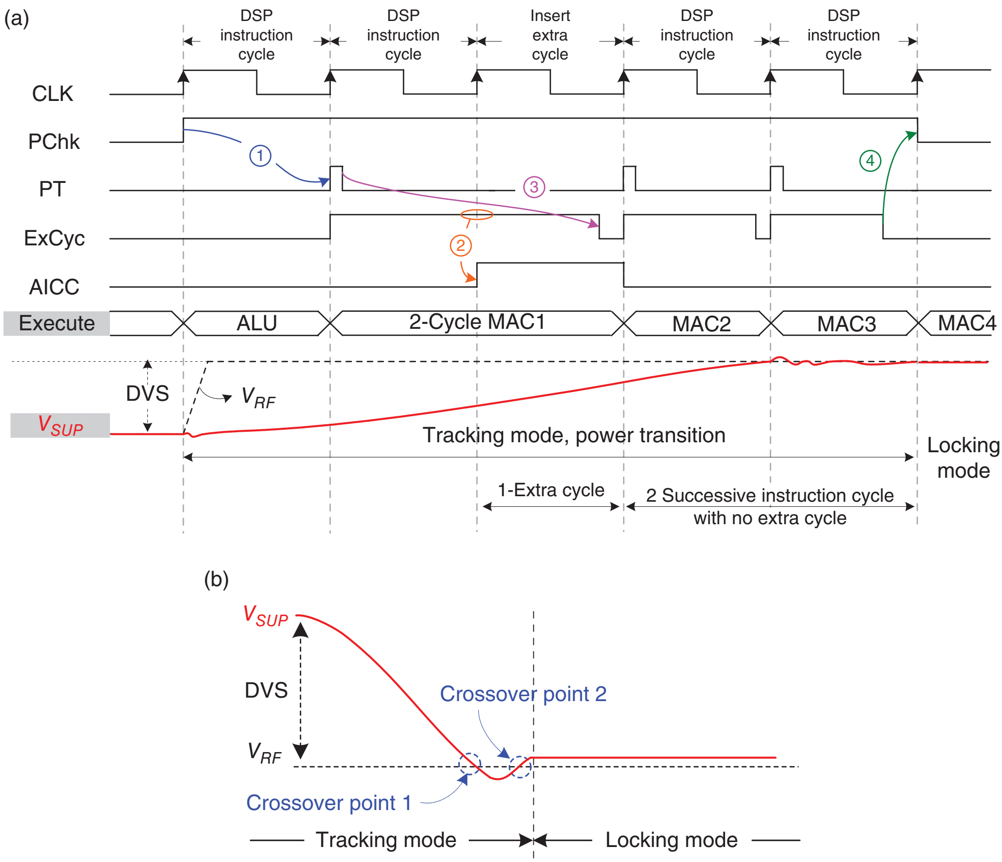

Figure 2.86 Timing diagram of iDVS operation. (a) Up-tracking condition. (b) Down-tracking condition

complete current instruction execution by setting AICC to high. According to Eq. (2.80) and Figure 2.82, no instruction is required to exceed two cycles for execution in the section of 1.4–1.8 V. Given the instruction for pre-decoding the pipeline structure and the fast transition response of the LASC D-LDO regulator, the iDVS-based DSP only requires one extra cycle during up-tracking voltage transition. If the iDVS controller detects a low level of AICC within two successive instruction cycles, then the supply voltage is well regulated and sufficient for instruction execution. The iDVS controller then withdraws the power check request signal PChk and returns to locking mode by setting LCKB to low.

During down-tracking voltage transition, the supply voltage V_{SUP} is sufficiently high to avoid blocking DSP execution. The control sequence of down-tracking voltage transition is shown as follows. First, the D-LDO regulator pulls the V_{SUP} to low. The iDVS controller then sends the group change signals GChg[N:1] to the AIC circuit and continuously monitors the comparison result of V_{SUP} with the reference voltage V_{RF}. Finally, the iDVS controller returns to locking mode by setting the LCKB to low until V_{SUP} and V_{RF} have two crossover points after signal AICC is maintained at low levels within two successive instruction cycles, as shown in

Figure 2.86(b). Simultaneously, the iDVS controller withdraws the power check request signal PChk. The supply voltage is adequate for executing instructions in locking mode. Therefore, the AIC mechanism achieves correct instruction execution during iDVS voltage transition without stopping the operation clock.

2.7 Switchable Digital/Analog-LDO (D/A-LDO) Regulator with Analog DVS Technique

2.7.1 ADVS Technique

In Soc applications, the sub-circuits can mainly be sorted into digital and analog sub-circuits. According to the analog/digital characteristics, different demands are made in supply voltage source qualities, such as driving capability, input/output voltage variation, transient response, noise immunity, and so on.

That is, power management is an essential design issue in Soc, which requires distributive voltage and current levels for distinct sub-circuits. Figure 2.87(a) shows the Soc with an analog sub-circuit and a digital sub-circuit, provided separately by supply voltages V_{OA} and V_{OB} from power management. Moreover, power reduction capability is another necessary feature in power management. DVS is the most efficient technique for power reduction. The DVS function is generally implemented in digital blocks, and thus the digital circuit power consumption can also be optimized by different operating instructions or tasks in Soc. To realize DVS-based power management, the power converter works with the system processor, which can send voltage indication signals to the power converter to adjust the output voltage. As shown in Figure 2.87(b), if Soc enters into data transmission operation, then the increase in V_{OB} satisfies the Soc speed requirement. The decrease in V_{OB} also conserves power when data transmission ends. Comparatively, analog circuits cannot use a similar DVS technique in digital circuits because the supply voltage for analog circuits should be kept constant to maintain some important performance. Moreover, when one SWR is used to provide V_{OA}, the switching voltage ripple at the V_{OA} will degrade accuracy, power supply rejection ratio (PSRR), and so on. By contrast, the LDO regulator can be utilized to suppress noise for high-quality supply voltage, sacrificing efficiency with a large dropout voltage when the LDO regulator directly converts V_{OA} from the input voltage V_{IN}. To reduce the power consumption on the dropout voltage of the LDO regulator, an LDO regulator is generally cascaded in series with a switching-based power converter. Consequently, complete conversion efficiency is expressed as:

$$\eta_{complete} = \eta_{switching} \cdot \eta_{LDO} \tag{2.81}$$

To maintain high efficiency and a high-quality supply voltage for analog sub-circuits, a switching-based power converter is responsible for a wide conversion ratio, whereas the LDO regulator is responsible for switching noise suppression with adequate dropout voltage. To prolong the battery life further using time for portable devices, we need to think whether a solution is available to improve the efficiency further. The dropout voltage of the LDO regulator remains a serious factor to limit complete conversion efficiency, although a switching-based power converter can benefit most complete conversion efficiencies because of the usage of the storage component, inductor, and capacitor. For example, the maximum $\eta_{complete}$ is

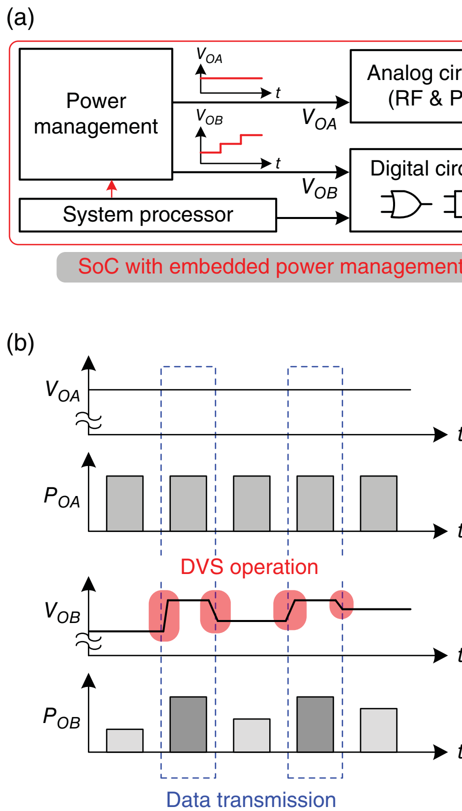

Figure 2.87 (a) Power management in Soc provides two distinct supply voltages, V_{OA} and V_{OB}, for digital and analog circuits, respectively. (b) DVS techniques at V_{OB} corresponding to the Soc operation compared with the fixed voltage at V_{OA}

approximately 81% when $\eta_{switching}$ is 90% and η_{LDO} is 90%. Consequently, the dropout voltage of the LDO regulator should be designed more carefully to improve efficiency.

Figure 2.88(a) illustrates the relationship between the dropout voltage V_{DPA} and the current of the power MOSFET in the analog LDO regulator. The performance of voltage ripple suppression depends on the dropout voltage across the pass transistor, which is regarded as a buffer between the input supply voltage and the output voltage. The power MOSFET in the triode region, which operates with a small dropout voltage, behaves as a 1 V controlled resistance. However, its PSR becomes worse than that in the deep saturation region. By contrast, a large dropout voltage in the deep saturation region improves the PSR but deteriorates the PCE, as shown in Figure 2.88(b). The definition of the ADVS technique for analog circuits is to scale the SWR output voltage dynamically, and simultaneously the dropout voltage V_{DPA} according to the output loading condition, and maintaining V_{OAR} constant for analog circuits. That is, the ADVS technique considers the trade-off between efficiency and output voltage ripple without affecting the performance of analog circuits in the Soc [23]. The cooperation between the SWR and the LDO becomes one design issue if the ADVS technique is applied in the Soc.

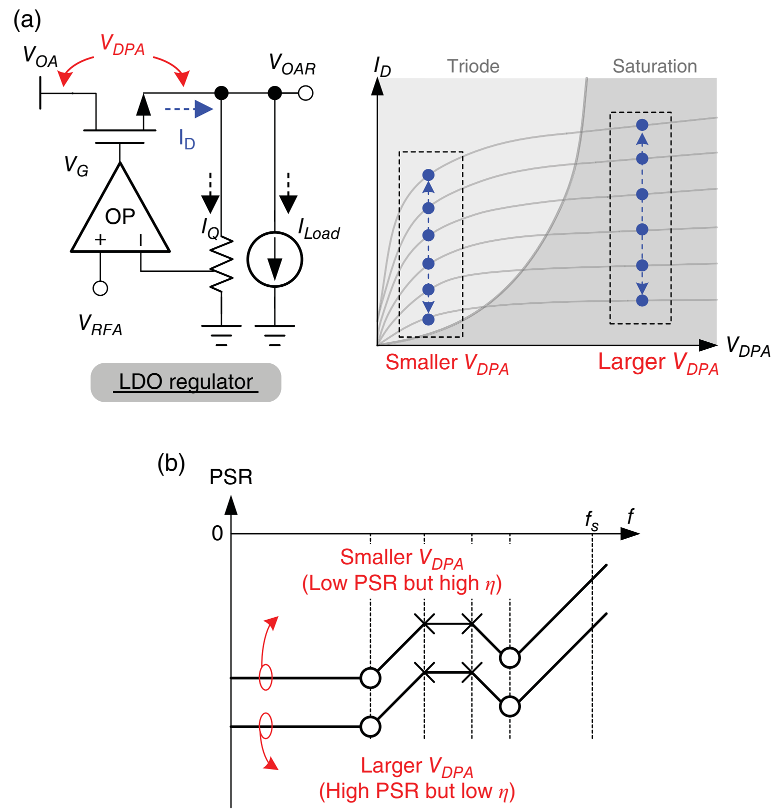

Figure 2.88 (a) Dropout voltage V_{DPA} of the LDO regulator. (b) Relationship between dropout voltage V_{DPA} and PSR

The drain current in the saturation region is shown in Eq. (2.82). The boundary condition between the triode region and the saturation region is defined as the V_{DS} that is equal to V_{OV} in Eq. (2.83). V_{DS} is proportional to the square of the drain current I_D if the channel length modulation parameter λ is neglected.

$$I_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) V_{OV}{}^2 (1 + \lambda V_{DS}) \text{ where } V_{DS} \geq V_{OV} \tag{2.82}$$

$$V_{DS} = V_{OV} \propto \sqrt{I_D} \tag{2.83}$$

As the loading current decreases, only a small dropout voltage is required to ensure proper functioning of the power MOSFET in the saturation region for good PSR. Therefore, a small dropout voltage can eliminate superfluous conduction power loss on the power MOSFET and enhance PCE.

Figure 2.89 compares the operations of both the DVS function in digital circuits and the ADVS function in analog circuits. When the Soc enters into operating mode, the DVS function that is indicated by the system processor is activated to optimize power consumption. The

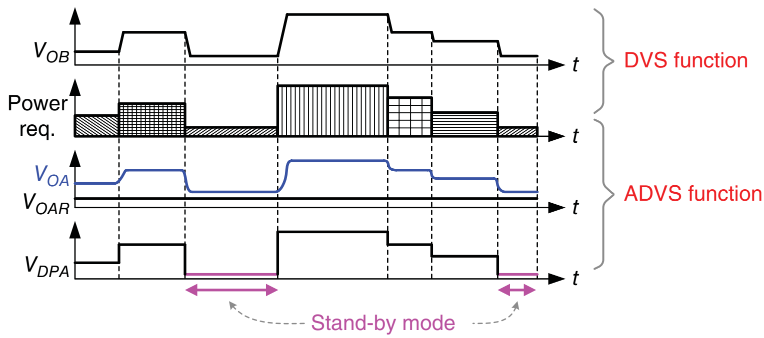

Figure 2.89 Operations of the DVS function in digital circuits and the ADVS function in analog circuits

ADVS function also enables dynamic dropout voltage adjustment of the LDO regulator to guarantee high PSR and PCE. This procedure can be realized by adjusting the output voltage level of V_{OA} corresponding to the loading current condition at V_{OAR}.

2.7.2 *Switchable D/A-LDO Regulator*

To realize versatile power management performance for Soc, reducing power loss in standby mode is a crucial consideration for prolonging battery life. As discussed earlier, the *C-free* LDO regulator encounters obstacles in stability under ultra-light load. In response to the disadvantages of the *C-free* LDO design, the output should dissipate a minimum load current to ensure stability but sacrifice power efficiency. Consequently, the switchable digital/analog low dropout (D/A LDO) regulator has been developed.

When the switchable D/A LDO regulator works in an analog operation, the power MOSFET is driven by an analog amplifier. Through the ADVS technique, the LDO regulator can exhibit good noise suppression. By contrast, the switchable D/A LDO regulator works in a digital operation if the Soc enters in silent or standby mode. The power MOSFET is driven by a digital controller, such as the D-LDO regulator. Although the capability of ripple suppression decreases, the sub-circuits in most applications can tolerate a large voltage variation in standby mode. Moreover, the dropout voltage is reduced to a relatively small value for an energy-efficient operation.

In prior studies, the current flowing through the resistor divider and the quiescent current are equal to tens or hundreds of microamperes. To reduce the loading current to a value lower than the minimum load current required by the conventional *C-free* LDO regulator, the D/A LDO regulator is switched from analog to digital operation. Thus, under ultra-light loads, the pole in the output moves toward the origin and stability decreases drastically. Without the assistance of the dummy load current, the power MOSFET is modulated by the digital controller rather than by the analog controller (the EA) to decrease the quiescent current significantly. Thus, the digital operation of the D/A LDO regulator breaks through the limitations of the minimum load current for the capacitor-free LDO regulator. The digitally operated LDO regulator confirms stability even under a no-load condition, while consuming only 50 nA quiescent current in the controller and 0.5 μA current in the resistor divider. One advantage of this regulator is that the digital controller features a low quiescent current and further enhances the PCE. In turn, a high current efficiency is achieved over a wide range of load current with a compact solution to Soc power management.

The concept of a switchable D/A LDO regulator is illustrated by the *I–V* characteristic curve in Figure 2.90. According to the loading current from the Soc power requirement, analog or digital operation is initiated to ensure voltage regulation and achieve high power efficiency. An analog operation uses the ADVS function to adjust the adaptive dropout voltage to maintain high efficiency.

The right section of Figure 2.90 illustrates the operating behavior of the power MOSFET M_Q in either analog or digital operation. In particular, the EA controls M_Q to determine an adequate gate voltage level that corresponds to the load current condition. Moreover, given the regulated output voltage, V_{OAR}, of the D/A LDO regulator, the loading current adjusts the dropout voltage to determine the adequate output voltage V_{OA} from the SWR. When the loading condition changes from heavy to light, V_{OA} changes from V_{OA_H} to V_{OA_L} and obtains a smaller dropout voltage while simultaneously decreasing the gate voltage level from V_{GH} to V_{GL} for an appropriate driving capability. If the loading current decreases under certain values, analog operation switches to digital operation to save power. During digital operation, the transistor M_Q is divided into several parallel SMUs. Portions of the SMUs are turned on and off by the digital controller according to the load current. The number of turned-on SMUs is denoted *m*. When the load current decreases continuously, the value of *m* shrinks. Thus, the switchable D/A LDO regulator selects suitable analog and digital operations to satisfy the demand from analog circuits in the Soc.

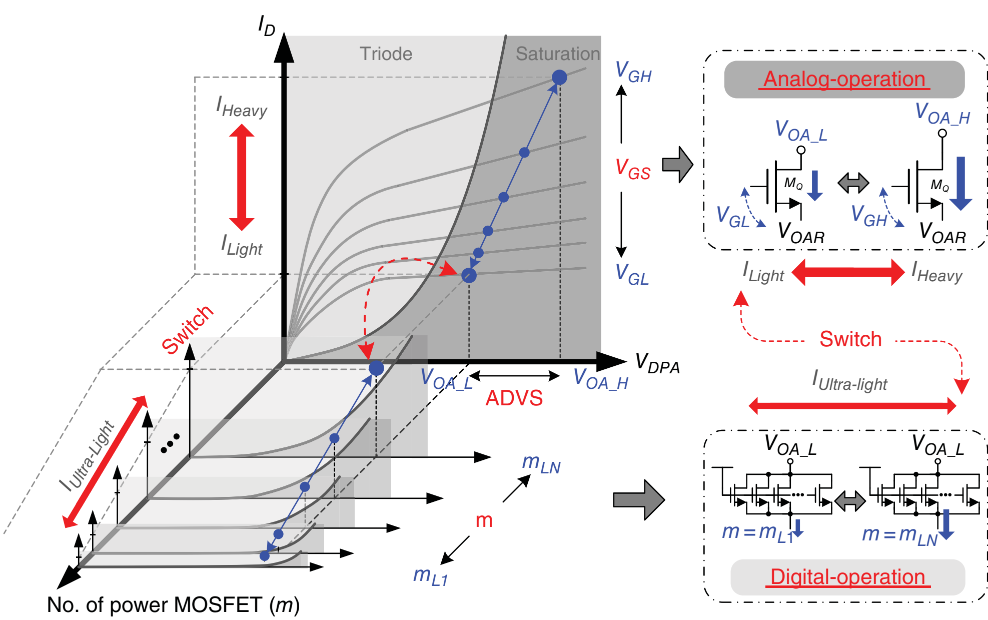

Figure 2.90 *I–V* characteristic curve of the switchable function in analog and digital operations

2.7.2.1 Switchable D/A LDO Regulator

A schematic of the switchable D/A LDO regulator is provided in Figure 2.91. Digital and analog operations indicate that SMUs are controlled by LASC [24] and the EA [25], respectively. The smooth switch technique (SST) decides whether digital or analog operation is used with respect to the loading current condition. The key point is a continuous and smooth switching procedure between digital and analog operations. The D/A selector is structured by an array of multiplexers and controlled by the load-dependent signals V_{AtoD} and V_{DtoA}. Pass transistors M_{Q1} to M_{QN} are controlled by the gate control signals V_{D1} to V_{DN} of the digital controller or the analog signal V_{GA} of the analog controller.

In analog operation, the EA generates the error signal V_{GA} to guarantee the output voltage V_{OAR} under different output loading current conditions, where each gate voltage, from V_{G1} to V_{GN}, of the pass transistors is connected to the error signal V_{GA}. By contrast, in digital operation, the LASC circuit generates thermometer control codes for each of the pass transistors when the Soc enters silent mode. The digital control method releases the minimum load limitations at only several microamperes for the system to conserve power effectively. Moreover, the frozen operation in the LASC circuit reduces the quiescent current to an ultra-low value for high efficiency. The switchable D/A LDO regulator achieves ripple suppression and energy-efficient operation in a distinct mode for high efficiency.

2.7.2.2 Smooth Switch Technique (SST)

The switchable D/A LDO regulator indicates that only one analog/digital operation can be enabled in case of a loading current change. The SST ensures a continuous and smooth takeover procedure between analog and digital operations to prevent undesirable oscillations that induce considerable output voltage ripples. The hysteresis window V_{hys} adjusts the reference voltages of the D/A LDO regulator during the switching procedure. The flowchart in Figure 2.92

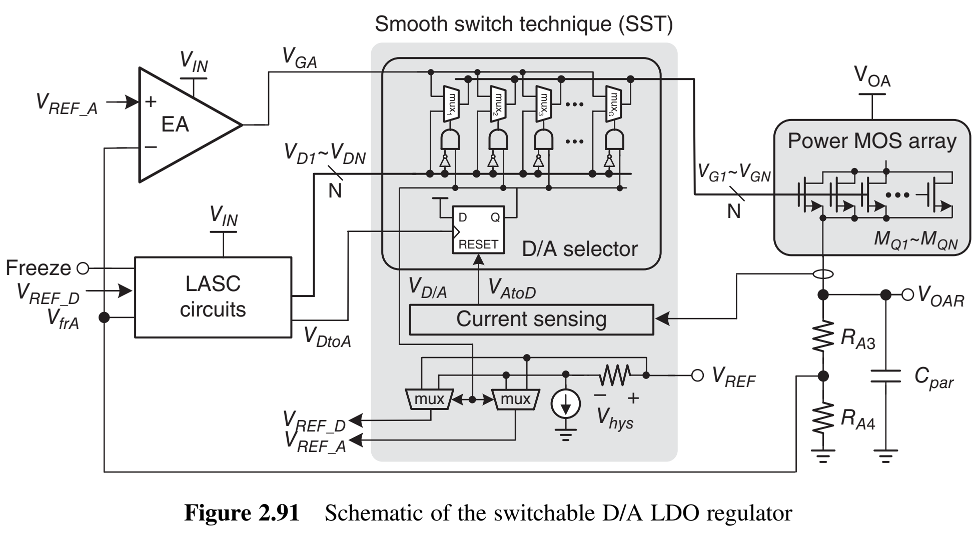

Figure 2.91 Schematic of the switchable D/A LDO regulator

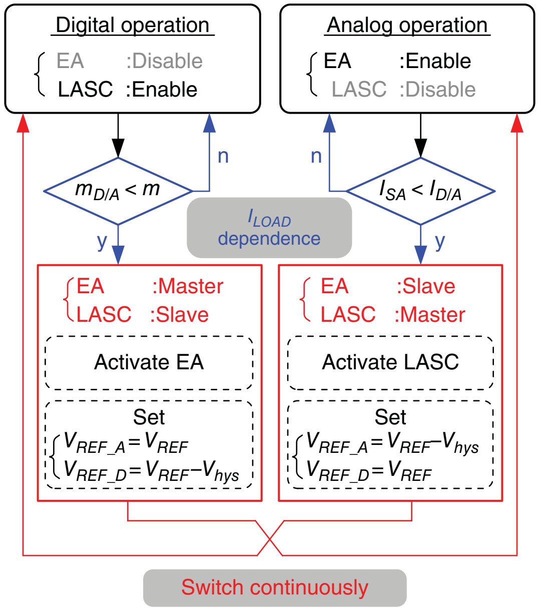

Figure 2.92 Flowchart of the switchable technique

describes the operation procedure during switching. The increasing value of m becomes larger than the predefined $m_{D/A}$, which triggers the operating procedure via the pulse signal V_{DtoA} when the LDO regulator switches from digital to analog operation as the loading current increases. Thus, the EA is activated and the reference voltage of LASC, that is V_{REF_D}, also changes from V_{REF} to the value of $V_{REF} - V_{hys}$. The EA and LASC temporarily operate simultaneously. Meanwhile, the EA gradually dominates control of the SMUs when the feedback voltage V_{frA} is larger than V_{REF_D}. The EA and LASC represent the master and the slave, respectively. Finally, the switching procedure is complete when the EA takes over the operation and disables LASC automatically. By contrast, when the LDO regulator switches from analog to digital operation during loading current decreases, the decreasing sensing current I_{SA} becomes smaller than the predefined $I_{D/A}$ and triggers the switching procedure by the pulse signal V_{AtoD} in Figure 2.91. The following switching procedure is complete via the aforementioned opposite procedure.

The operation waveforms of the SST are shown in Figure 2.93. The currents I_{LDO_D} and I_{LDO_A} flow through the SMUs and are controlled by LASC and the EA, respectively. In cases of increasing loading current, digital operation switches to analog operation, as illustrated in Figure 2.93(a). At $t = t_1$, I_{LDO_D} increases corresponding to the increasing number of turned-on SMUs. At $t = t_2$, if m is larger than $m_{D/A}$ then the EA is enabled and the reference voltage of LASC, V_{REF_D}, changes from V_{REF} to the value of $V_{REF} - V_{hys}$. Consequently, the EA and the LASC function have a master–slave relationship. During this period, the EA controls the remaining switched-off SMUs and I_{LDO_A} subsequently increases to regulate the output voltage V_{OAR}. Once the feedback voltage V_{frA} increases to a value higher than V_{REF_D} at $t = t_3$, I_{LDO_D} is reduced to zero and LASC is automatically disabled. Finally, the switching

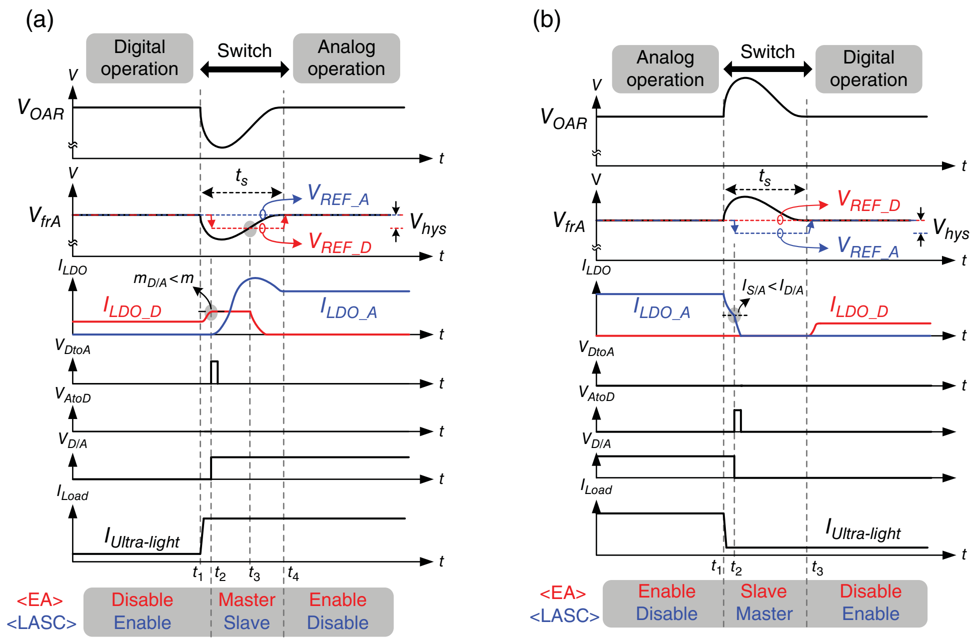

Figure 2.93 Control mechanism of the switchable technique between digital operation and analog operation. (a) Switching from digital operation to analog operation. (b) Switching from analog operation to digital operation

procedure is complete after V_{OAR} recovers at $t = t_4$. Analog operation takes over and enables only the EA. LASC is shut down to enable single analog loop control work.

Analog operation can be switched back to digital operation if the loading current decreases to a sufficiently low value, as shown in Figure 2.93(b). At $t = t_1$, the I_{LDO_A} controlled by the EA decreases continuously. At $t = t_2$, once the condition of I_{SA} is smaller than that of $I_{D/A}$, LASC is enabled and the reference voltage of the EA, V_{REF_A}, changes from V_{REF} to the value of $V_{REF} - V_{hys}$. Thus, I_{LDO_D} gradually increases while I_{LDO_A} decreases to zero. The switching procedure is complete when V_{OAR} is recovered at $t = t_3$.

According to the load condition, the SMUs are controlled by the analog or digital controller determined by the SST. Approximately 10% of the SMUs are driven by the digital controller under light loads. Meanwhile, 90% of the SMUs are turned off. During the D/A switching procedure under light-to-heavy load changing, the analog controller controls a portion of the power MOS units, while the digital controller controls other portions. The analog controller controls all power MOSFET units at the end of the load change. Finally, none of the SMUs is controlled by the digital controller. Consequently, the energy delivered to the output is continuous to retain a low output voltage variation. By reflecting the variation of the output voltage through the feedback network, the condition $m_{D/A} < m$ or $I_{SA} < I_{D/A}$ implies which controller is suitable for use. Thus, the SMUs are adequately controlled even if the loading condition changes faster than the switching time. Upon setting the reference voltage to $V_{REF} - V_{hys}$, stability is confirmed if a moderate or long period is used as the hysteresis window.

2.7.2.3 ADVS PMU with Combined SWR and Switchable D/A LDO Regulator for Soc

Among currently available advanced power management unit (PMU) designs, the single-inductor dual-output (SIDO) converter is used to further reduce PMU size, where the two output voltages are V_{OA} and V_{OB} for analog and digital sub-circuits in the Soc, respectively, as shown in Figure 2.94. The advantage of the SIDO converter is its compact size, which can be attributed to the usage of only one off-chip inductor compared with others that use multiple off-chip inductors [7,19,26–28]. Detailed design guidelines are provided in the next chapter.

For high efficiency, the ADVS technique can be implemented in the PMU design. That is, the combination of SWR and the switchable D/A LDO regulator can provide ADVS PMU for Soc. With regard to PMU design using the dual DVS techniques in Soc, the SIDO converter shown in Figure 2.95 has two outputs. The first output is for digital sub-circuits and the second, with the switchable D/A LDO regulator, is for analog sub-circuits.

With only one off-chip inductor, the power stage of the SIDO converter is composed of four power switches from M_1 to M_4. These switches transfer the energy from the input battery to both outputs, V_{OA} and V_{OB}. V_{OB} supplies digital circuits directly to the Soc. In DVS operation, the DVS indicator receives the control signal V_{Bit} from the system processor to generate the reference voltage V_{RFB} for V_{OB}. By contrast, the switchable D/A LDO regulator is cascaded in series with the V_{OA} of the SIDO converter to guarantee supply quality for analog circuits.

When the D/A LDO regulator is operated using the analog function, the load current condition at V_{OA} is reflected to the reference adjuster of ADVS modulation to adjust the dropout voltage dynamically, which subsequently leads to voltage ripple suppression or power conservation. If the output load condition at V_{OA} continuously decreases to an ultra-light condition, then the switchable D/A LDO regulator is switched to digital operation. Thus, the quiescent current decreases because the digital controller and the dropout voltage are minimized to enhance power efficiency further. The load-dependent technique indicates the switching procedure between analog and digital operations in the switchable D/A LDO regulator. Therefore, with the feedback control loop in the SIDO power module, the ADVS and DVS functions are achieved at V_{OA} and V_{OB}, respectively. The SIDO controller uses the current-programmed

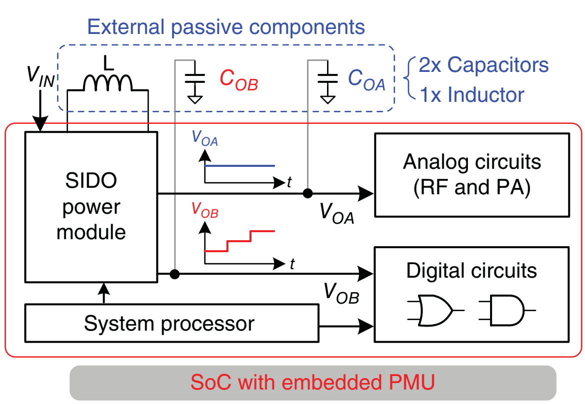

Figure 2.94 SIDO converter used in PMU design to deliver the advantage of a compact size solution

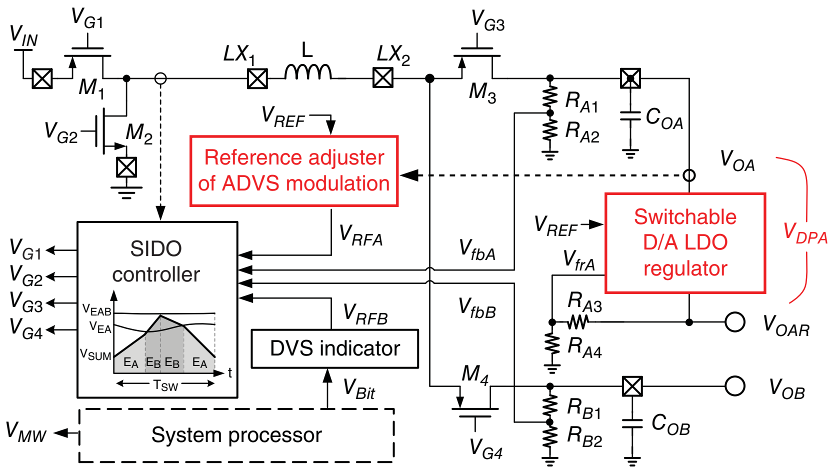

Figure 2.95 Structure of the SIDO power module for dual DVS functions

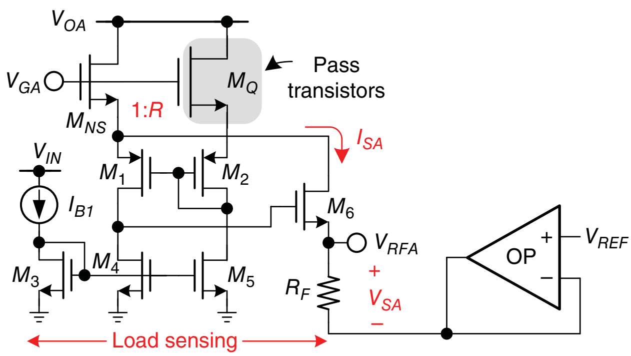

Figure 2.96 Reference adjuster in the ADVS technique

control scheme [28] to realize the energy delivery function at the power stage and ensure high power efficiency.

A schematic of the reference adjuster in ADVS is presented in Figure 2.96. The current flowing through the pass transistors must be monitored to adjust the dropout voltage dynamically in the analog LDO regulator. The load-sensing circuit obtains the supply current information. Then, the replica-sensed current I_{SA} generates the reference voltage V_{RFA} by resistor R_F to modulate the output voltage level of V_{OA} properly and achieve an optimal dropout voltage in the switchable D/A LDO regulator, which operates in the analog operation. V_{SA} reflects the dropout voltage corresponding to the loading current condition in the analog LDO regulator. That is, an adequate dropout voltage is maintained in the analog LDO regulator with regard to the loading current condition. High efficiency and PSR can be maintained in the analog LDO regulator. By contrast, the D/A switchable LDO regulator is switched to digital operation if the loading current condition is under light loads. The major reason for this procedure is to break through the limitation requested in the *C-free* LDO regulator. The digital operation of the D/A

LDO regulator reduces the minimum load current requirement to near zero [7]. Meanwhile, the dropout voltage is continuously minimized and the quiescent current can be reduced further to maintain high efficiency even under ultra-light loads. The switchable D/A LDO regulator ensures high efficiency over a wide load range and implements the ADVS technique in the PMU design.

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